



US012416878B2

(12) **United States Patent**
Takahashi et al.

(10) **Patent No.:** **US 12,416,878 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **INFORMATION STORAGE SYSTEM
INCLUDING A PLURALITY OF TERMINALS**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/243,109**

(22) Filed: **Sep. 7, 2023**

(65) **Prior Publication Data**
US 2023/0418181 A1 Dec. 28, 2023

Related U.S. Application Data

(60) Continuation of application No. 17/869,795, filed on
Jul. 21, 2022, now Pat. No. 11,768,448, which is a
(Continued)

(30) **Foreign Application Priority Data**

Jun. 11, 2010 (JP) 2010-134560
Mar. 22, 2011 (JP) 2011-062216
(Continued)

(51) **Int. Cl.**
G03G 21/16 (2006.01)
B41J 2/175 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G03G 15/0863** (2013.01); **B41J 2/17526**
(2013.01); **B41J 2/1753** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC G03G 15/0863; G03G 15/0865; G03G
15/0872; G03G 15/0875; G03G 15/0886
(Continued)

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No. 2019-231553, 2 pages.

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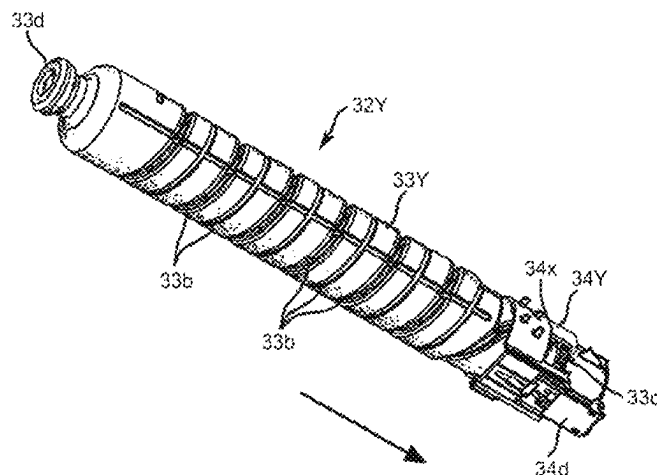
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(57) **ABSTRACT**

A toner container includes a container body, an opening, a
cover, a toner discharge port, a shutter, a cylindrical struc-
ture, protrusions, and an identifier. The shutter is attached to
the cover and includes a main shutter covering the toner
discharge port. The main shutter moves between a closed
position to close the toner discharge port and an open
position to open the toner discharge port. The protrusions are

(Continued)



on both sides of the cover and protrude in a direction orthogonal to a toner container mounting direction. The identifier is on the cover to indicate a toner color. The cylindrical structure's leading end is downstream of leading ends of the protrusions in the mounting direction. The main shutter's leading end is downstream of the protrusions' leading ends in the mounting direction when the main shutter is in the closed position. The protrusions are downstream of the identifier in the mounting direction.

15 Claims, 58 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/110,348, filed on Dec. 3, 2020, now Pat. No. 11,429,036, which is a continuation of application No. 16/929,446, filed on Jul. 15, 2020, now Pat. No. 11,188,007, which is a continuation of application No. 16/584,982, filed on Sep. 27, 2019, now Pat. No. 10,725,398, which is a continuation of application No. 15/970,427, filed on May 3, 2018, now Pat. No. 10,754,275, which is a continuation of application No. 15/430,261, filed on Feb. 10, 2017, now Pat. No. 9,989,887, which is a continuation of application No. 14/750,679, filed on Jun. 25, 2015, now Pat. No. 9,599,927, which is a continuation of application No. 14/047,755, filed on Oct. 7, 2013, now Pat. No. 9,110,402, which is a division of application No. 13/691,023, filed on Nov. 30, 2012, now Pat. No. 8,660,441, which is a continuation of application No. 13/448,987, filed on Apr. 17, 2012, now Pat. No. 8,346,105, which is a continuation of application No. PCT/JP2011/063993, filed on Jun. 13, 2011.

(30) Foreign Application Priority Data

Mar. 22, 2011 (JP) 2011-062283
Apr. 6, 2011 (JP) 2011-084820
Apr. 11, 2011 (JP) 2011-087786

(51) Int. Cl.

G03G 15/00 (2006.01)
G03G 15/08 (2006.01)
G03G 21/00 (2006.01)
G03G 21/18 (2006.01)

(52) U.S. Cl.

CPC **B41J 2/17546** (2013.01); **G03G 15/0865** (2013.01); **G03G 15/0872** (2013.01); **G03G 15/0875** (2013.01); **G03G 15/0886** (2013.01); **G03G 15/0891** (2013.01); **G03G 15/50** (2013.01); **G03G 15/5066** (2013.01); **G03G 15/55** (2013.01); **G03G 15/553** (2013.01); **G03G 21/00** (2013.01); **G03G 21/16** (2013.01); **G03G 21/1652** (2013.01); **G03G 21/1661** (2013.01); **G03G 21/1676** (2013.01); **G03G 21/1875** (2013.01); **G03G 21/1885** (2013.01); **G03G 2215/0697** (2013.01); **G03G 2221/166** (2013.01); **G03G 2221/1823** (2013.01); **Y10T 29/49002** (2015.01)

(58) Field of Classification Search

USPC 399/252, 262, 269
See application file for complete search history.

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FIG.1

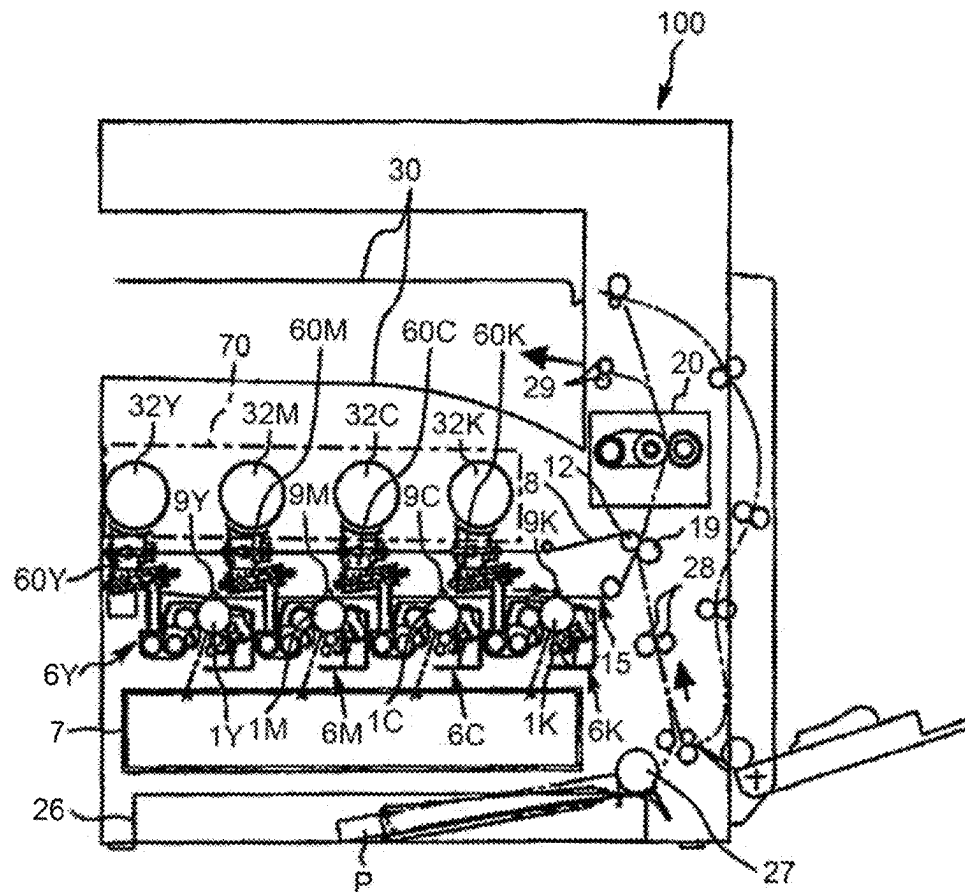


FIG.2

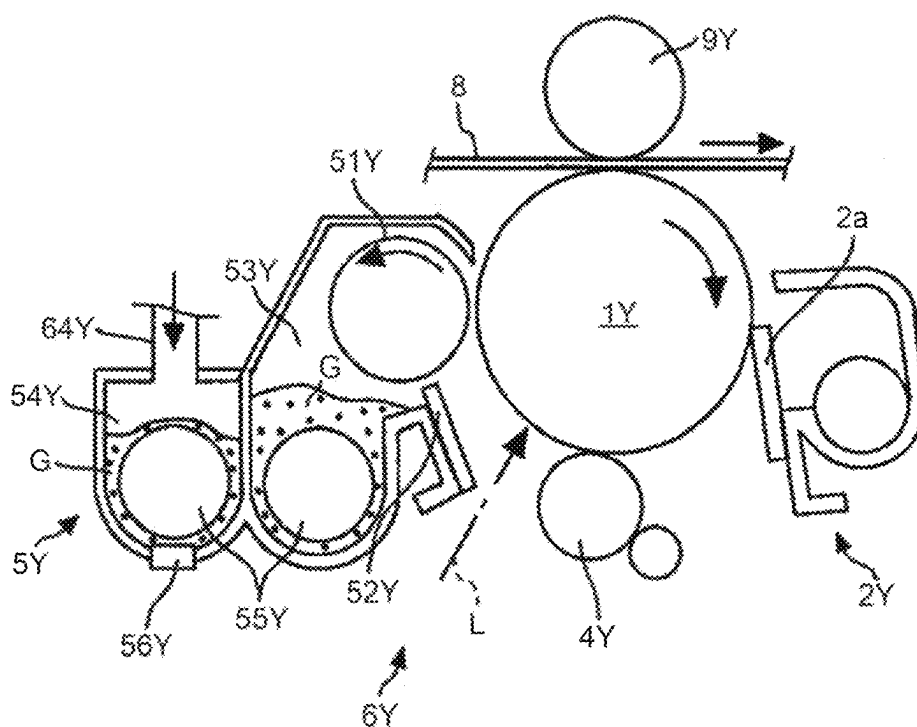


FIG.3

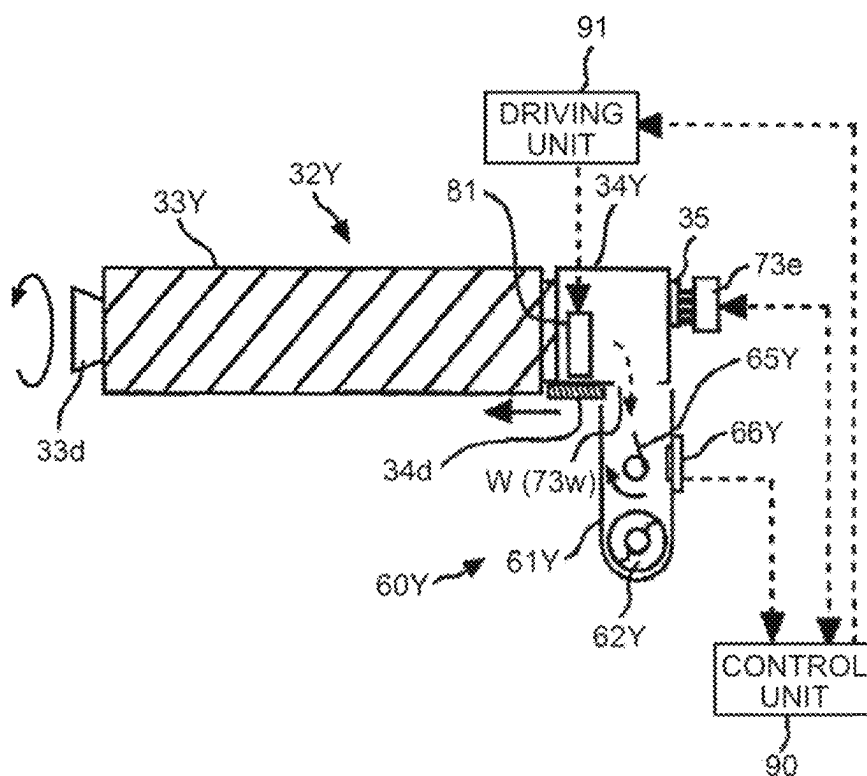


FIG.4

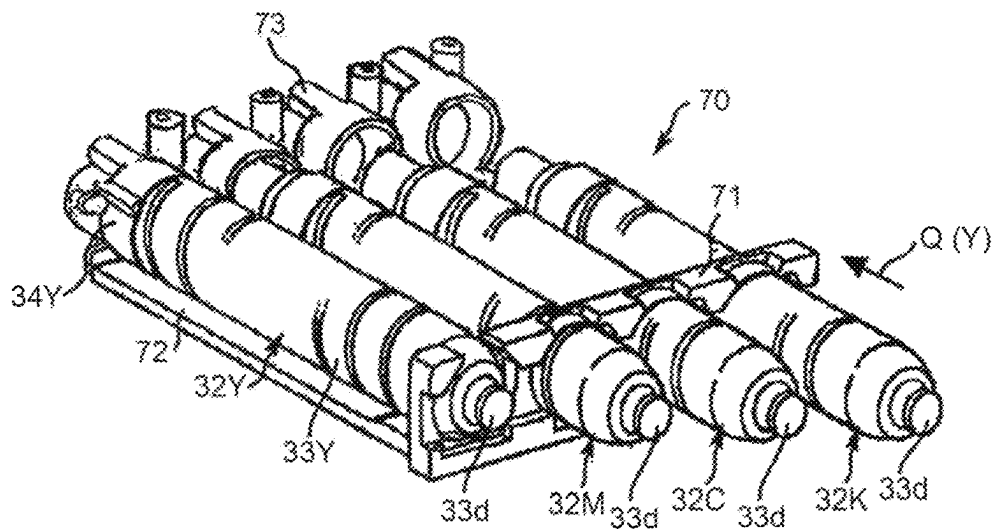


FIG.5

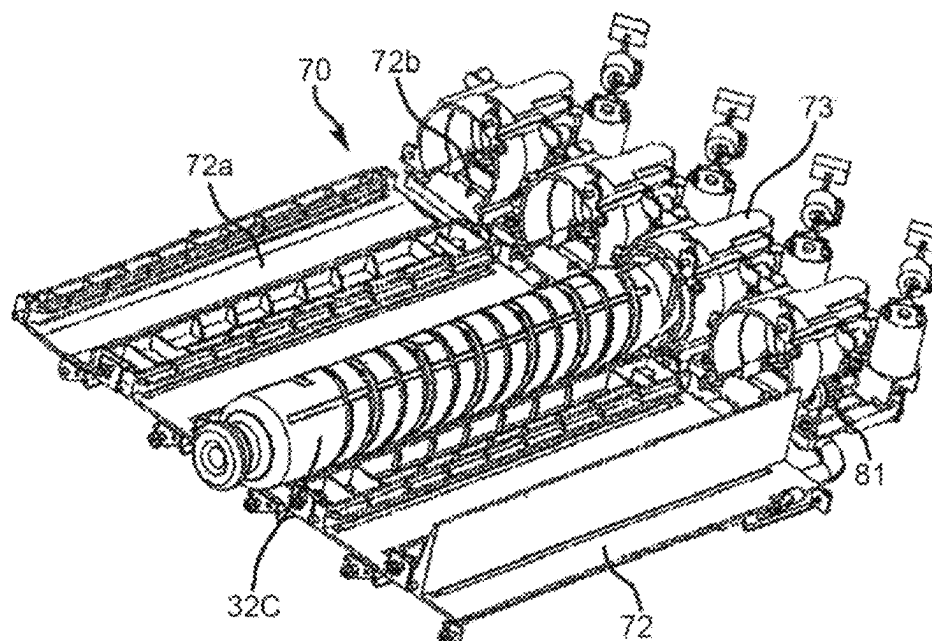


FIG.6

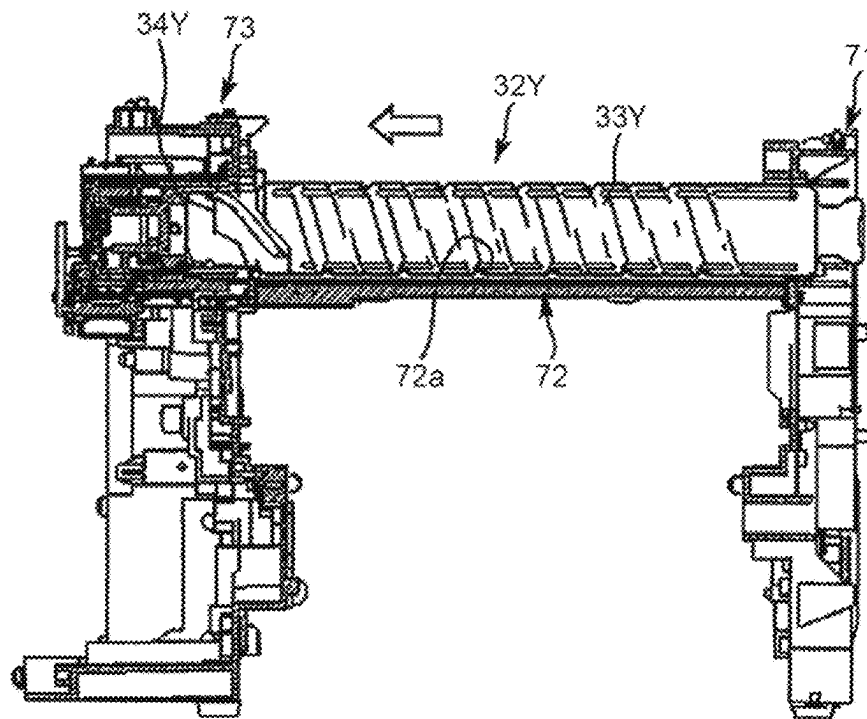


FIG.7

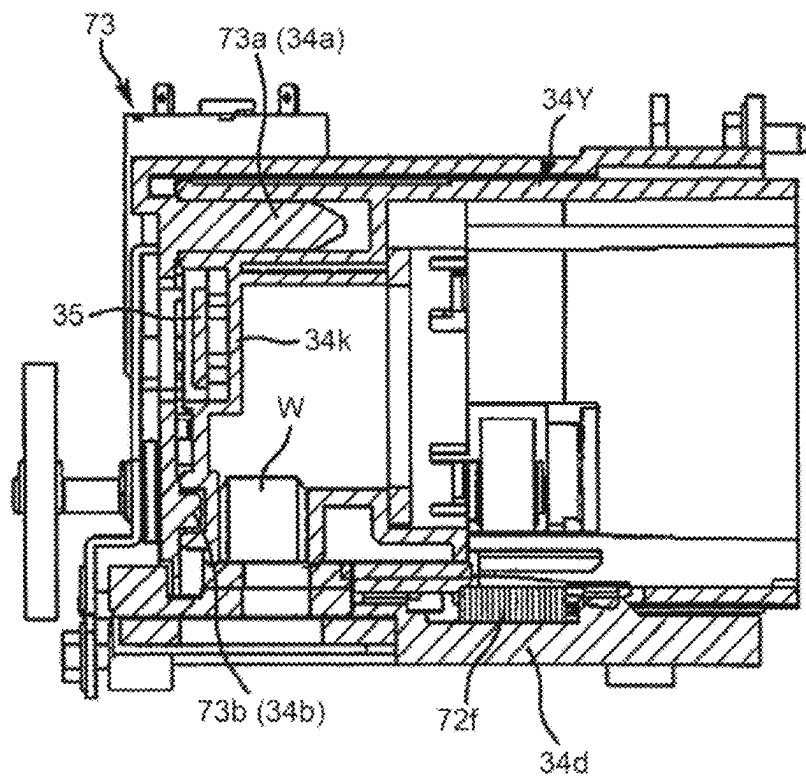


FIG.8

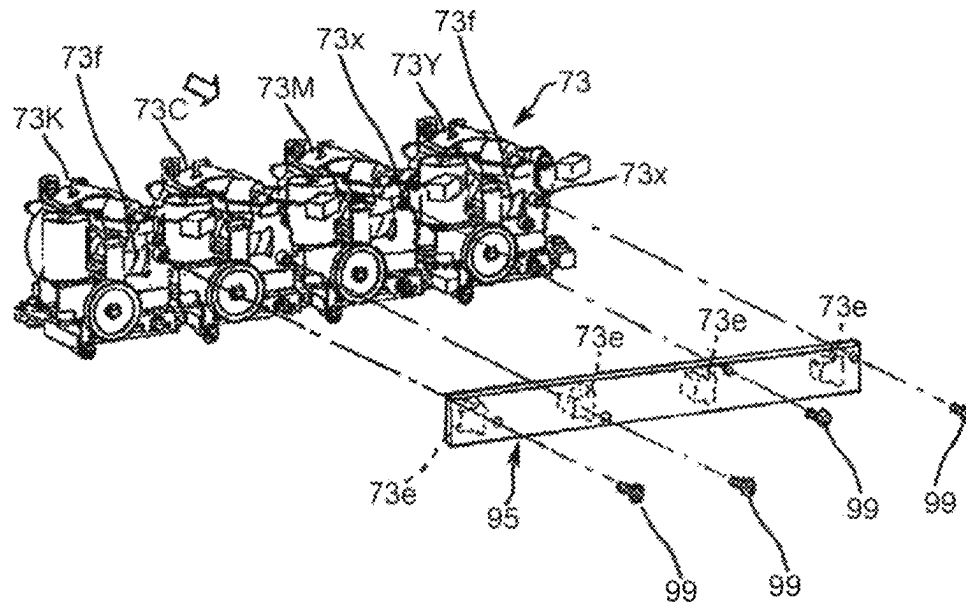


FIG.9

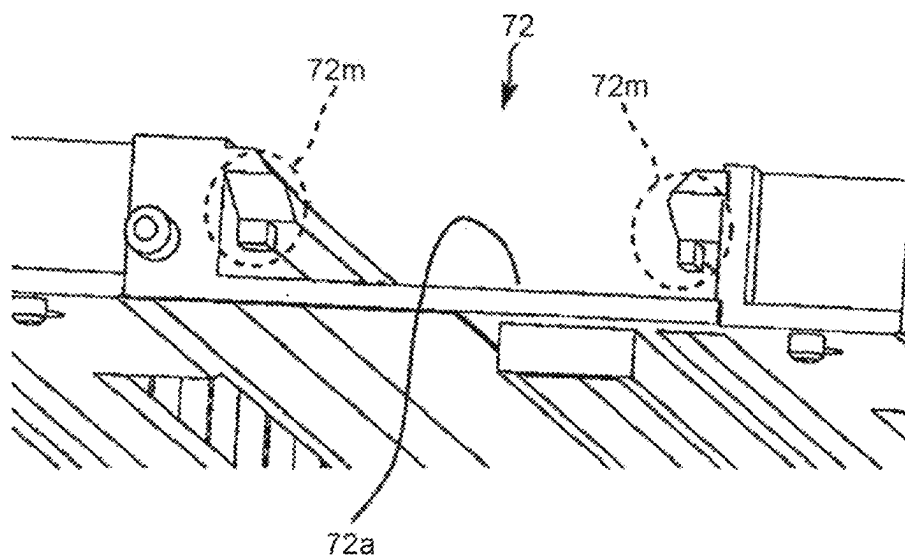
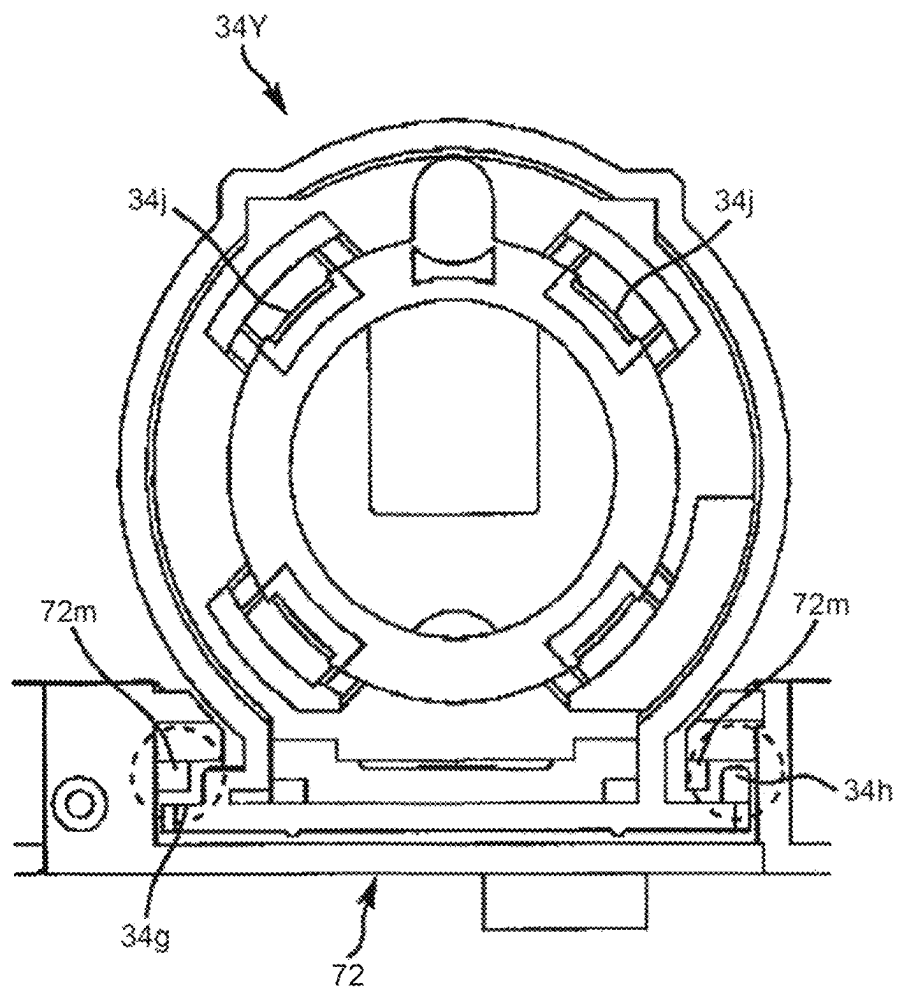


FIG.10



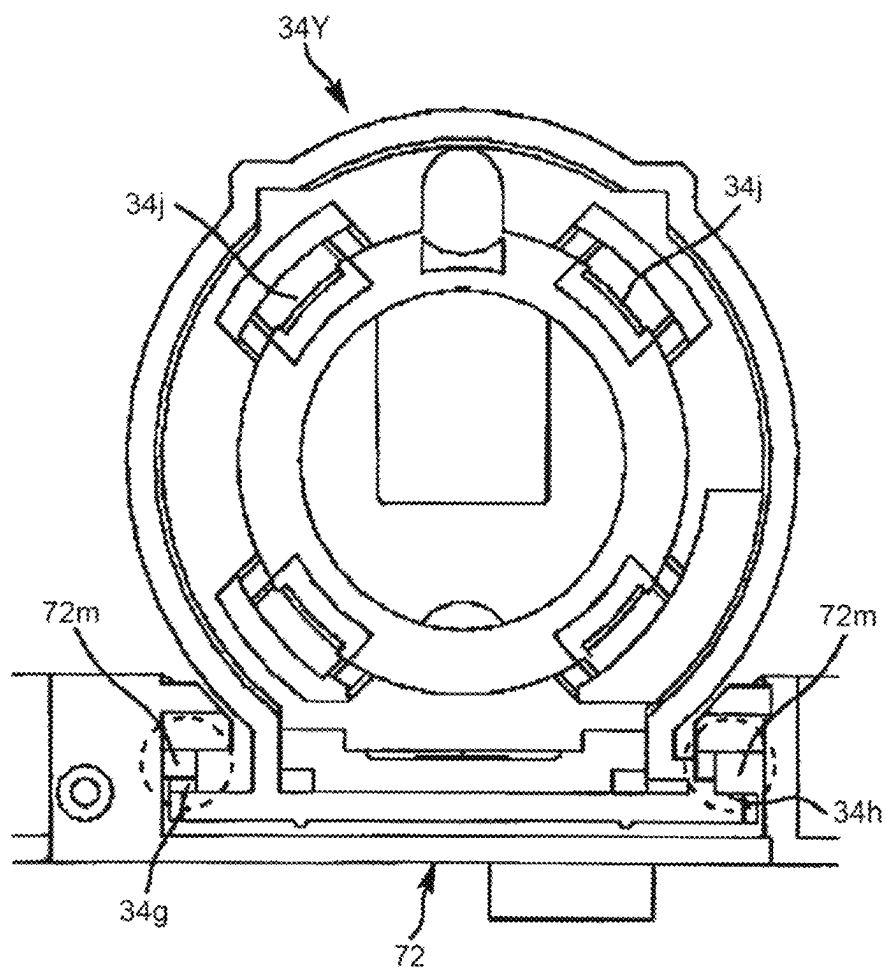


FIG.12

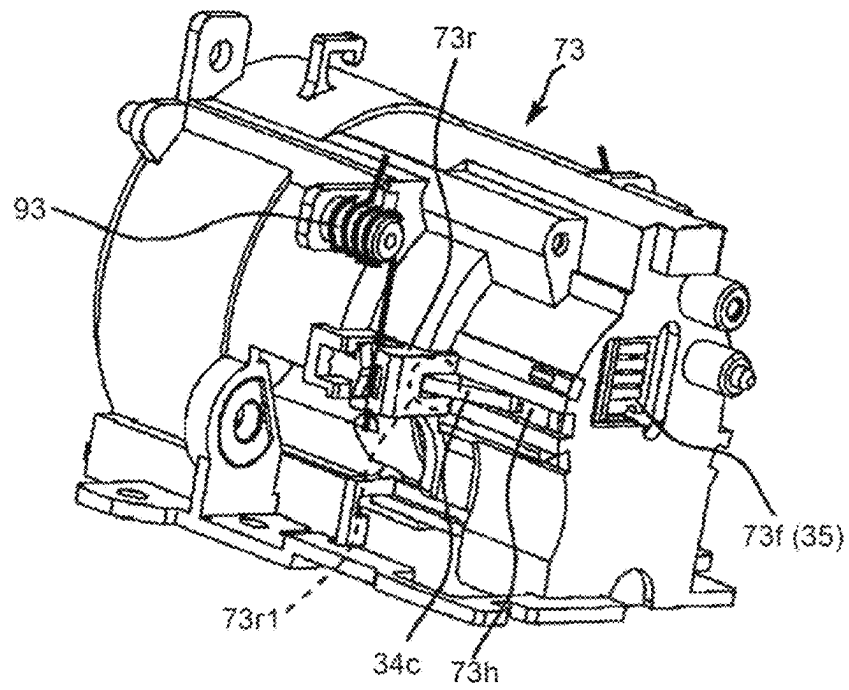


FIG.13

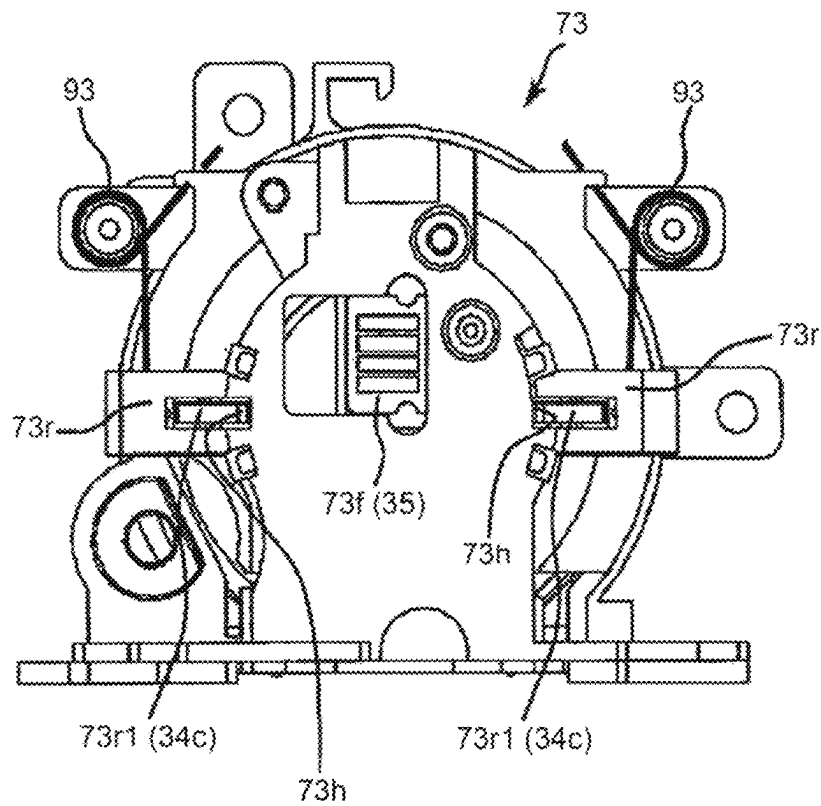


FIG.14A

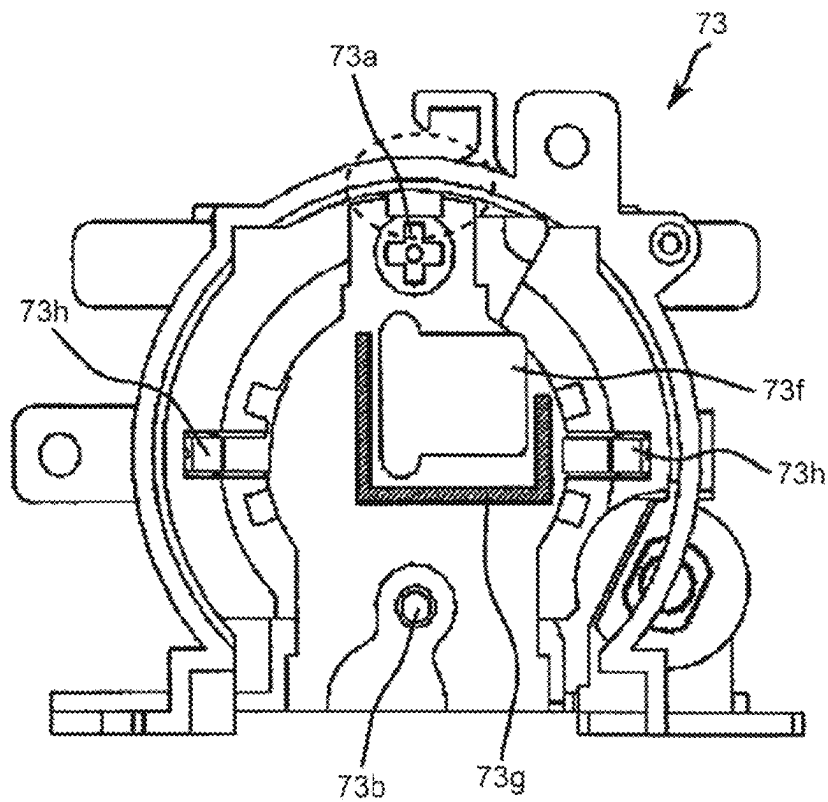


FIG.14B

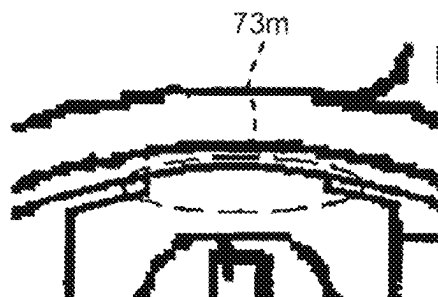


FIG. 15

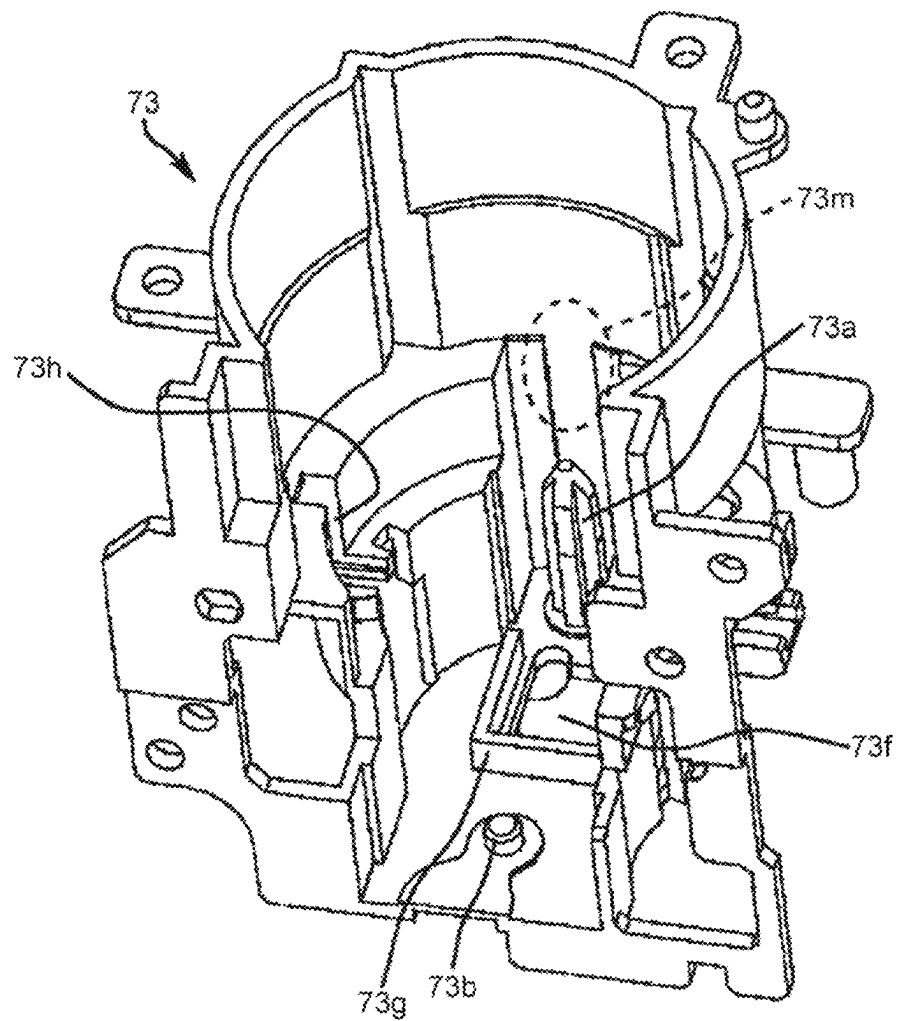


FIG.16

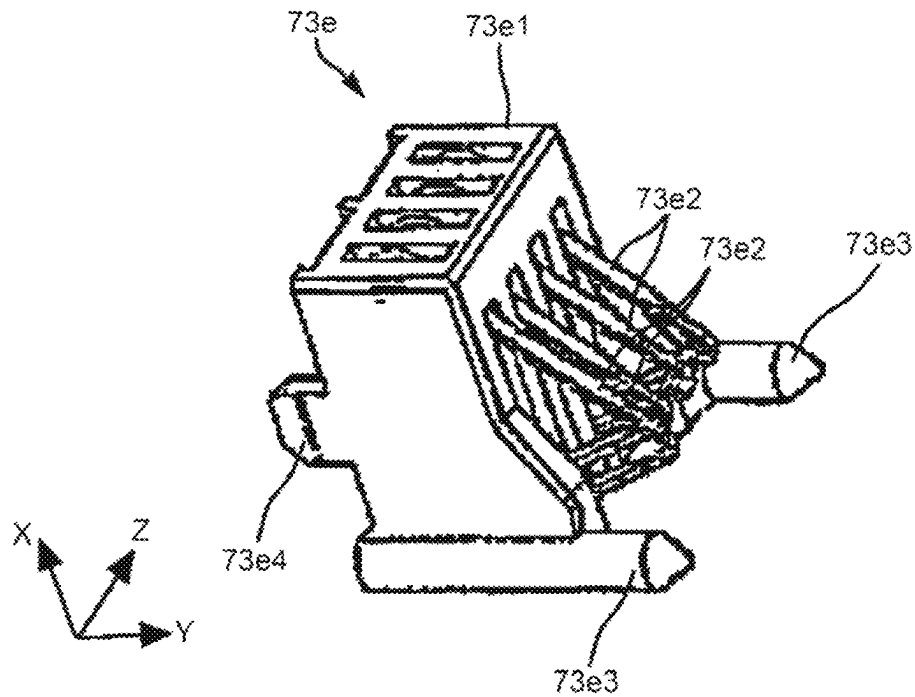


FIG.17

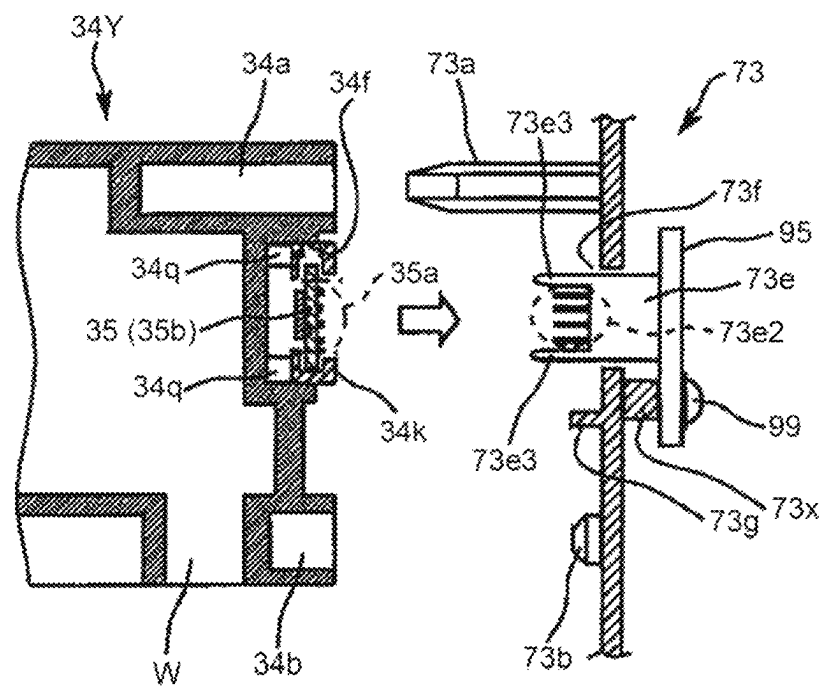


FIG.18

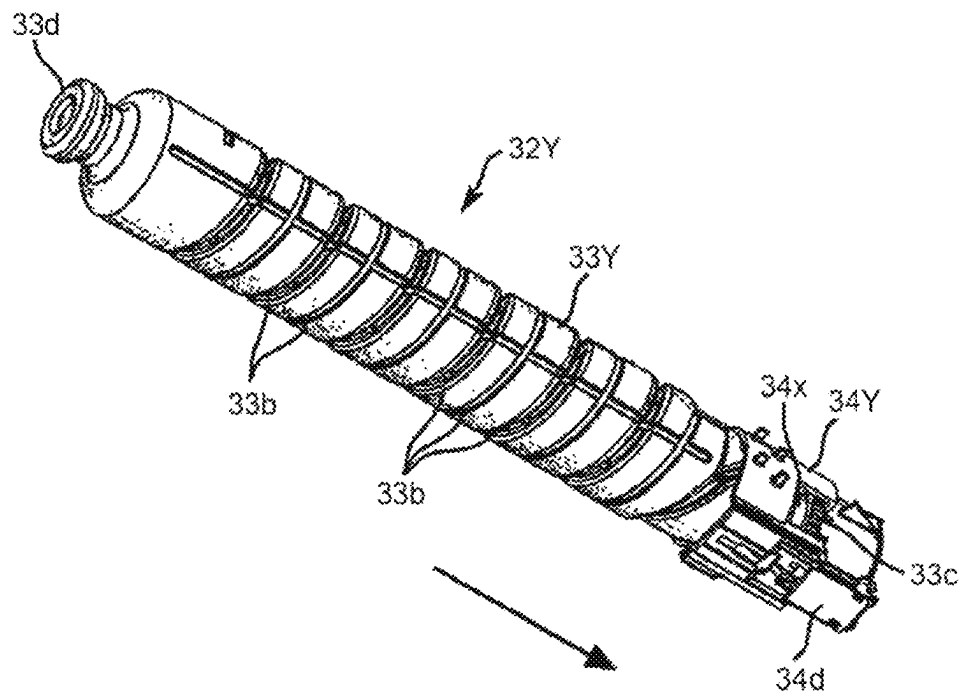


FIG.19

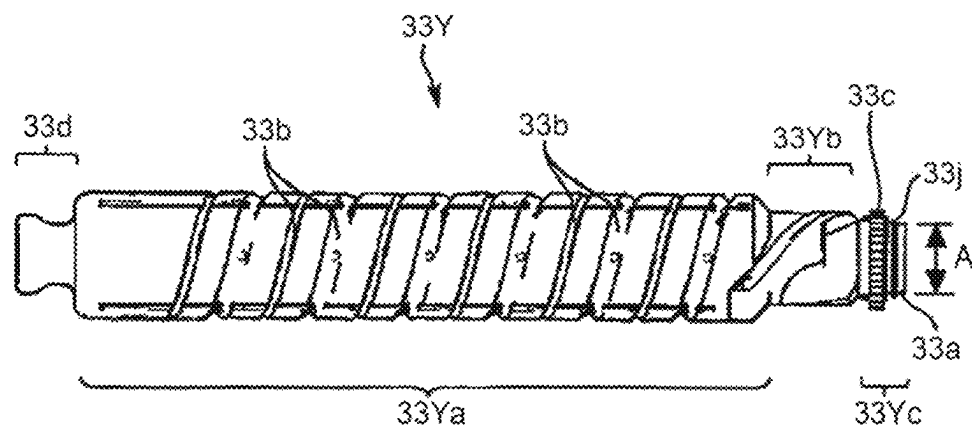


FIG.20

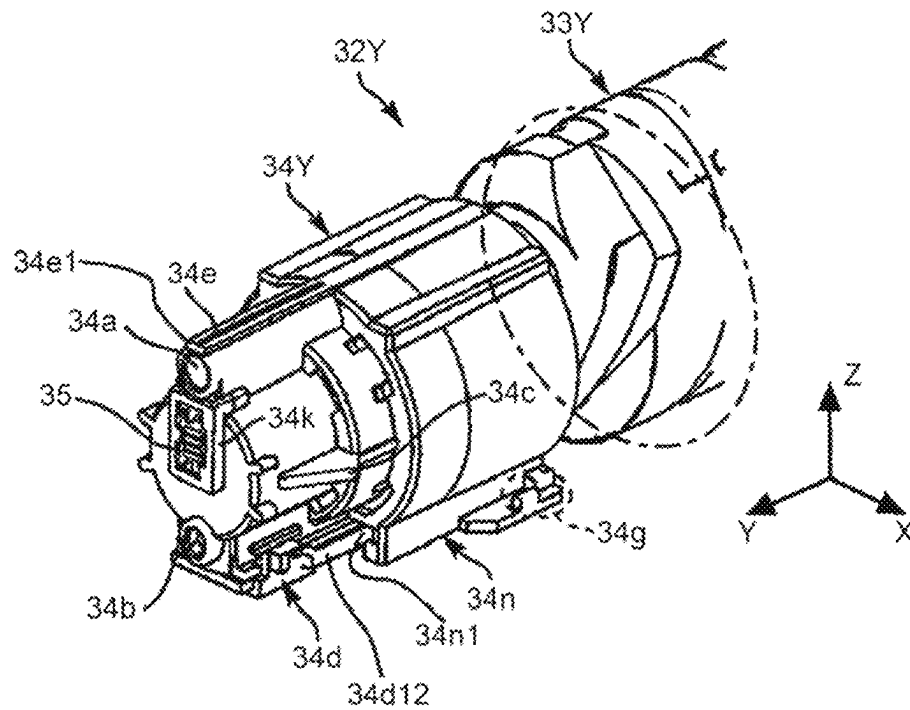


FIG.21

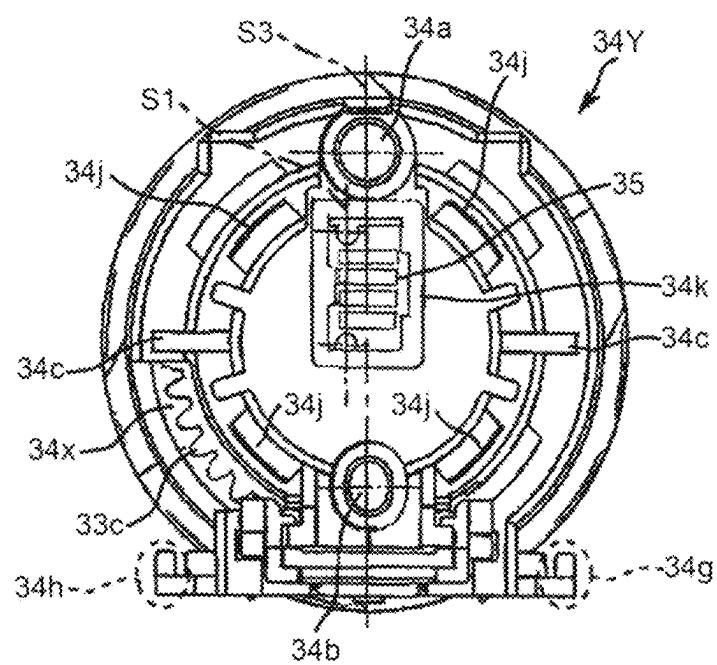


FIG.22

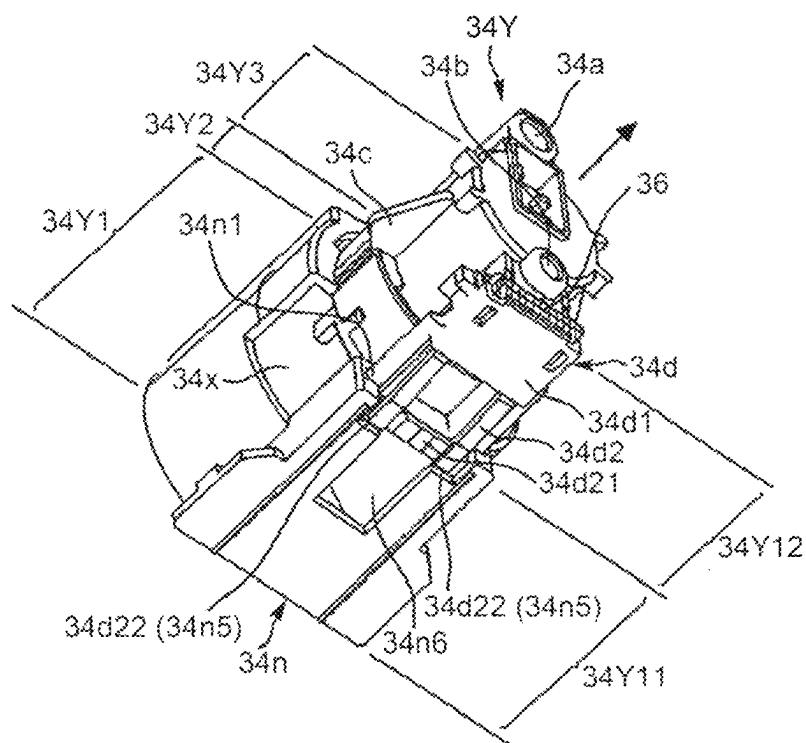
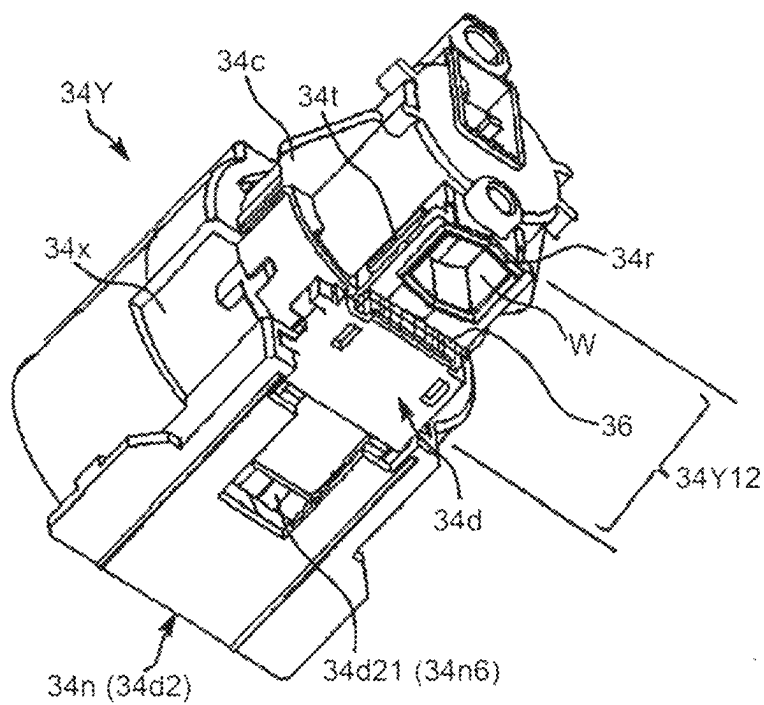


FIG.23



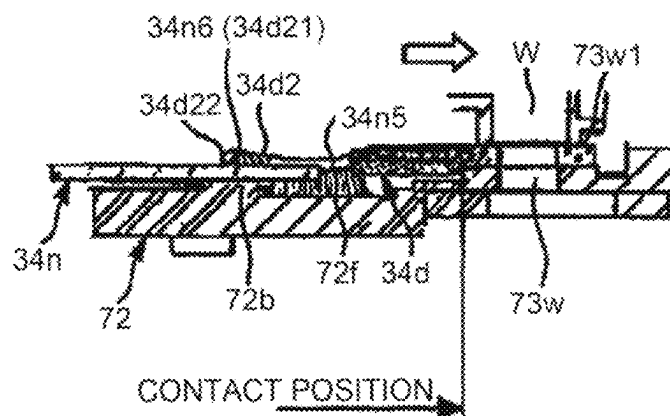


FIG.25

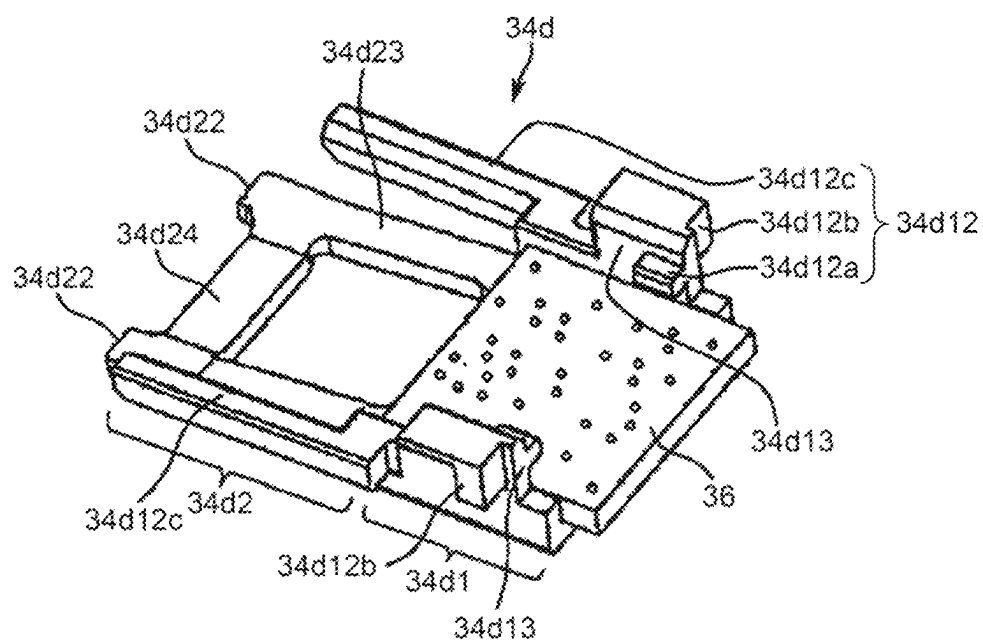


FIG.26

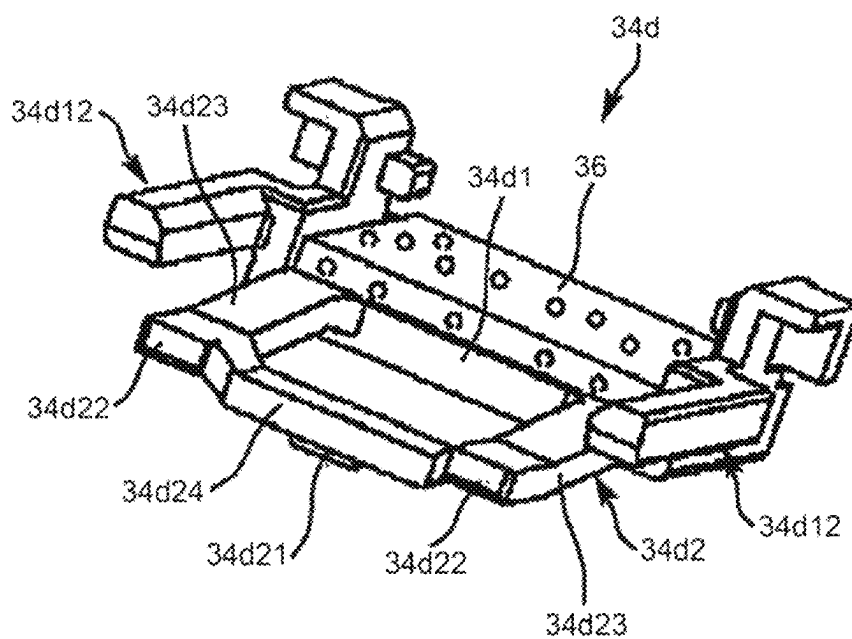


FIG.27

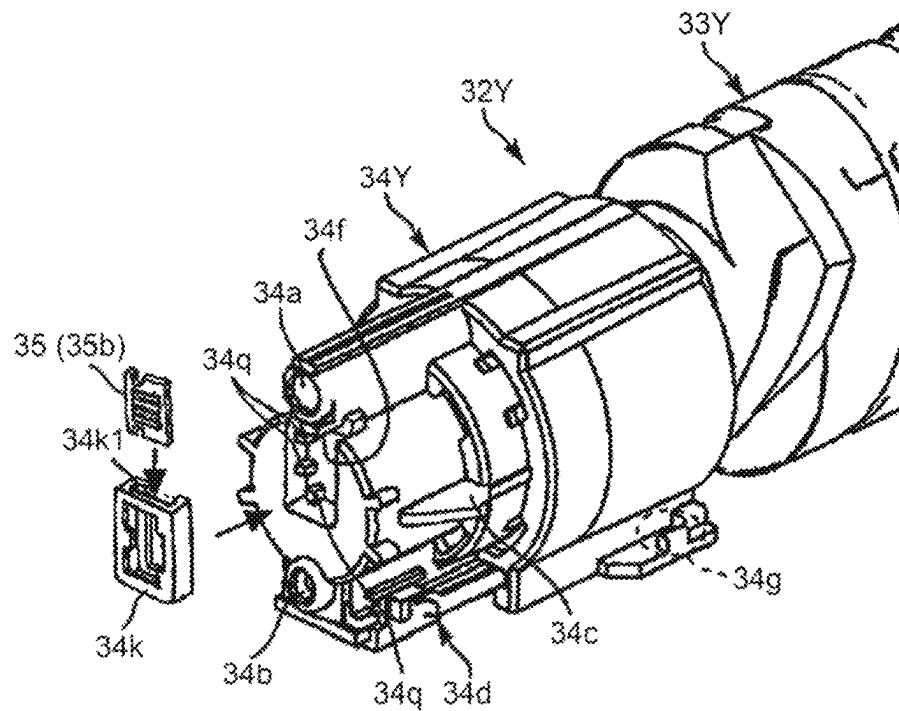


FIG.28

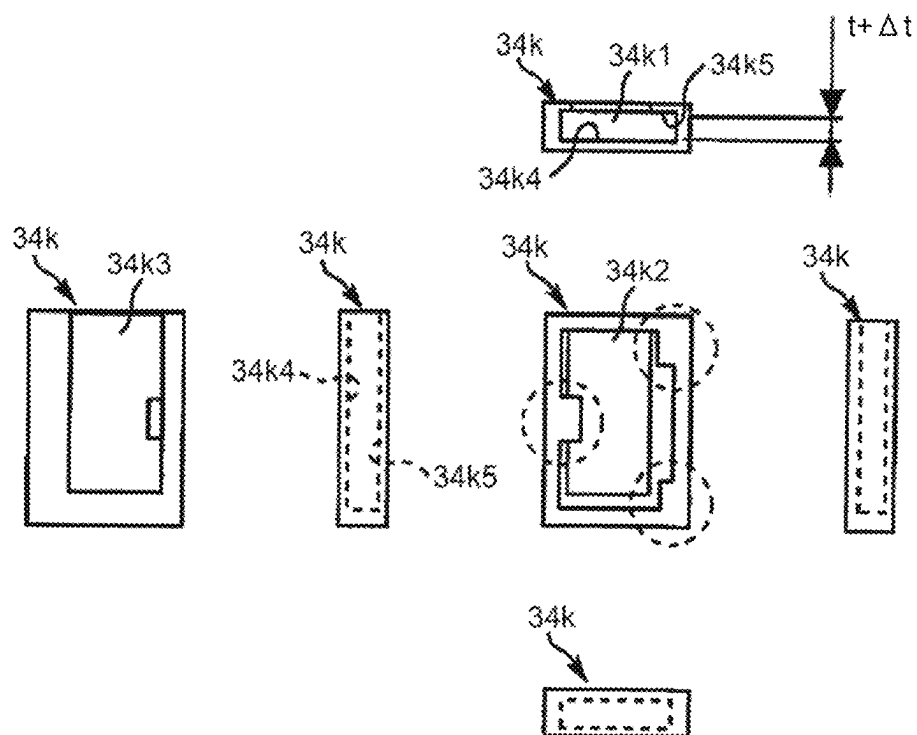


FIG.29

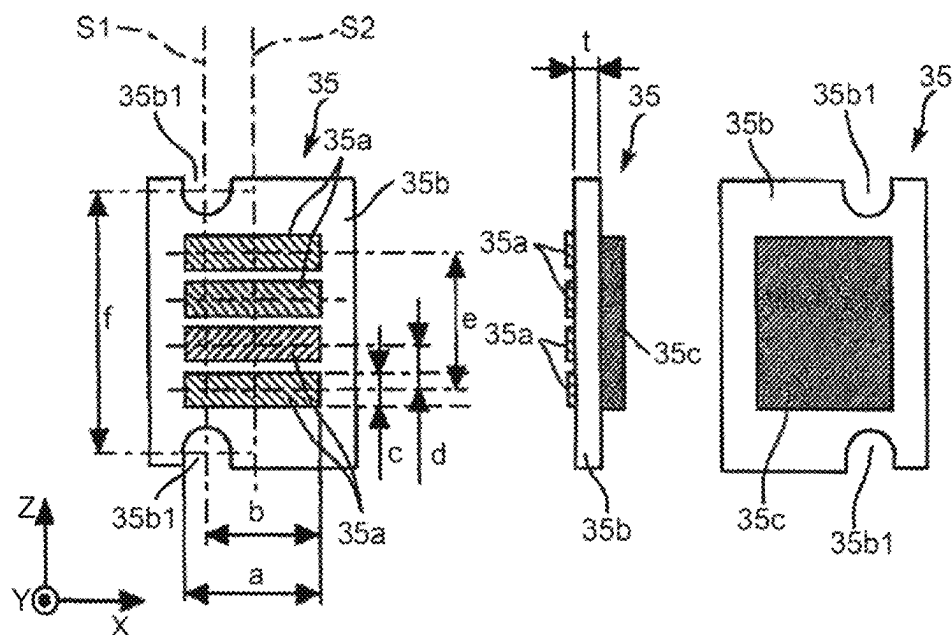


FIG.30

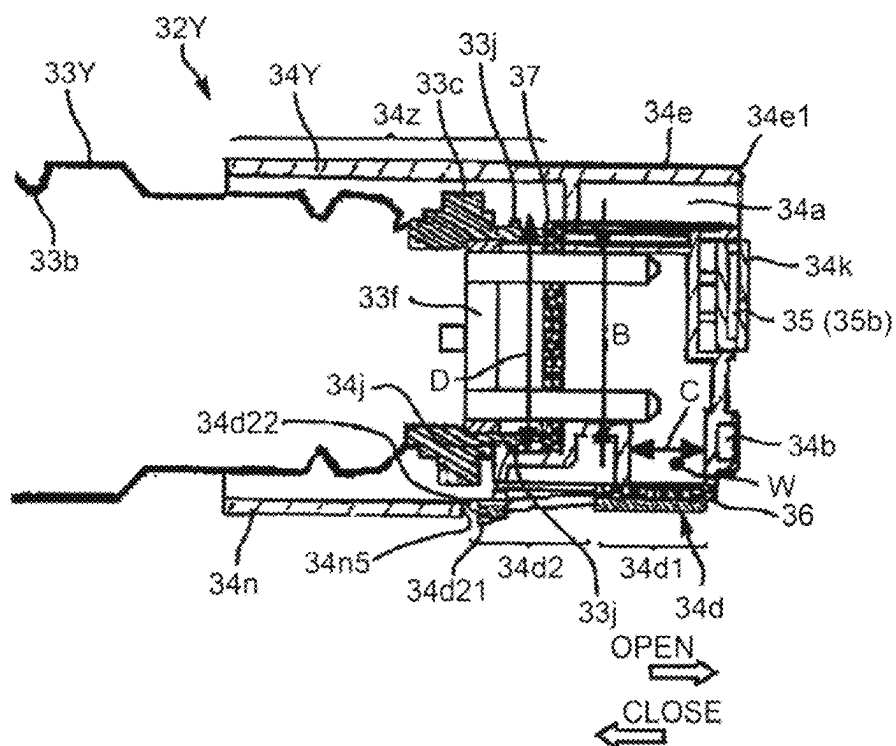


FIG.33

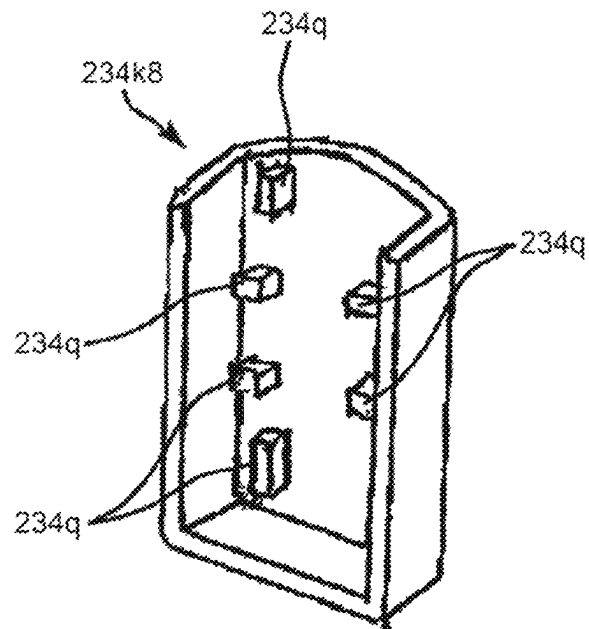


FIG.34

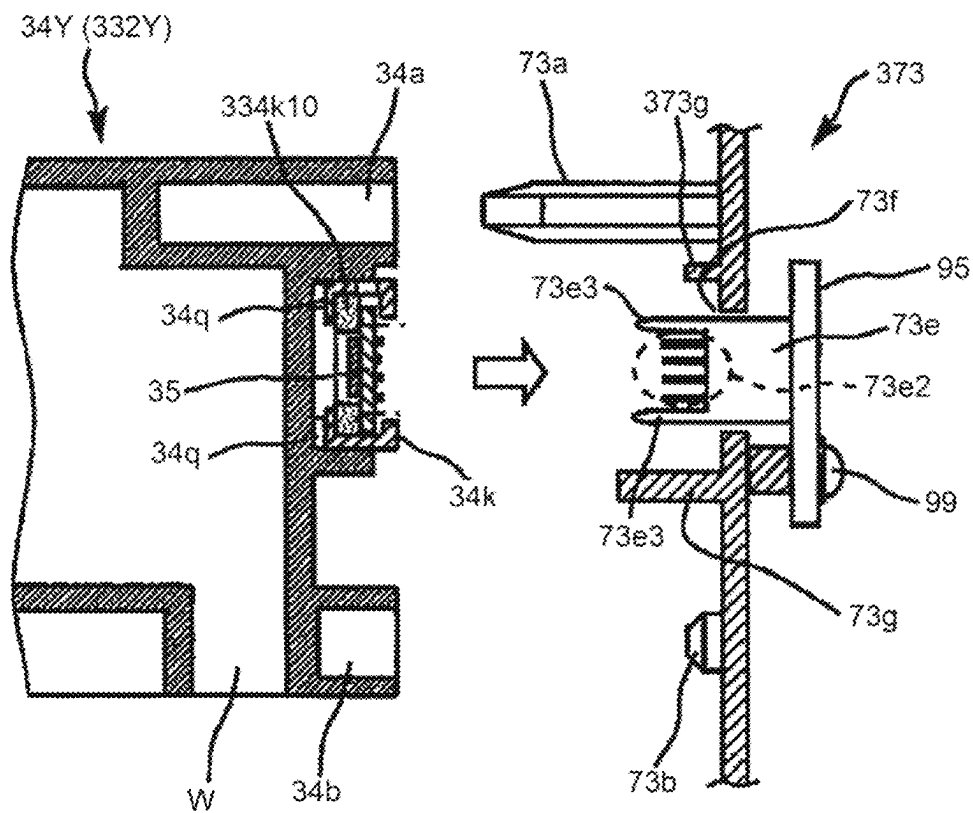


FIG.35

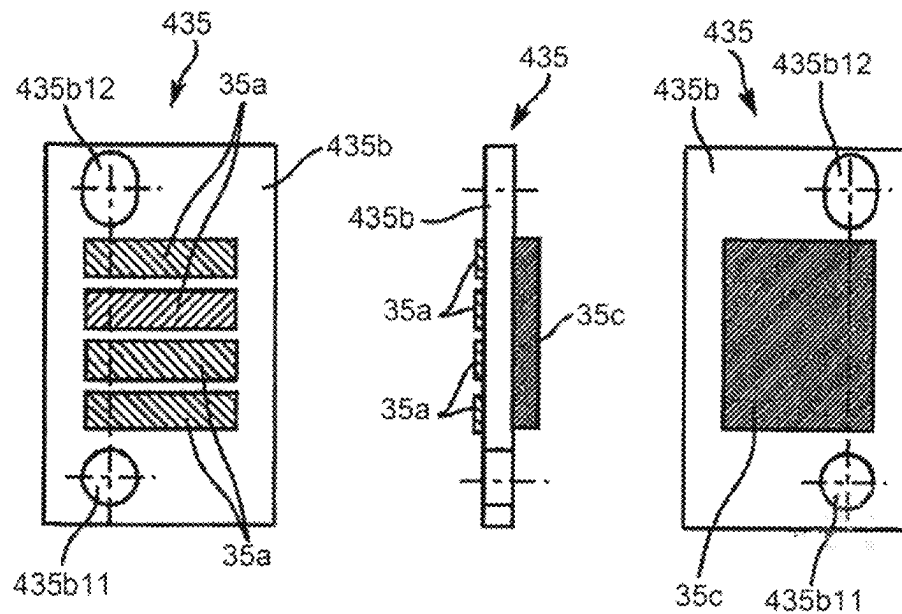


FIG.36

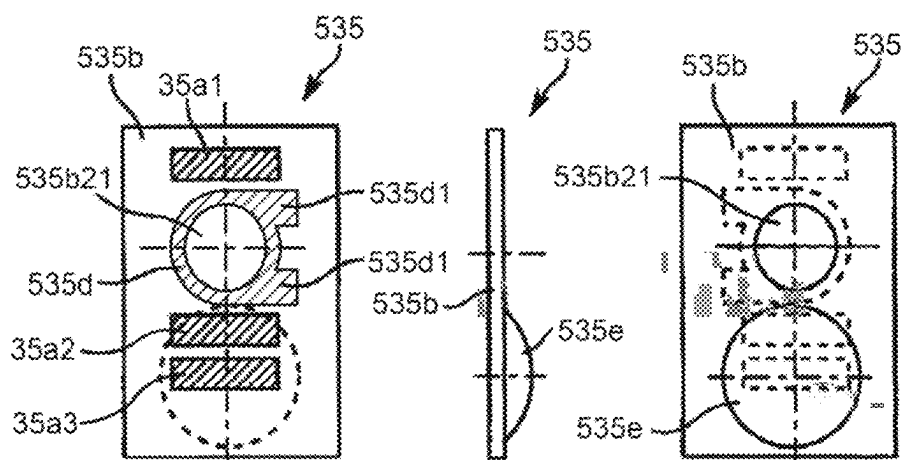


FIG.37

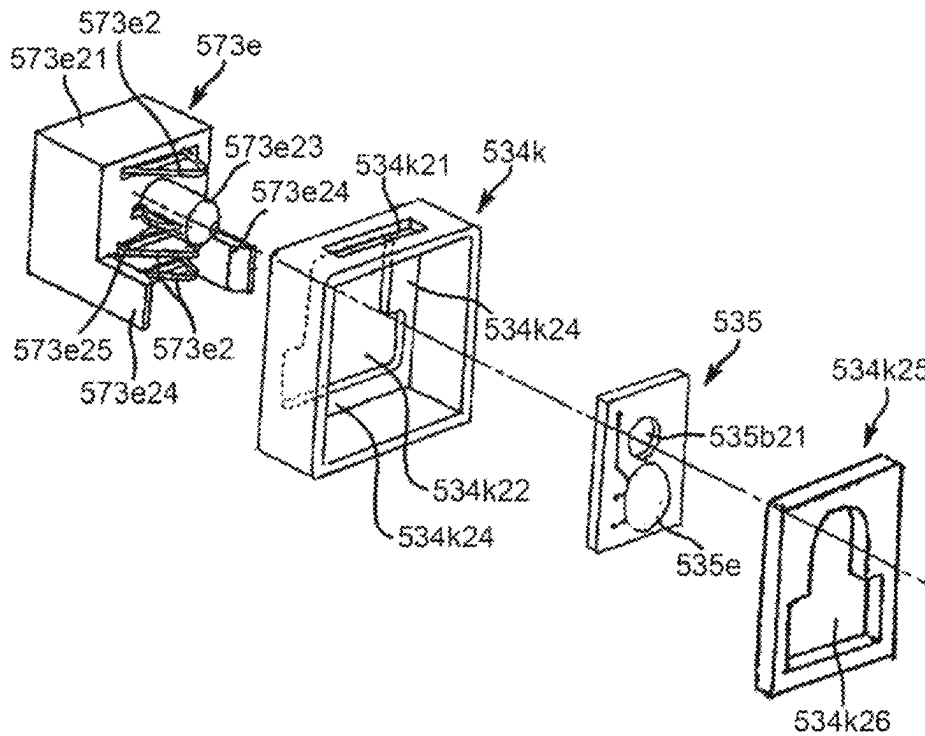


FIG.38

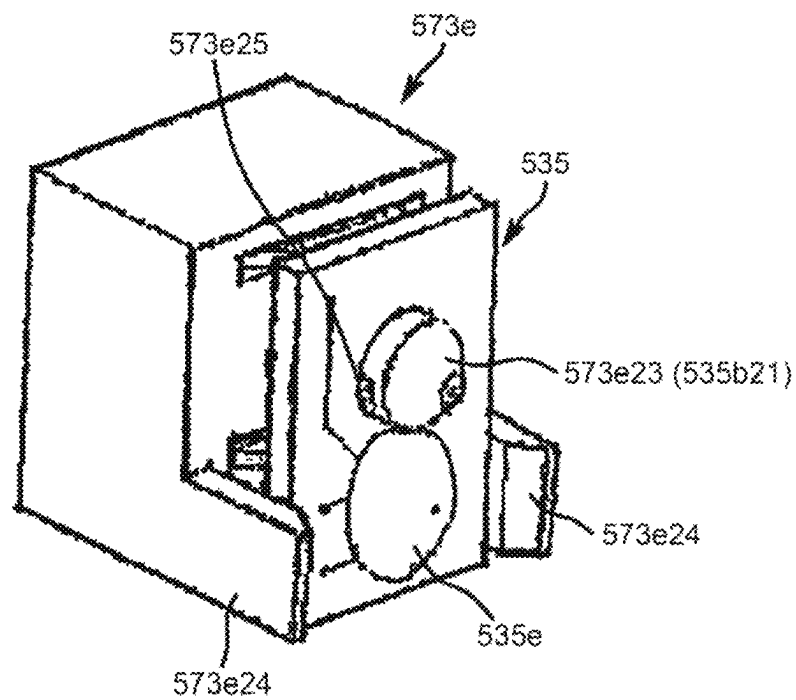


FIG.39A

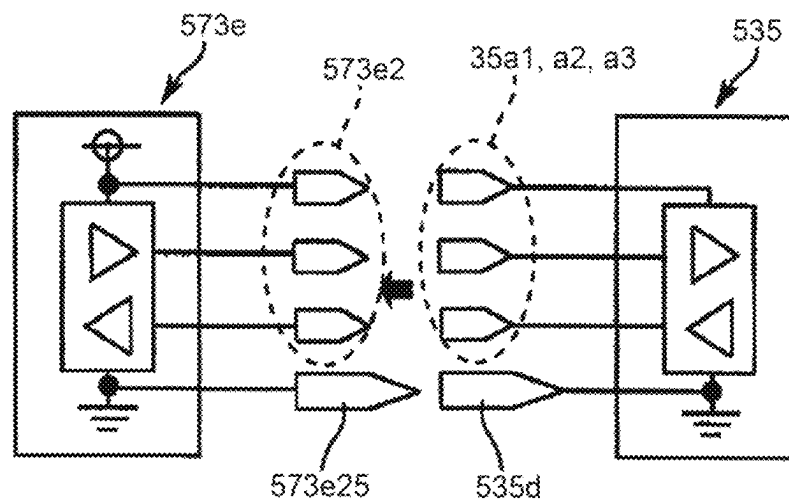


FIG.39B

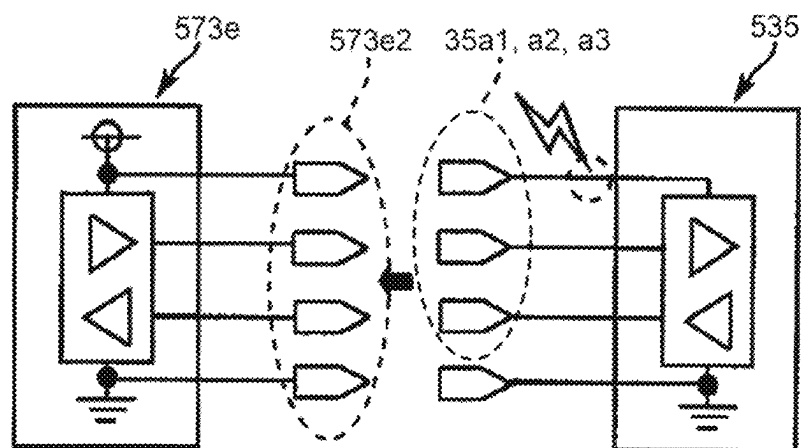


FIG.40A

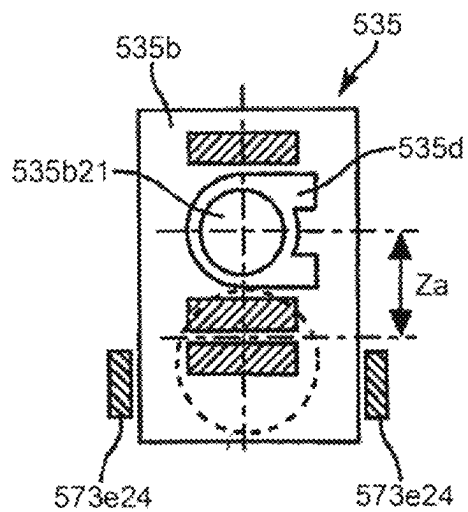


FIG.40B

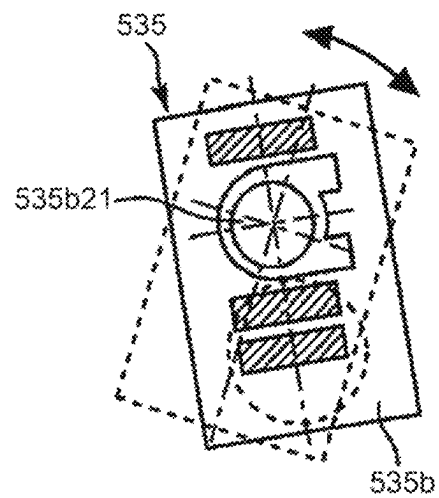


FIG.41

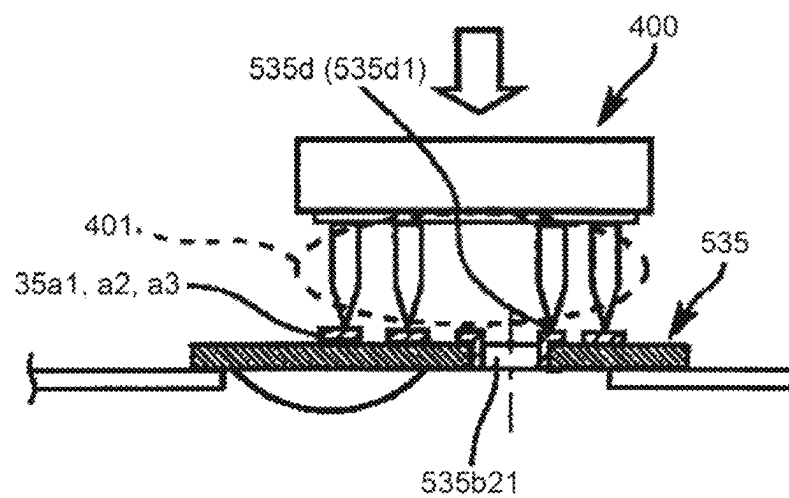


FIG.42A

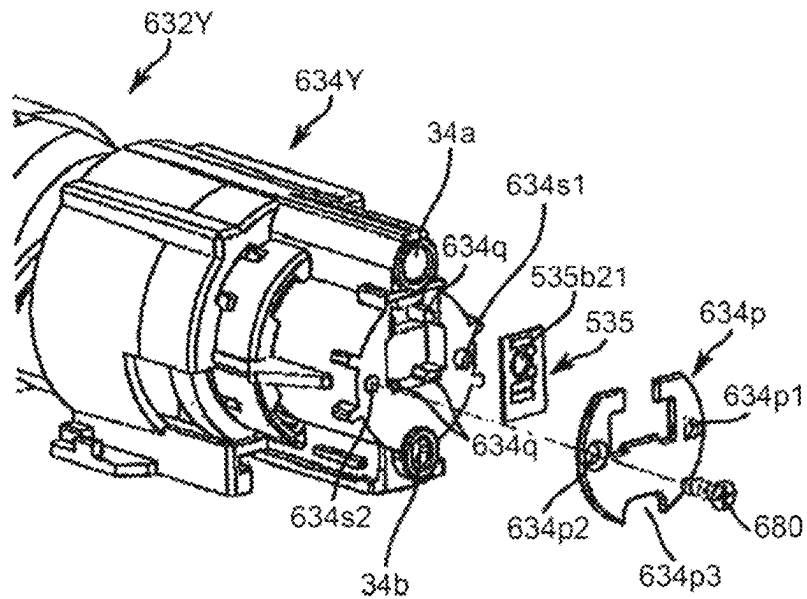


FIG.42B

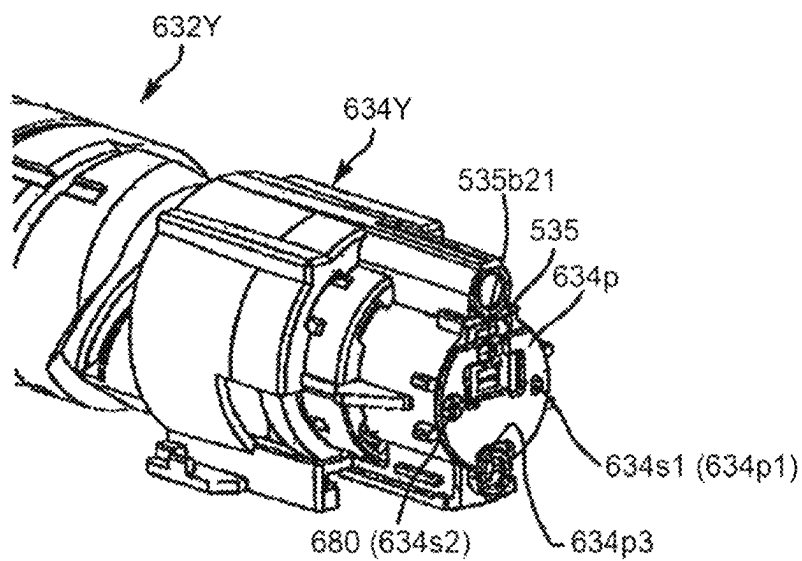


FIG.43

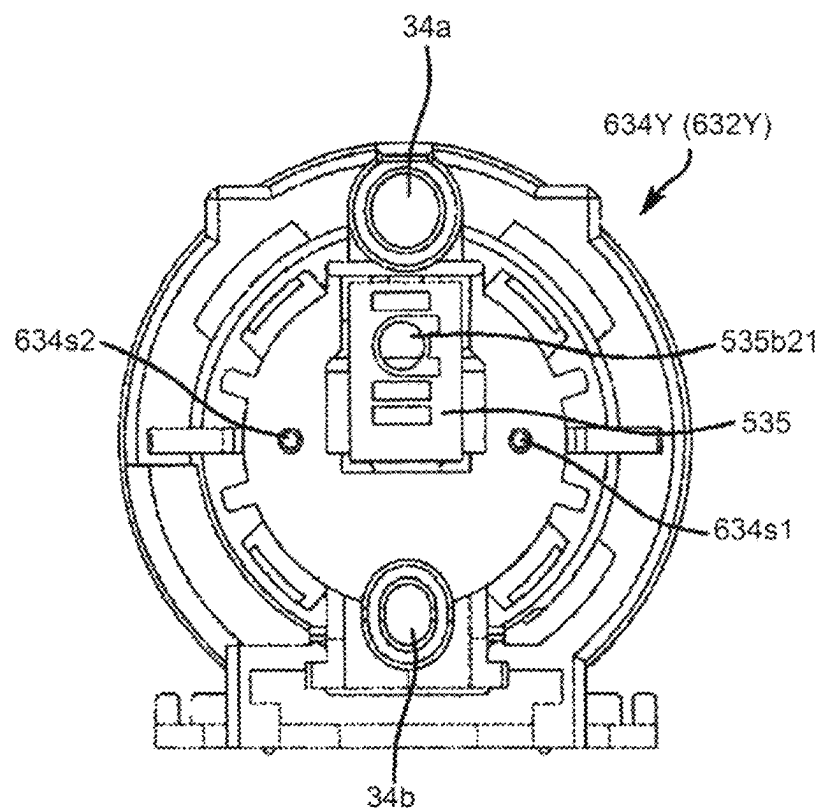


FIG.44

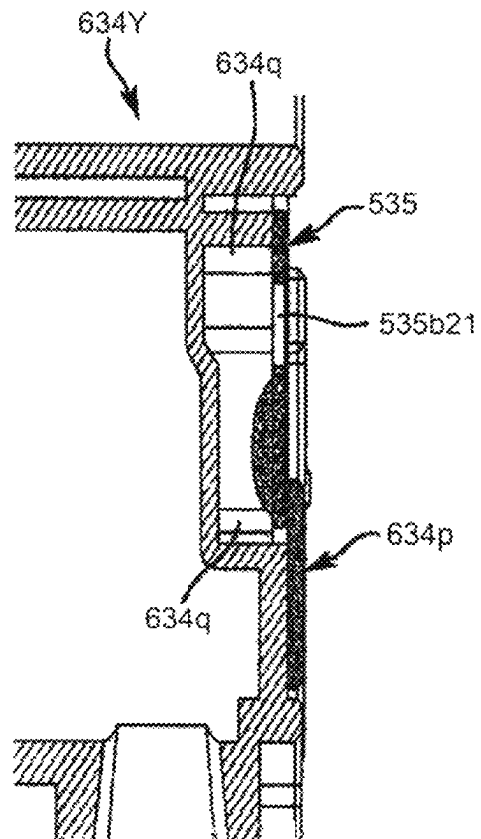


FIG.45

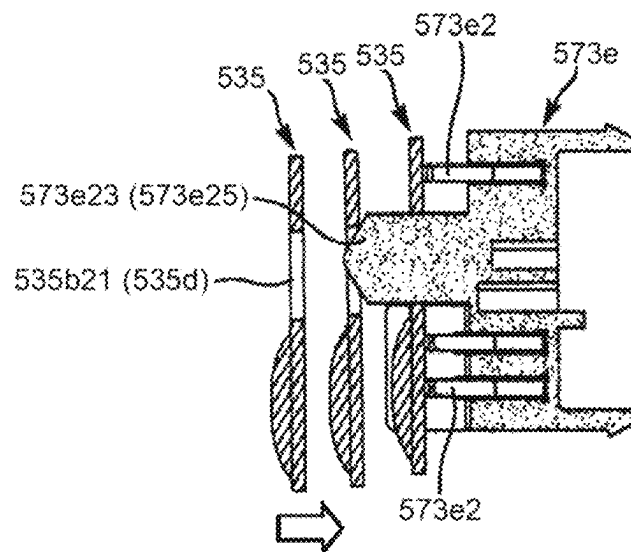


FIG.46A

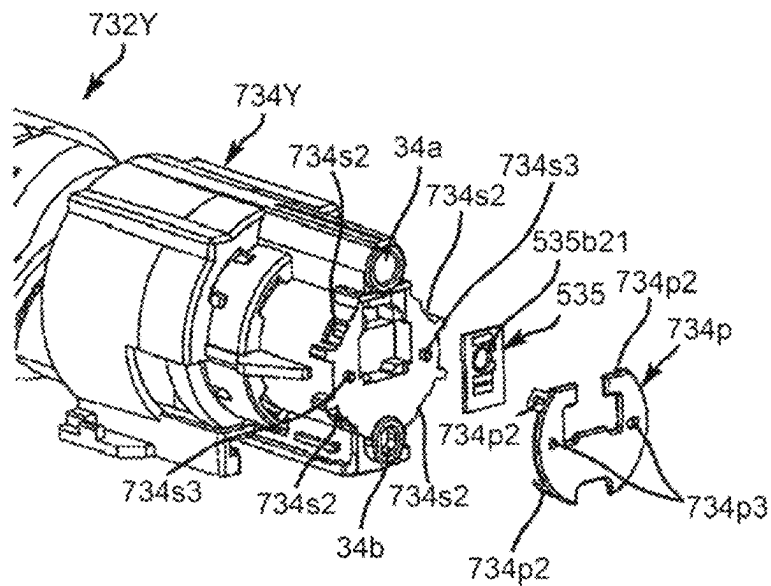


FIG.46B

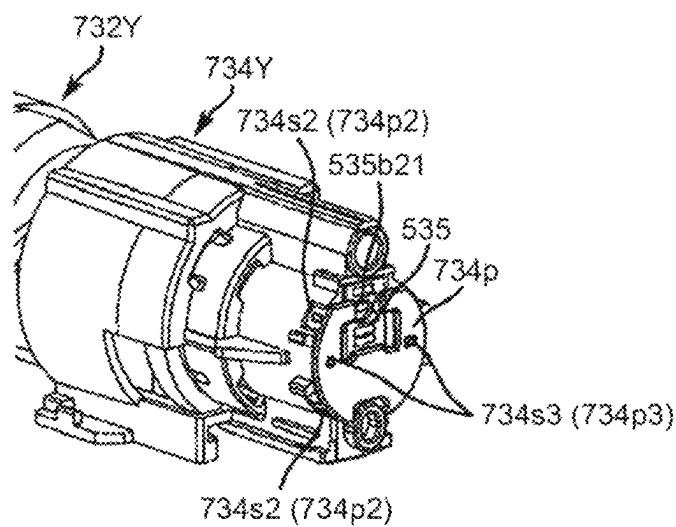


FIG.47A

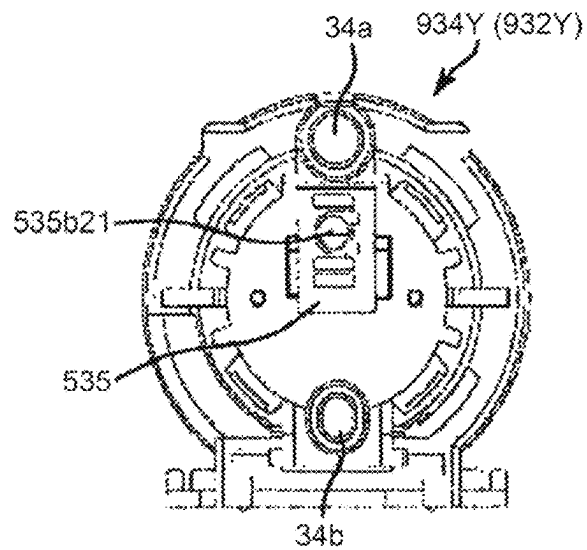


FIG.47B

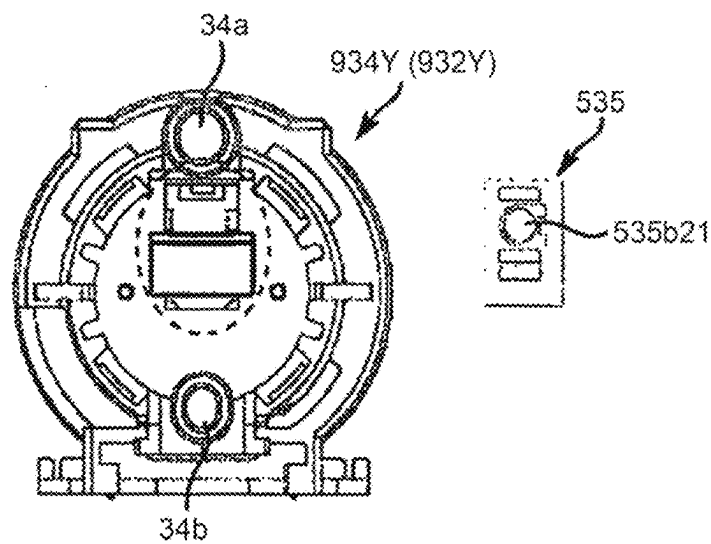


FIG.47C

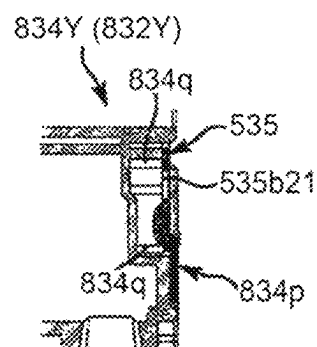


FIG.48

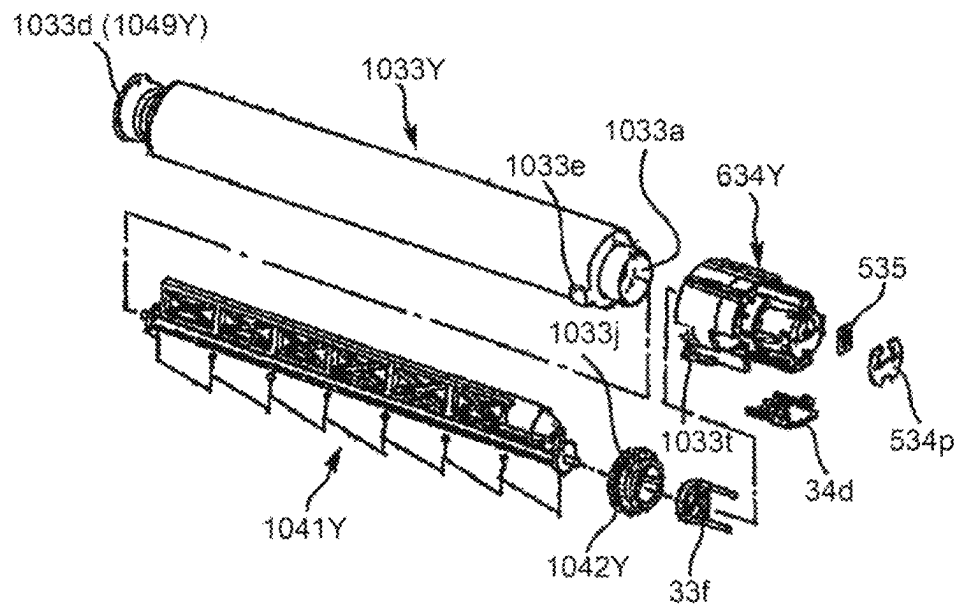


FIG.49

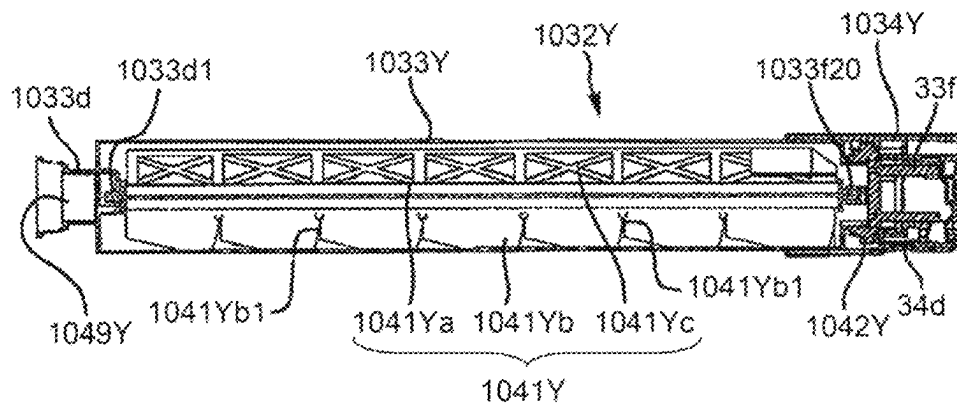


FIG.50

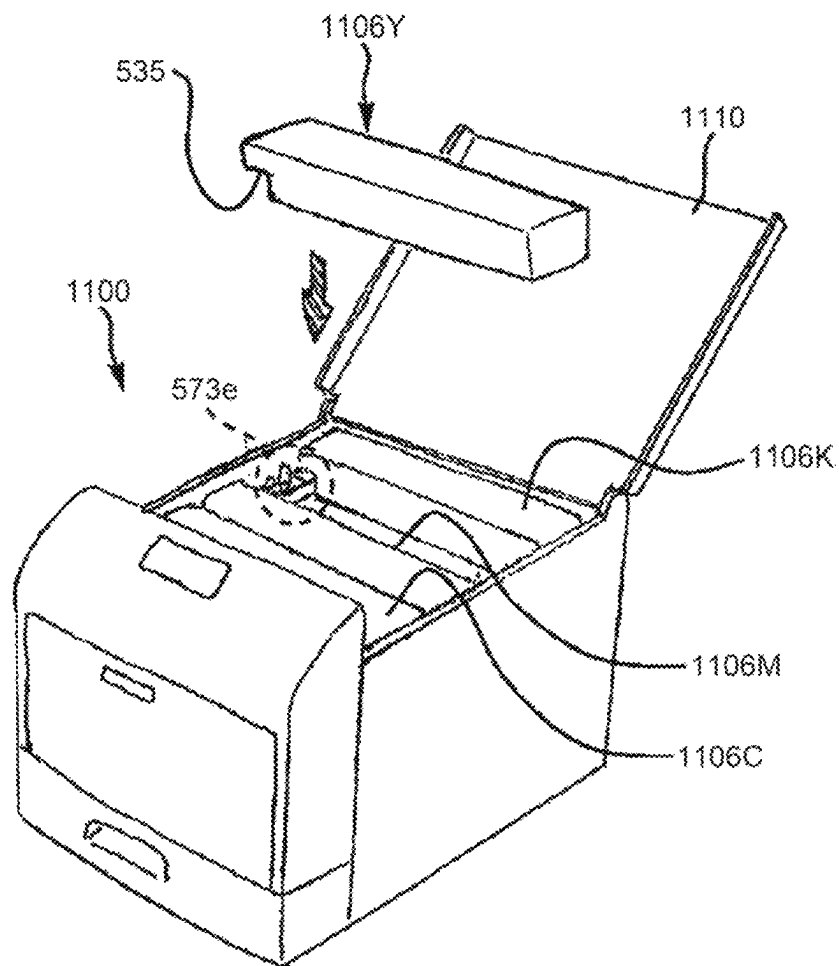


FIG.51A

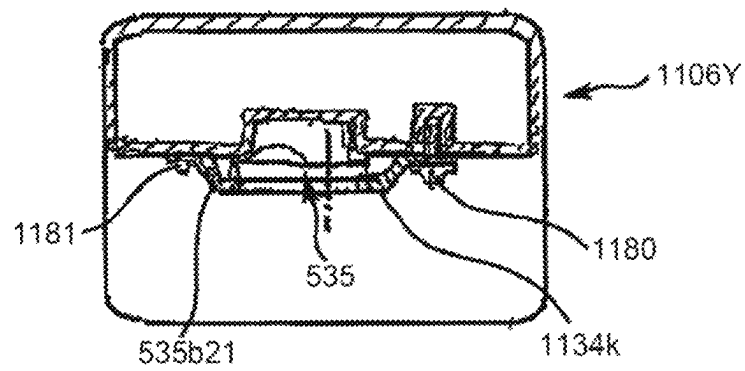


FIG.51B

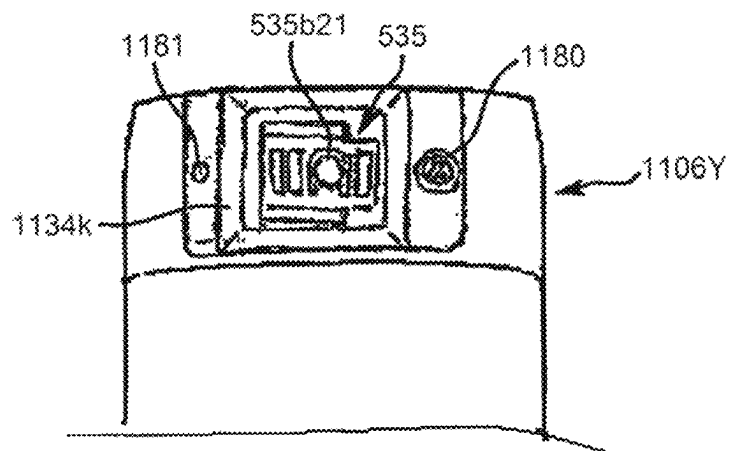


FIG. 52

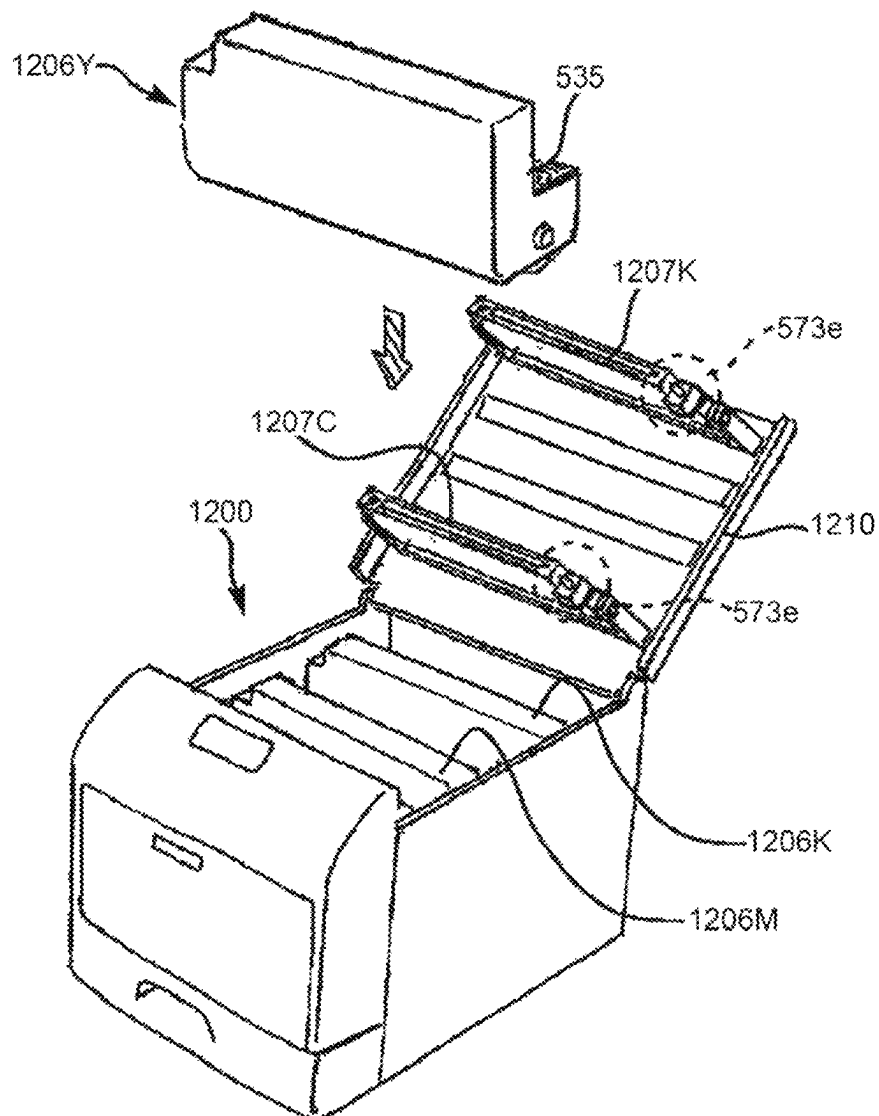


FIG.53

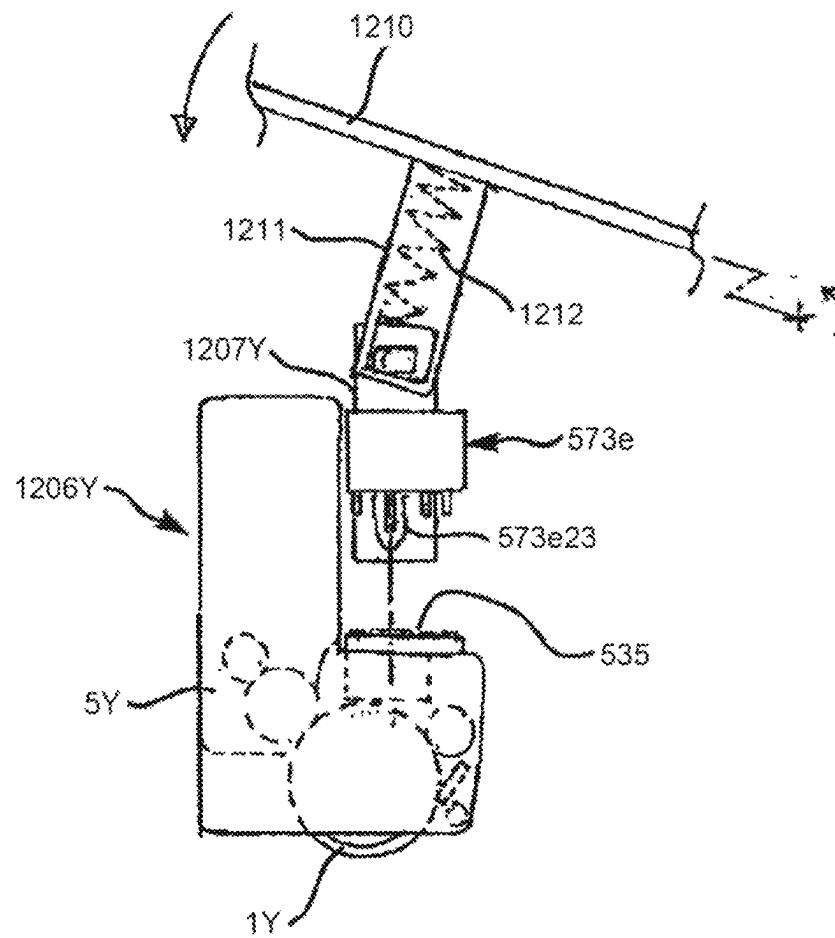


FIG.54

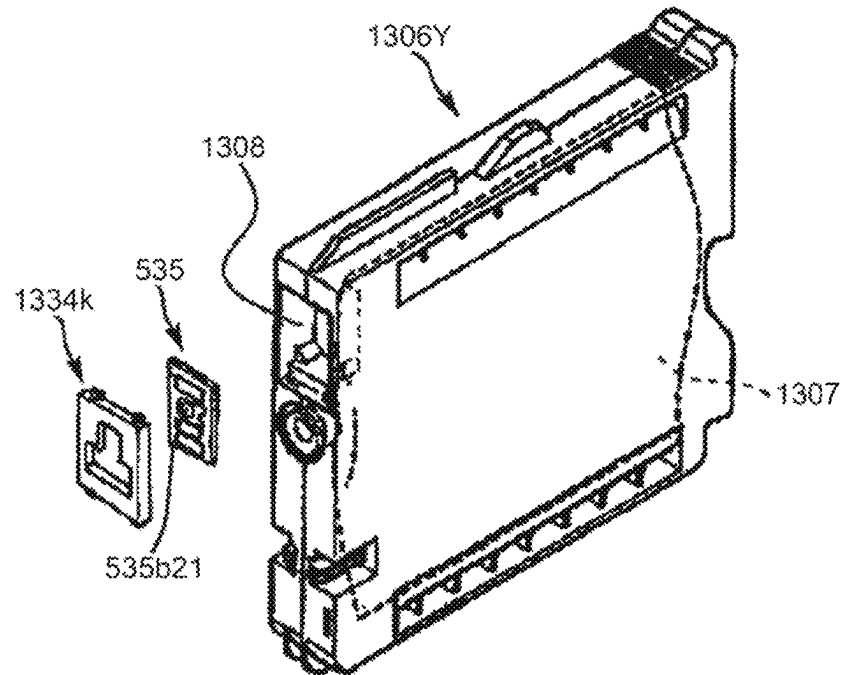


FIG.55

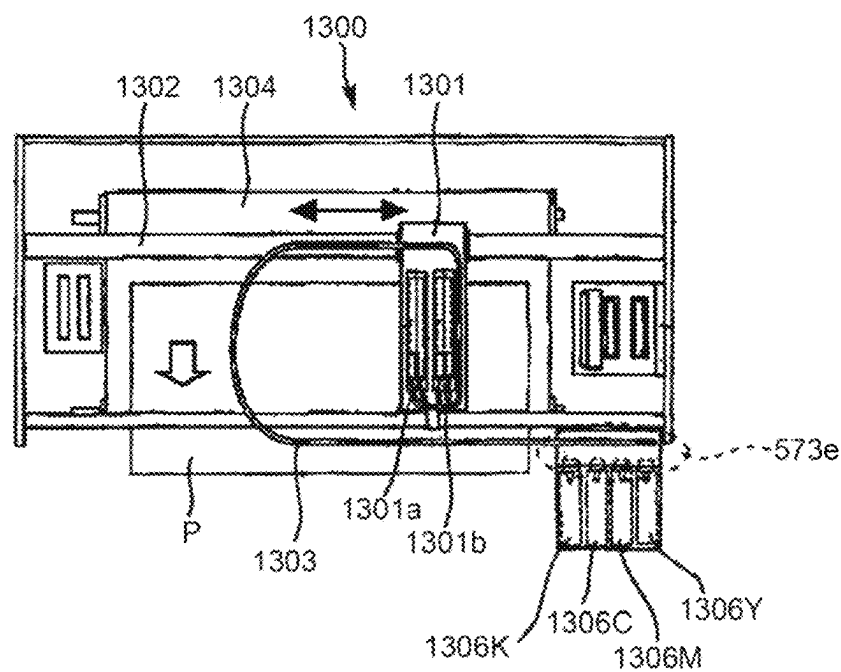


FIG.56

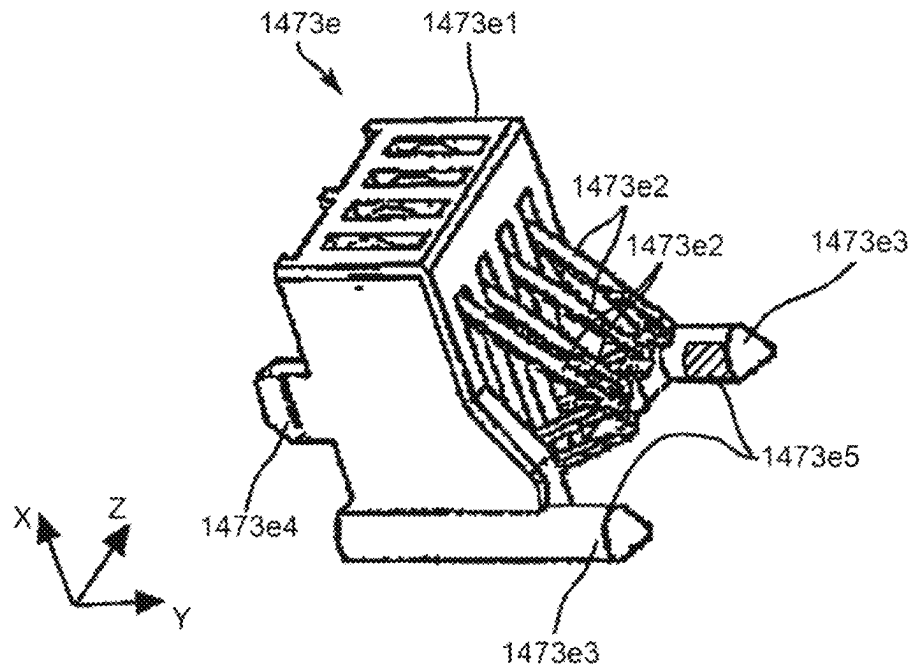


FIG.57

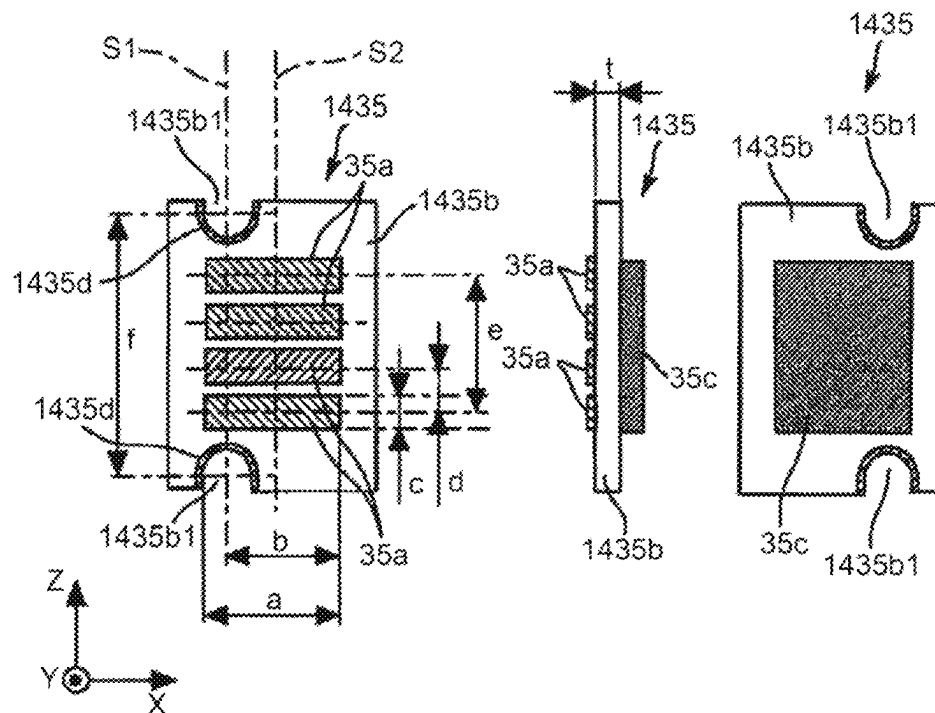


FIG.58

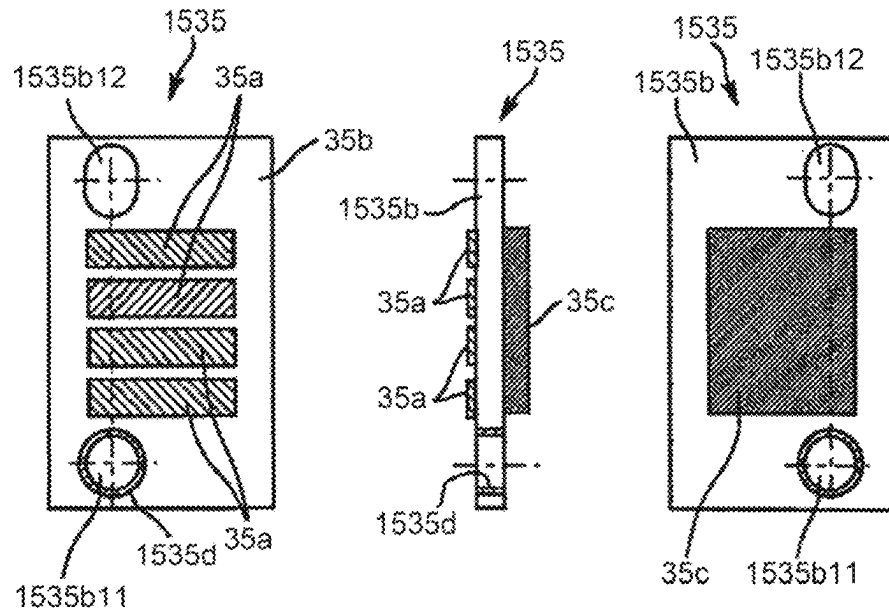


FIG.59

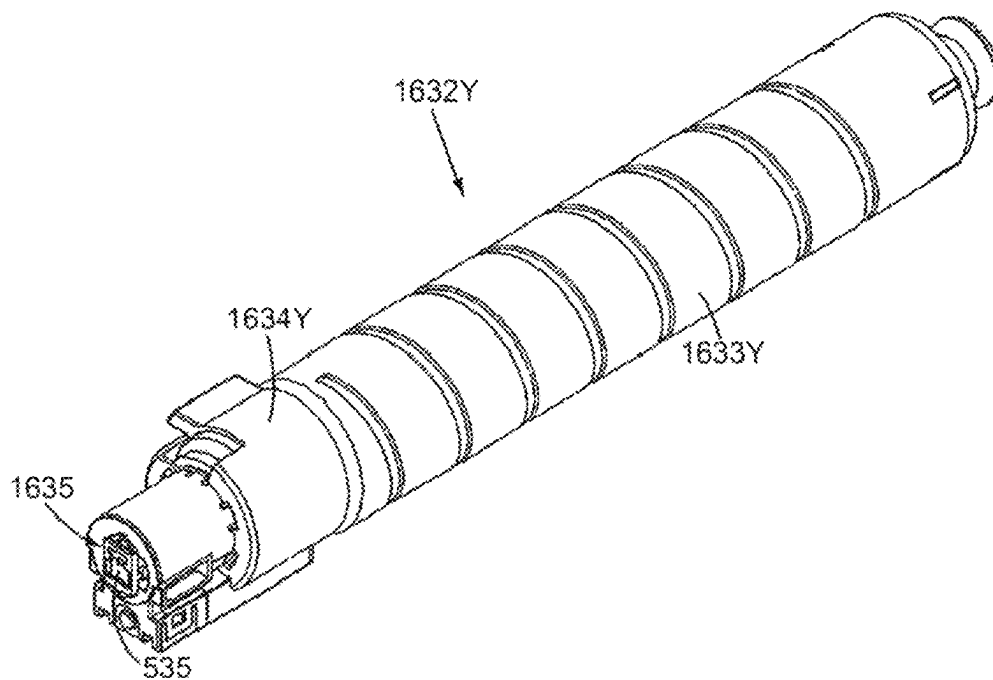


FIG.60

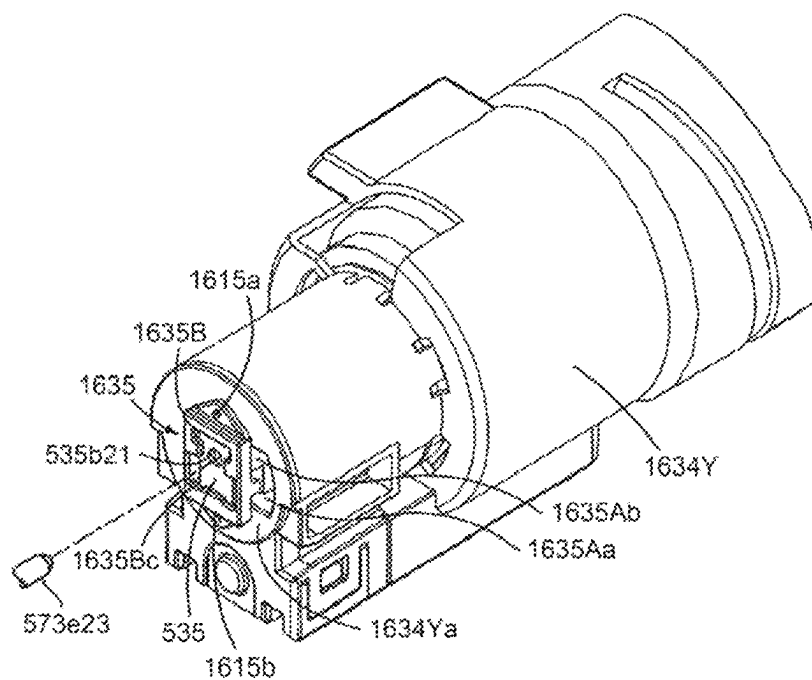


FIG.61

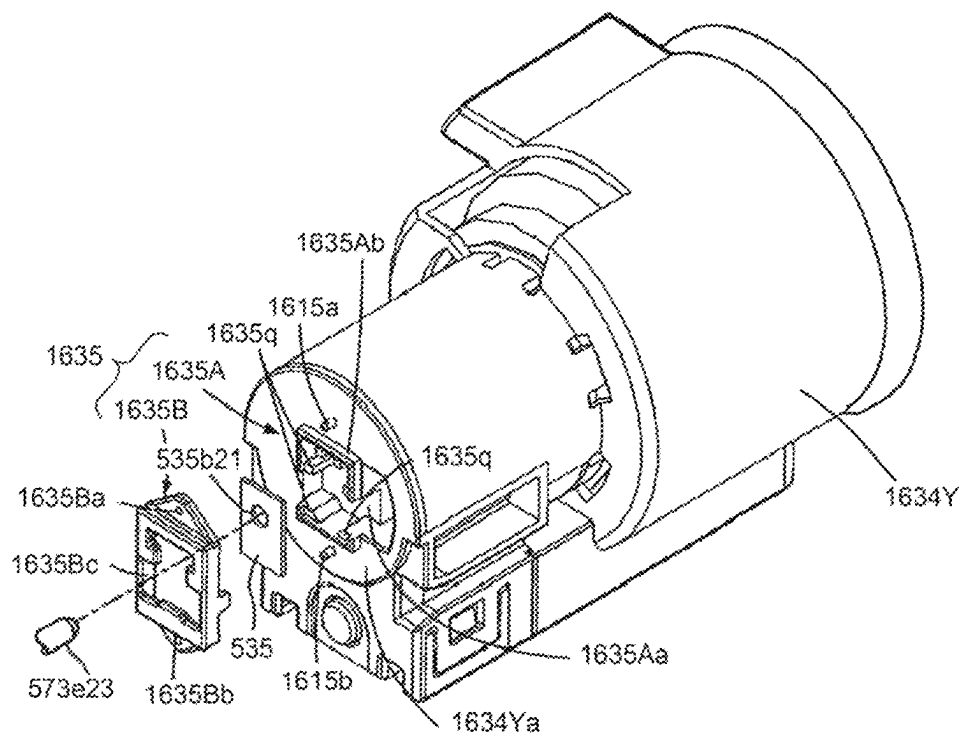


FIG.62

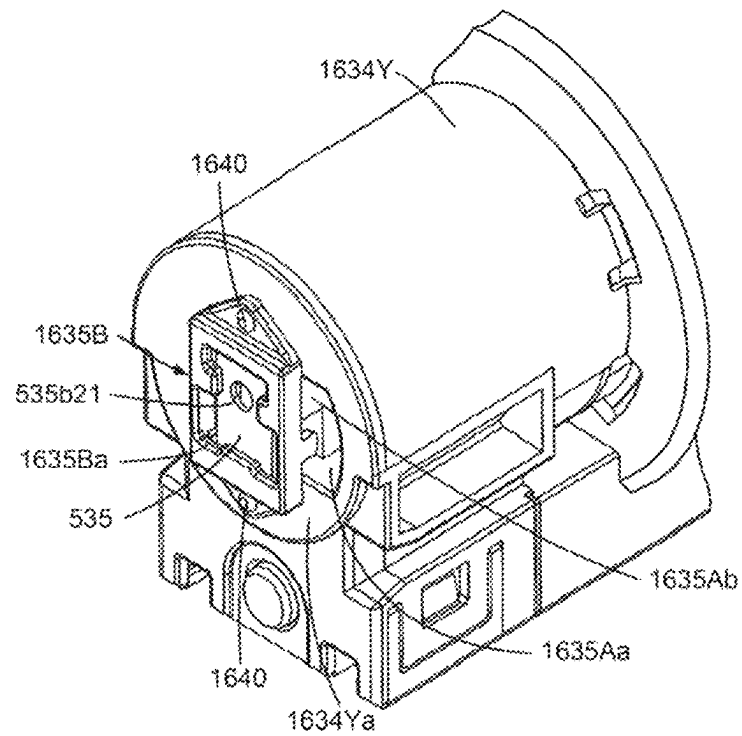


FIG.63

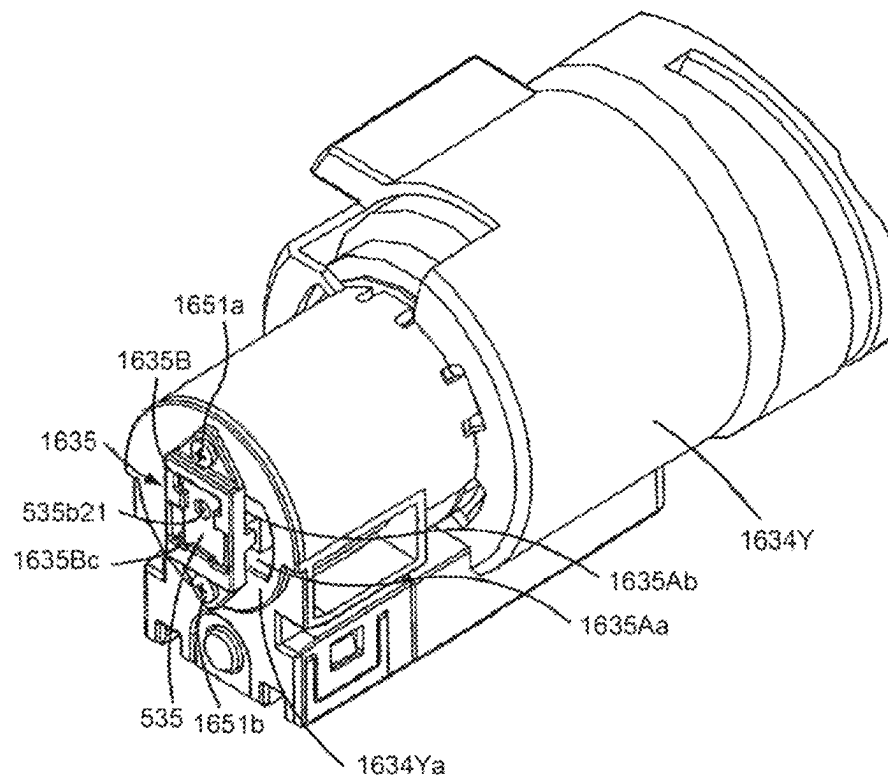


FIG.64

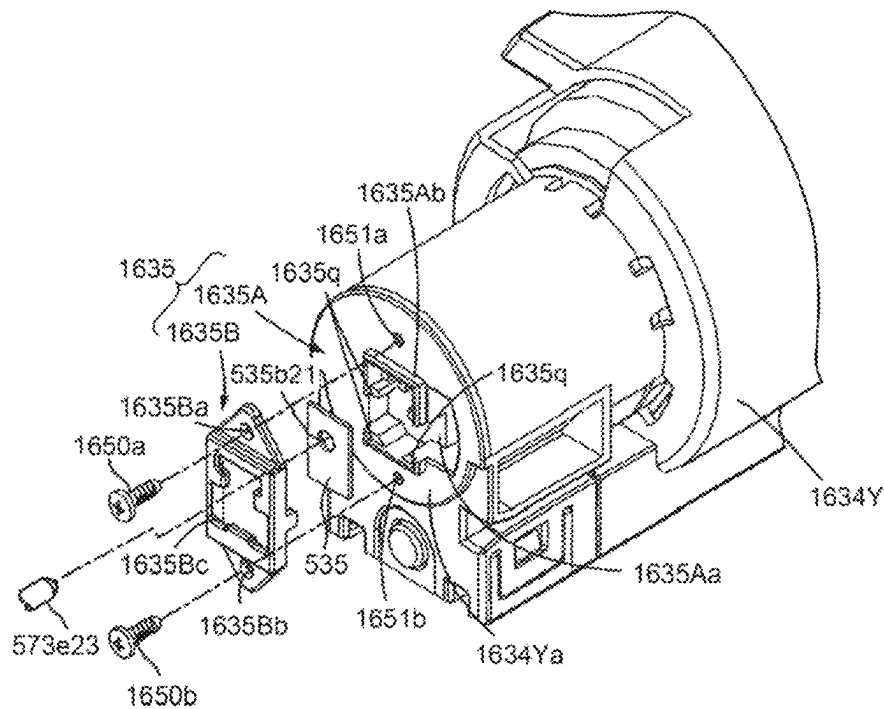
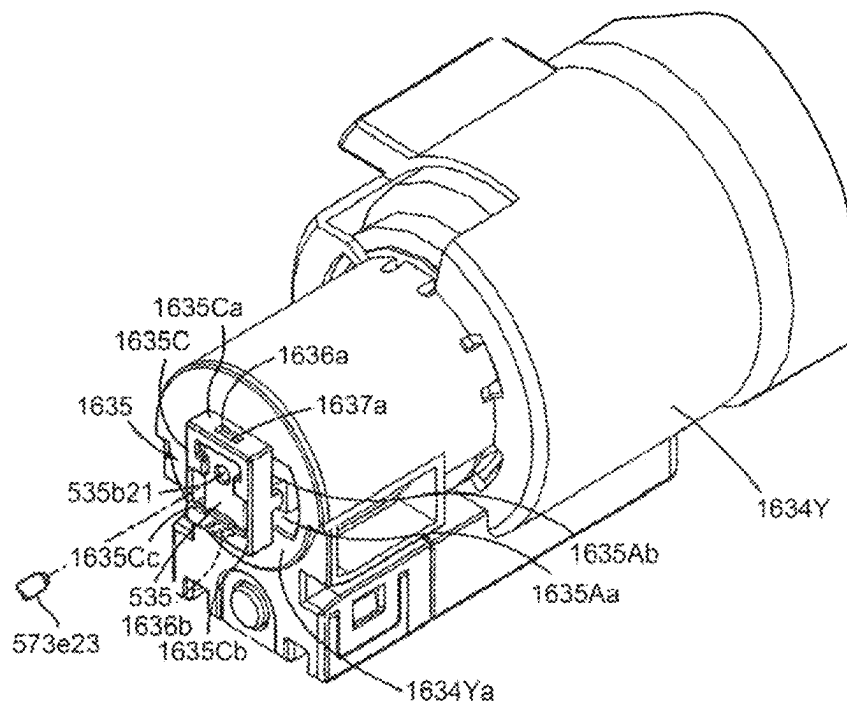


FIG.65



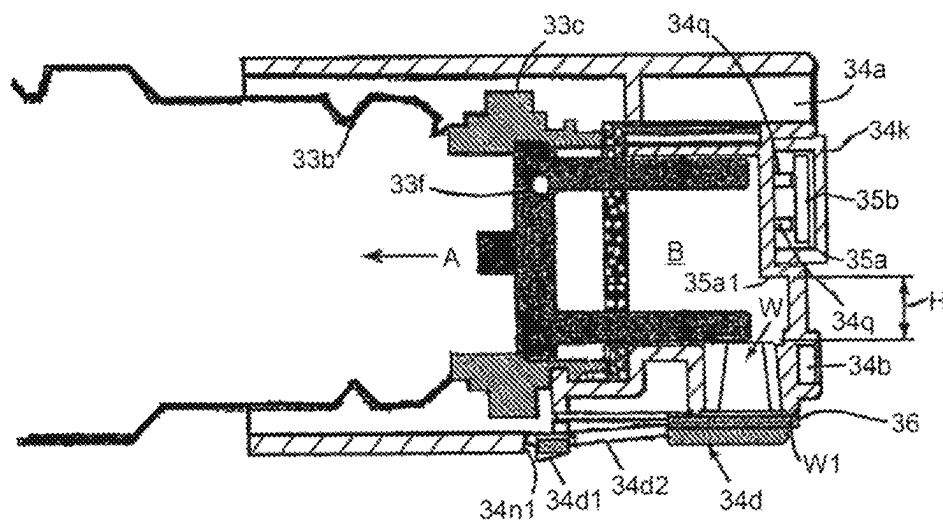


FIG. 68

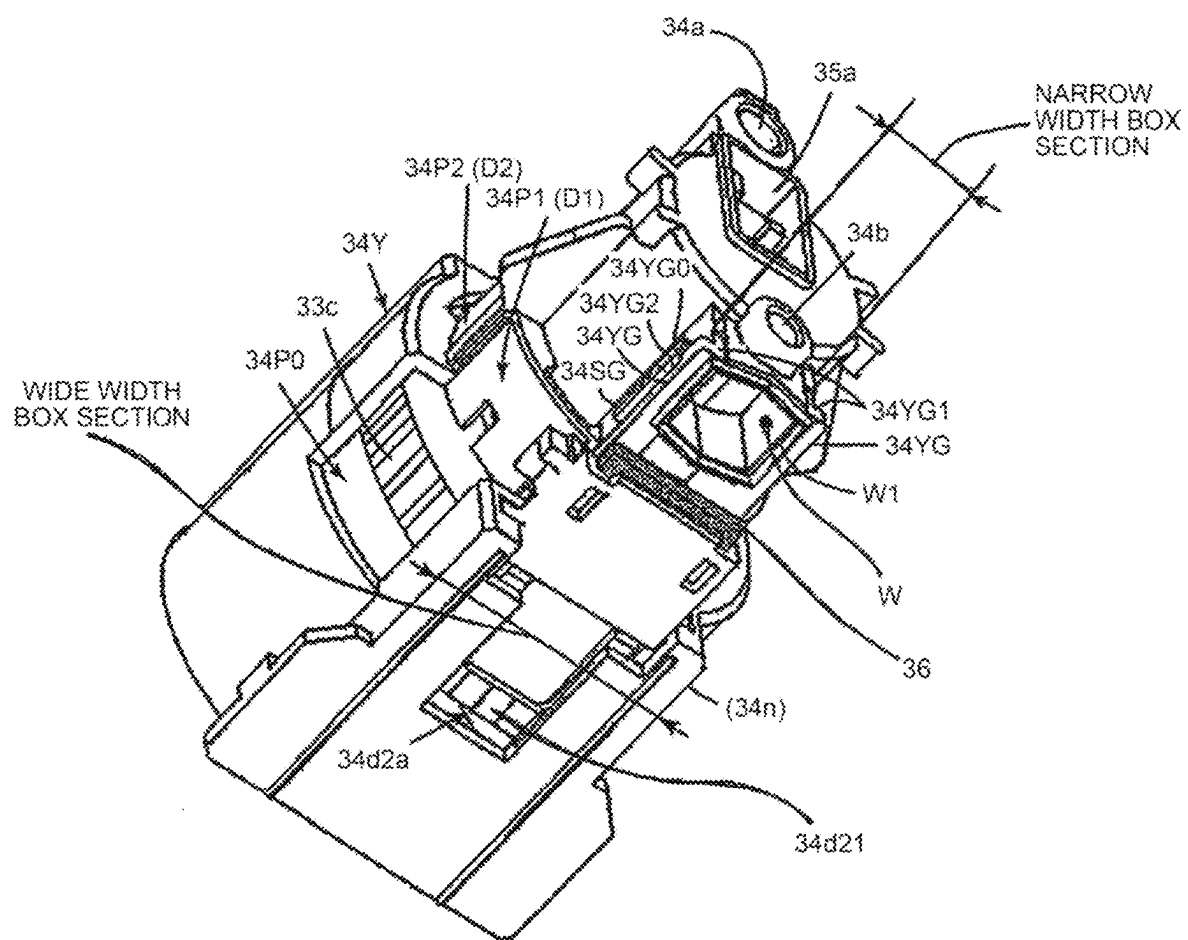


FIG.69A

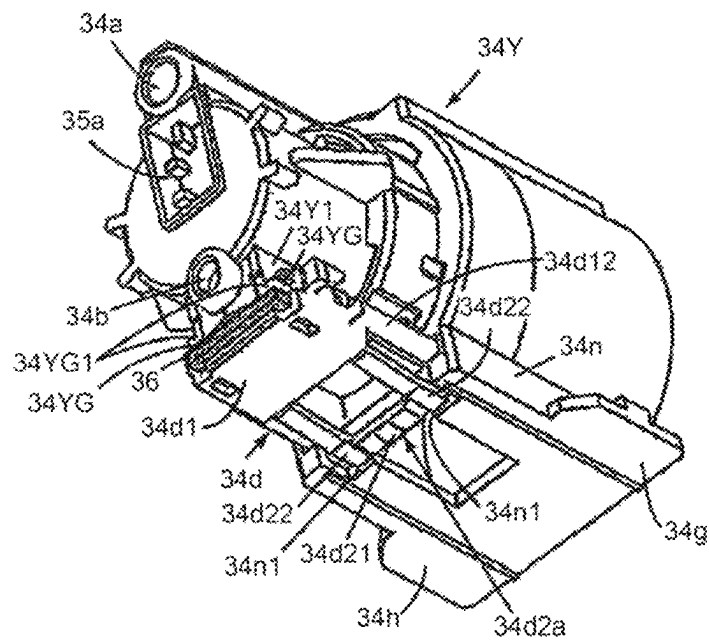
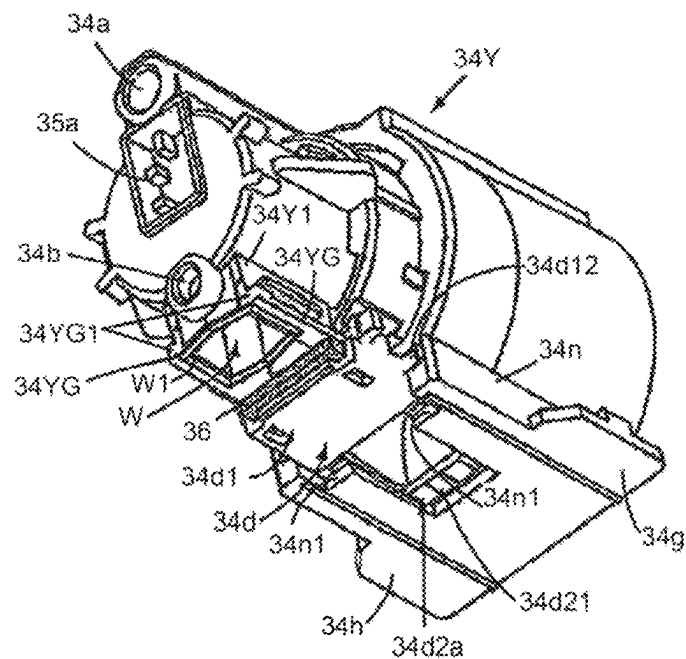


FIG.69B



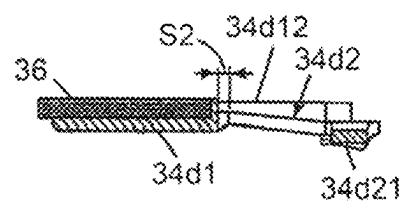


FIG.71A

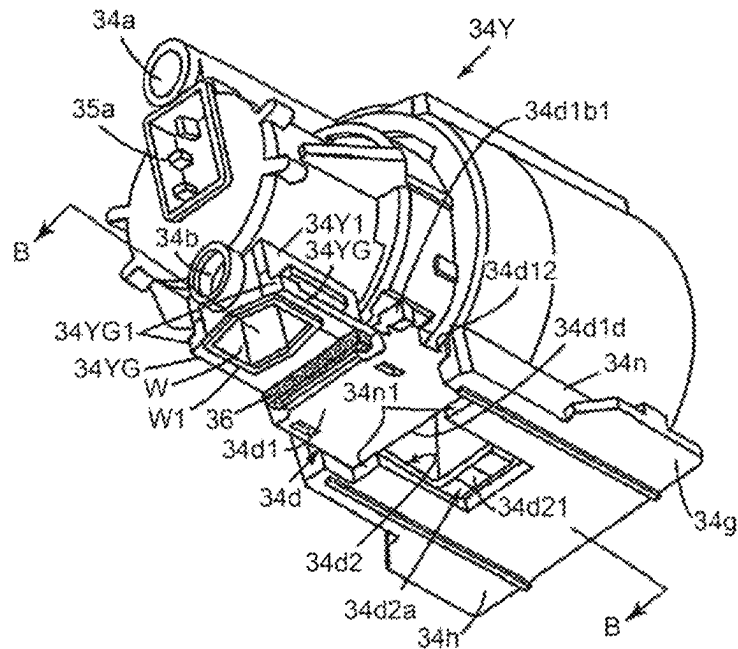


FIG.71B

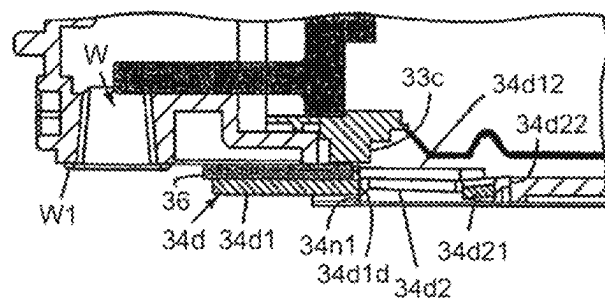


FIG.71C

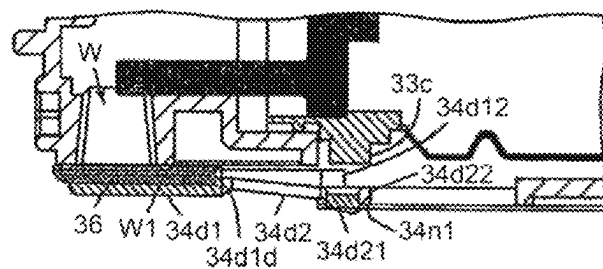


FIG.72

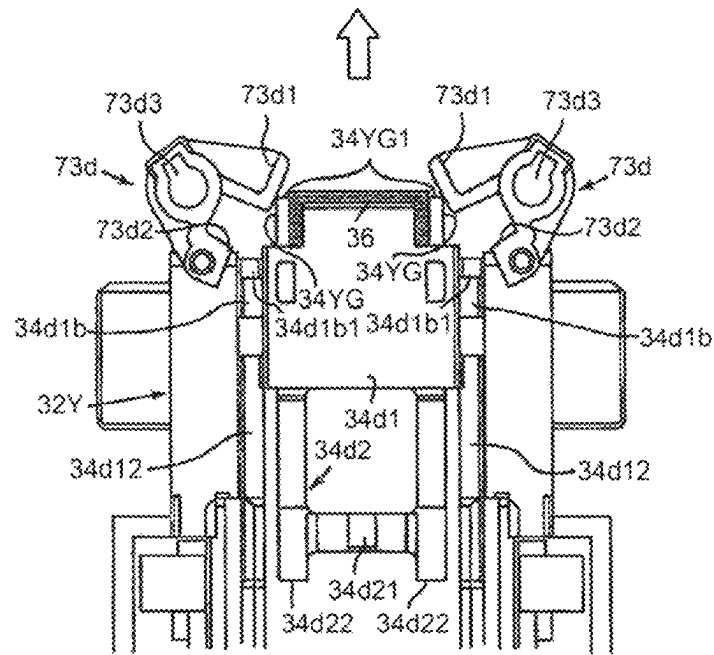


FIG.73

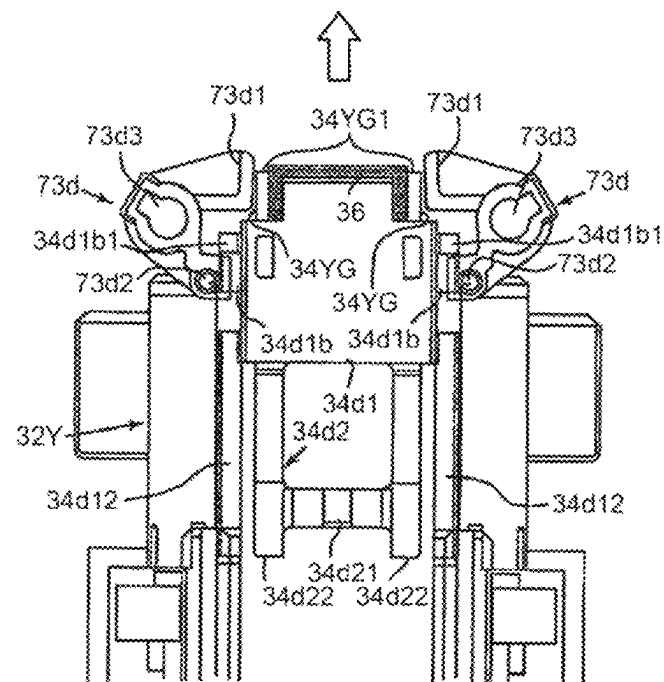


FIG. 74

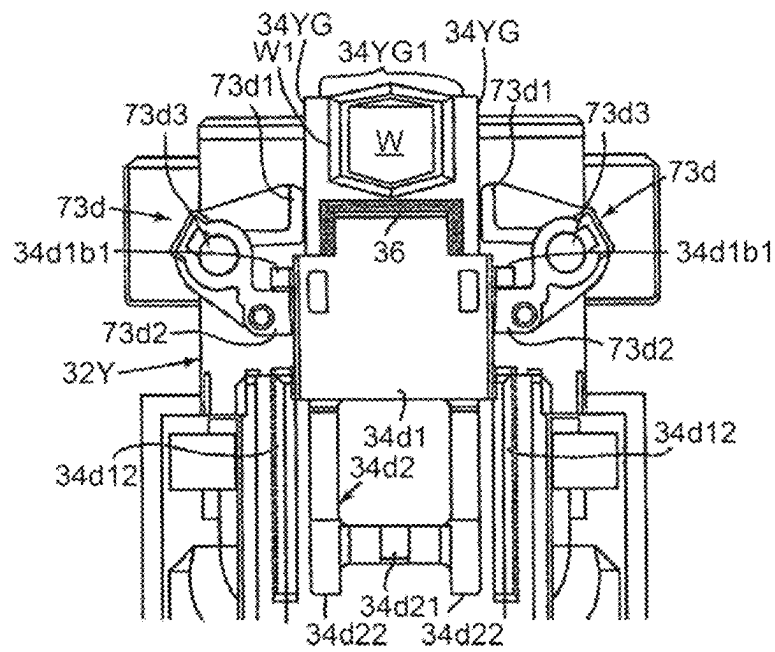


FIG.75A

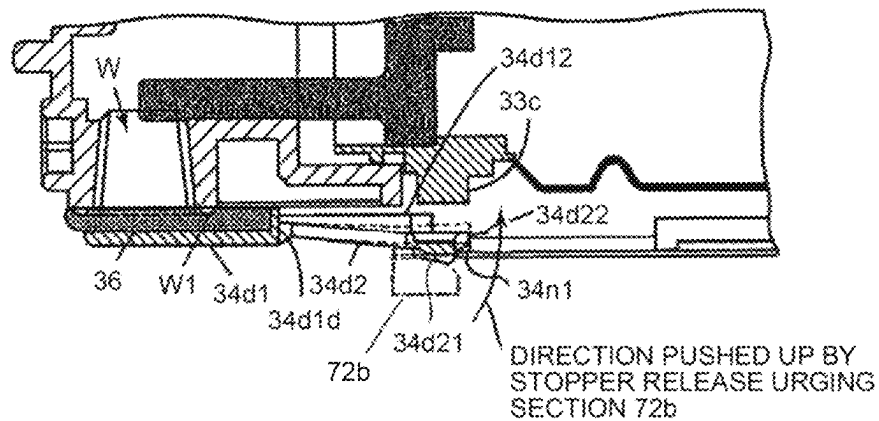


FIG.75B

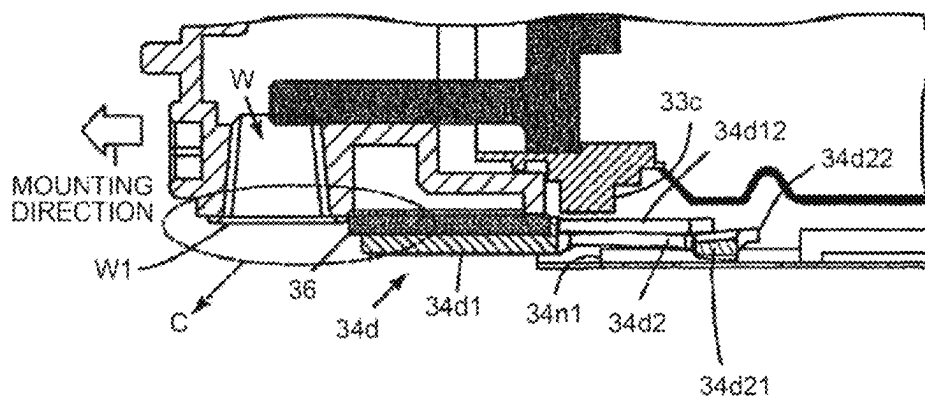


FIG.75C

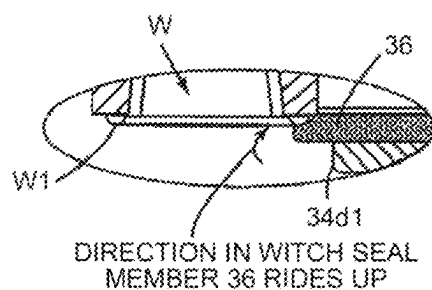


FIG.75D

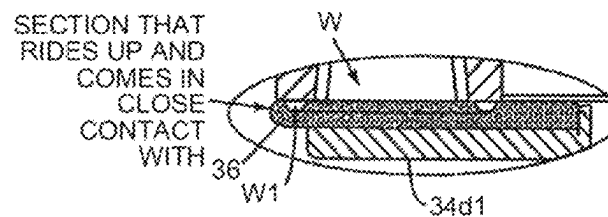


FIG.76A

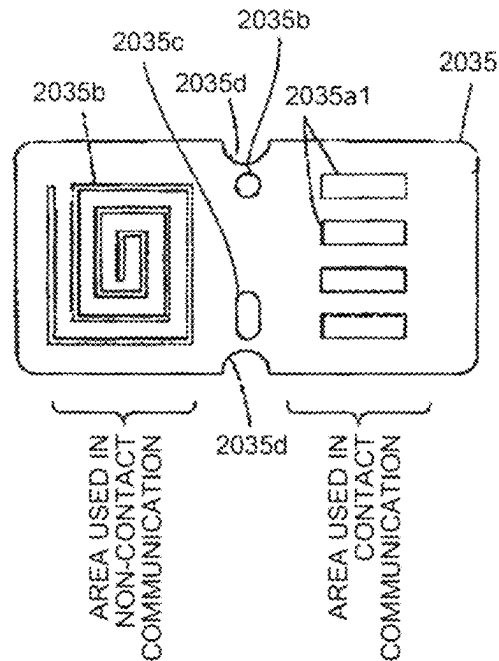


FIG.76B

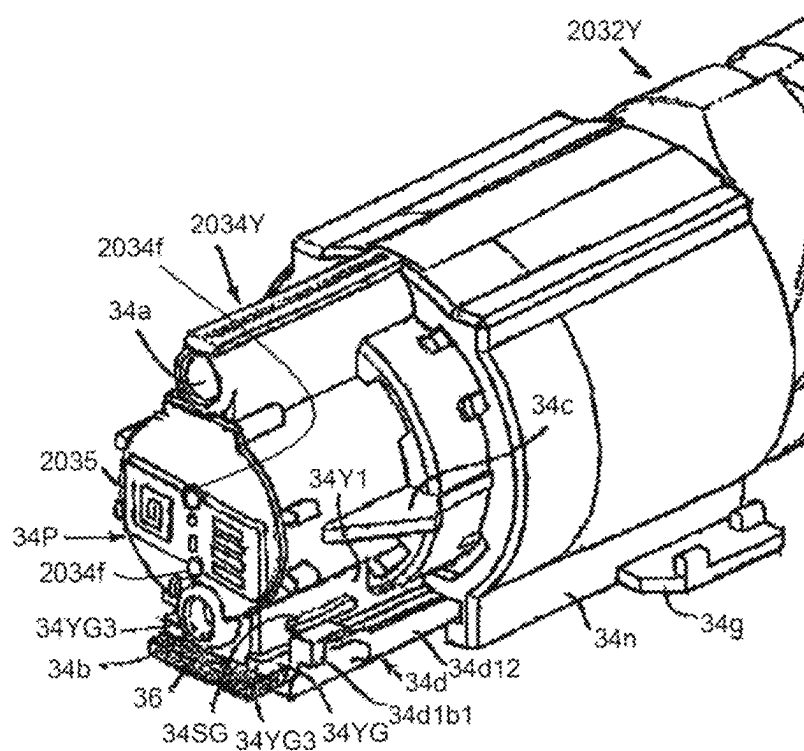


FIG.77

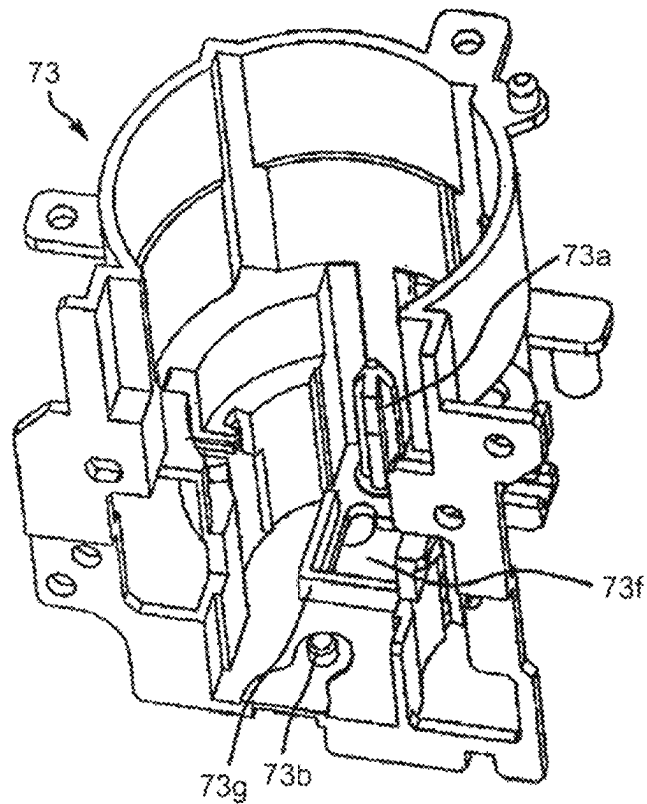


FIG.78

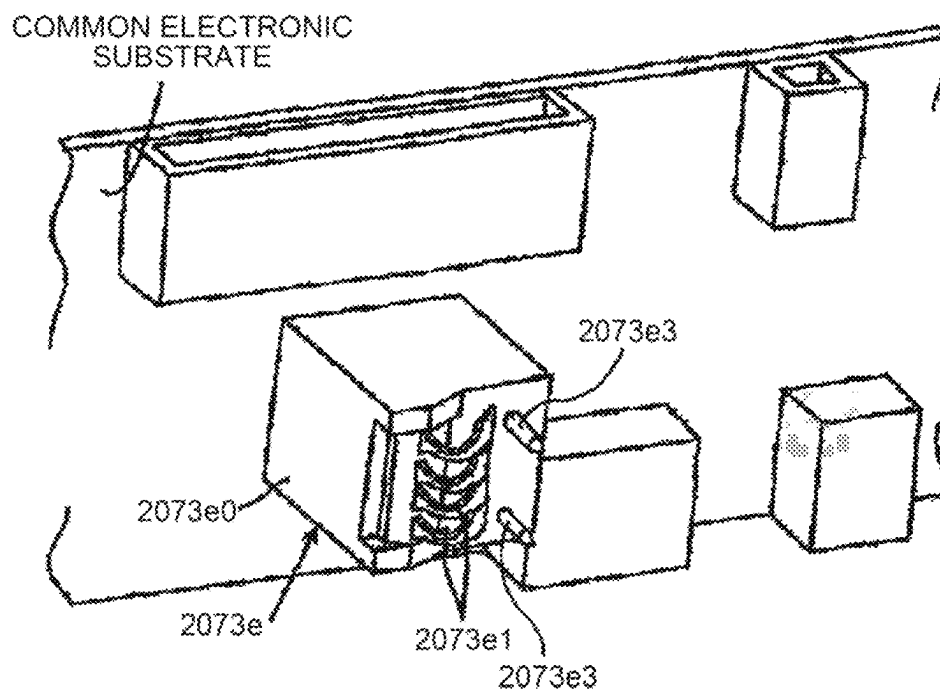


FIG.79

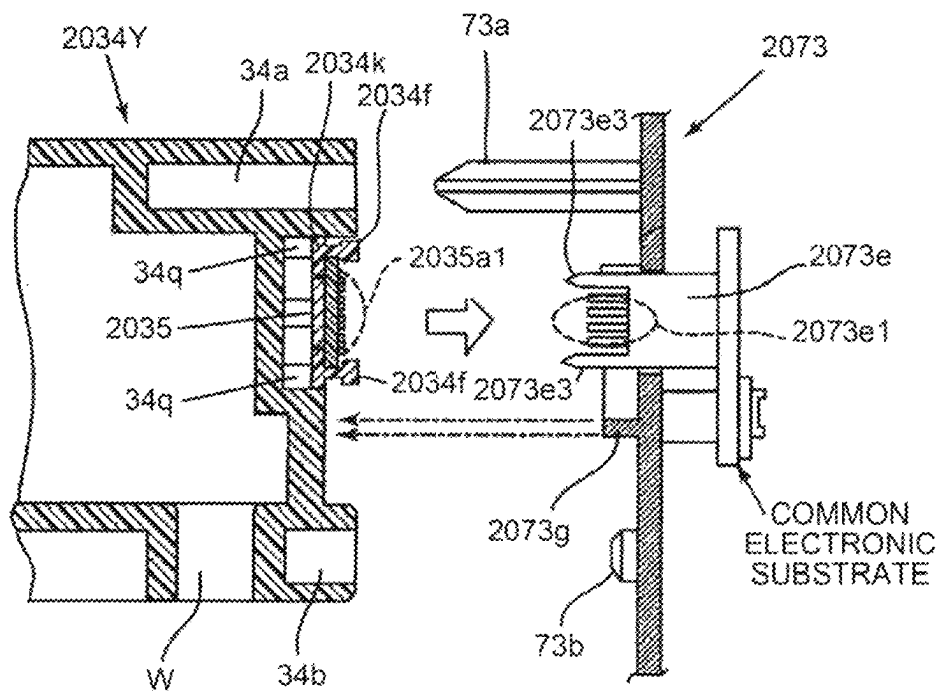


FIG.80

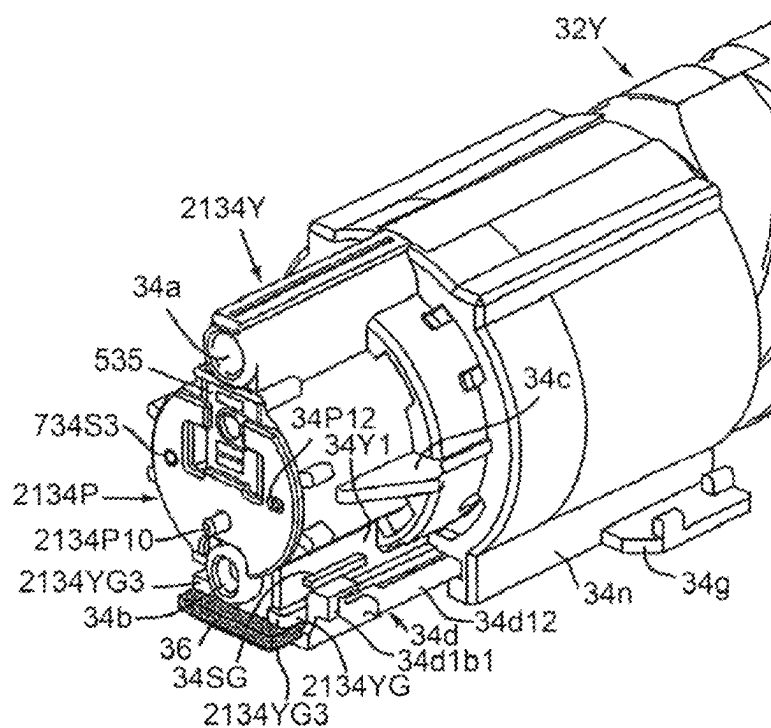


FIG.81

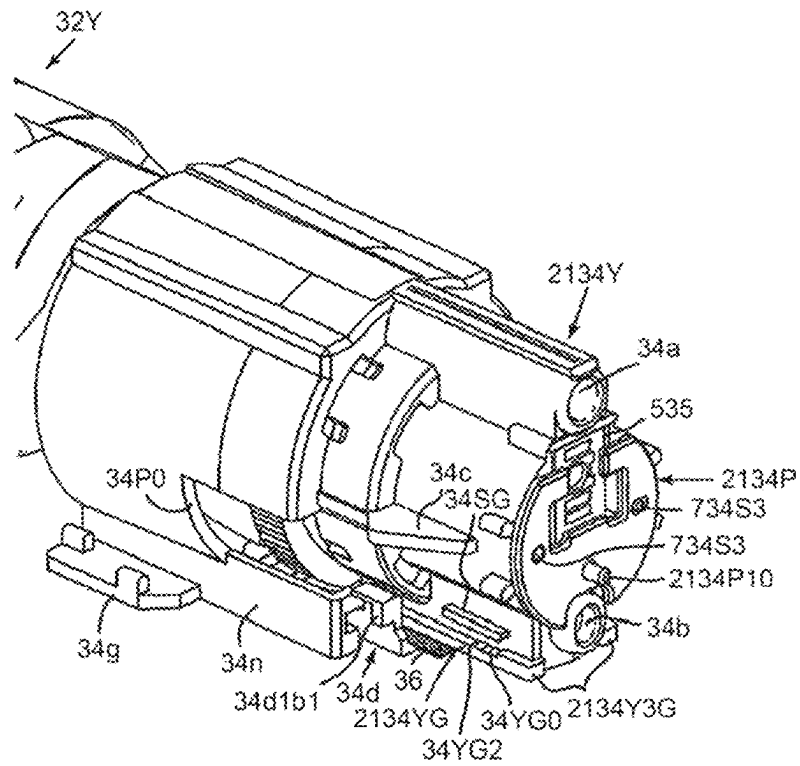


FIG.82

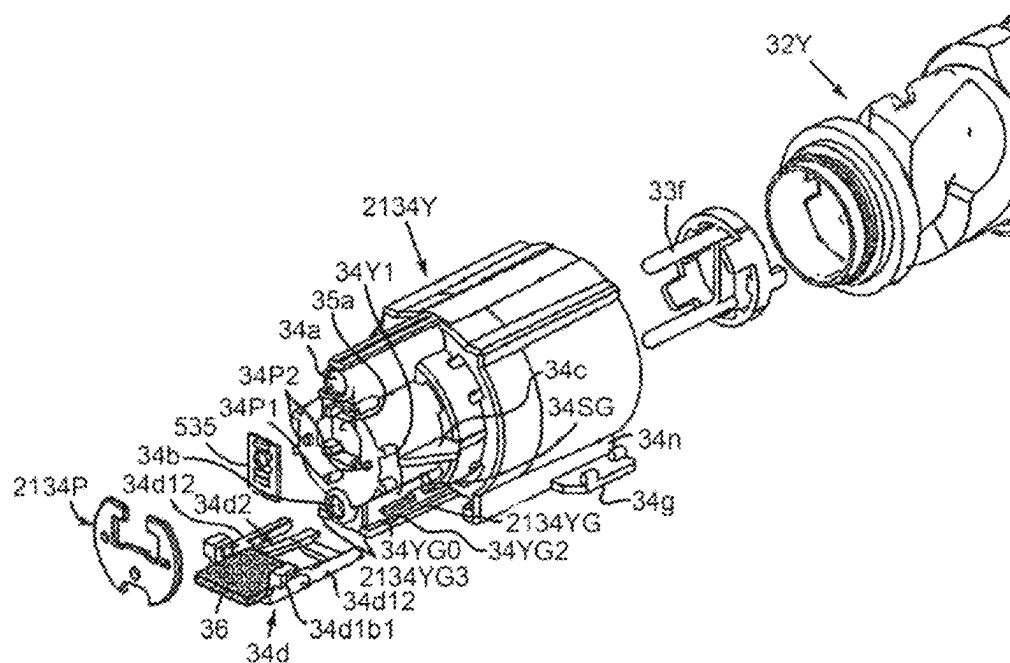


FIG.83

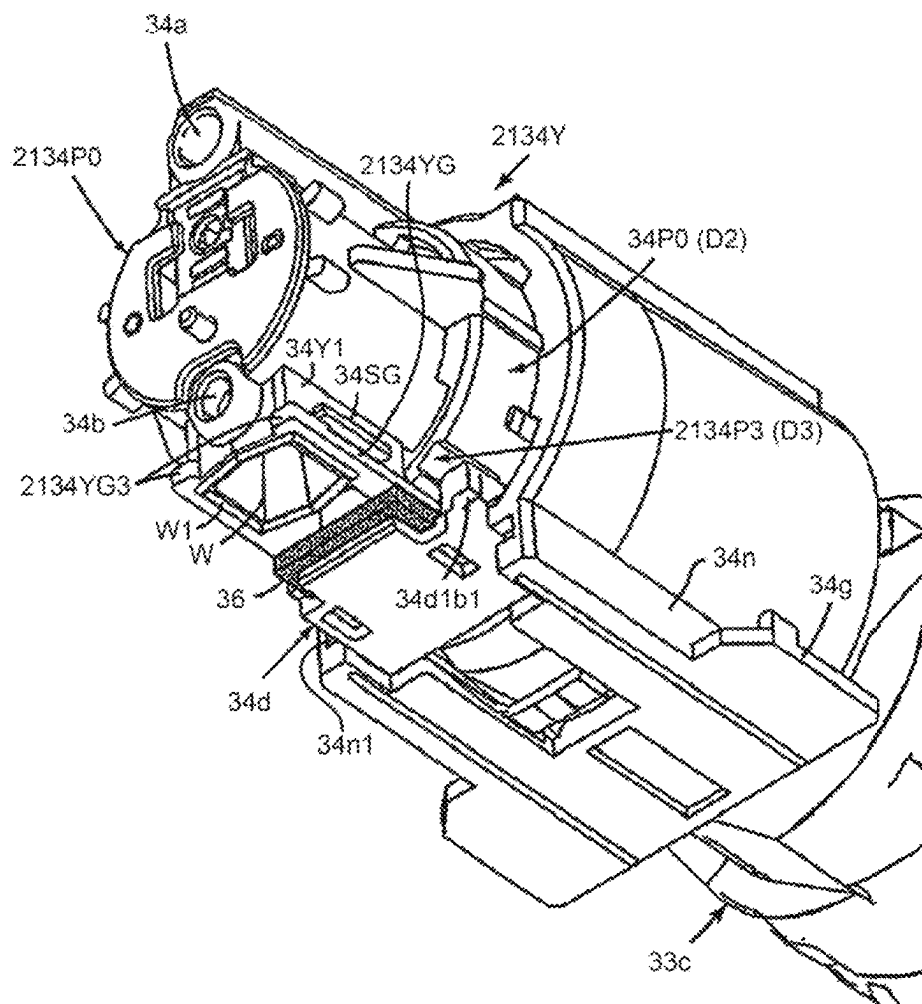


FIG.84

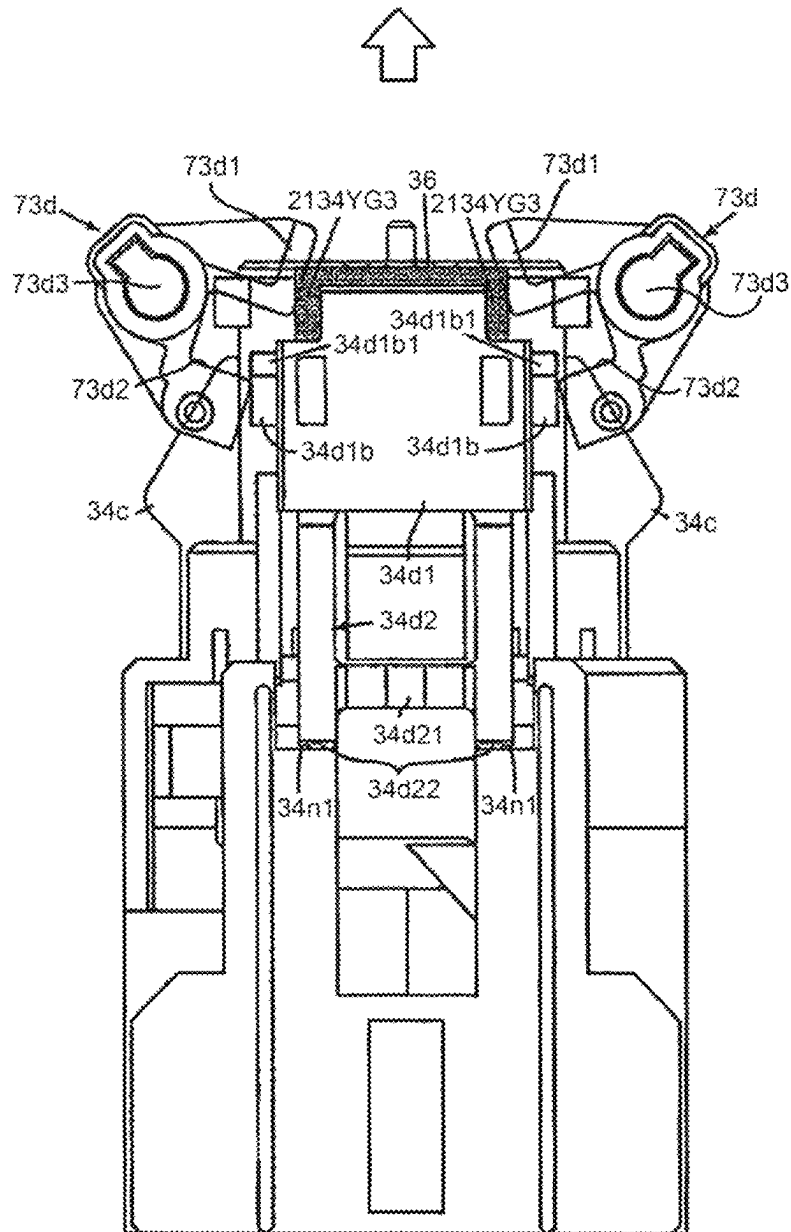


FIG.85

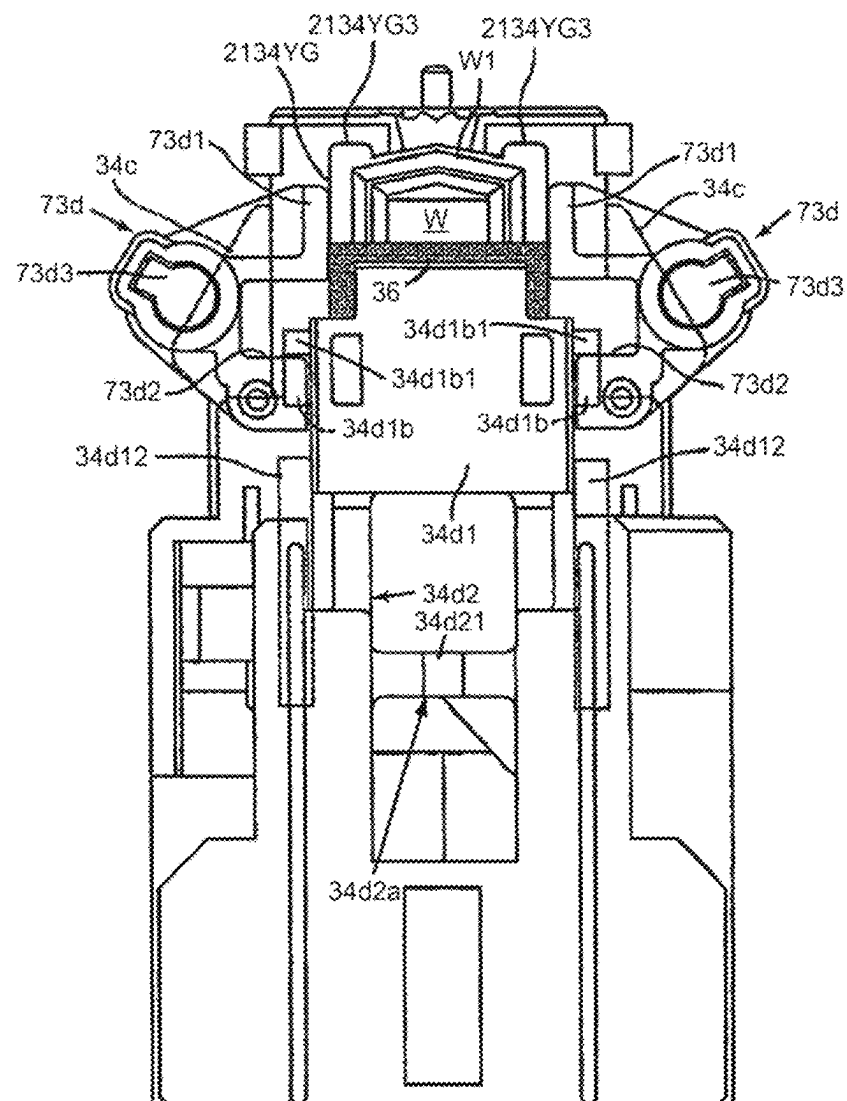


FIG.86

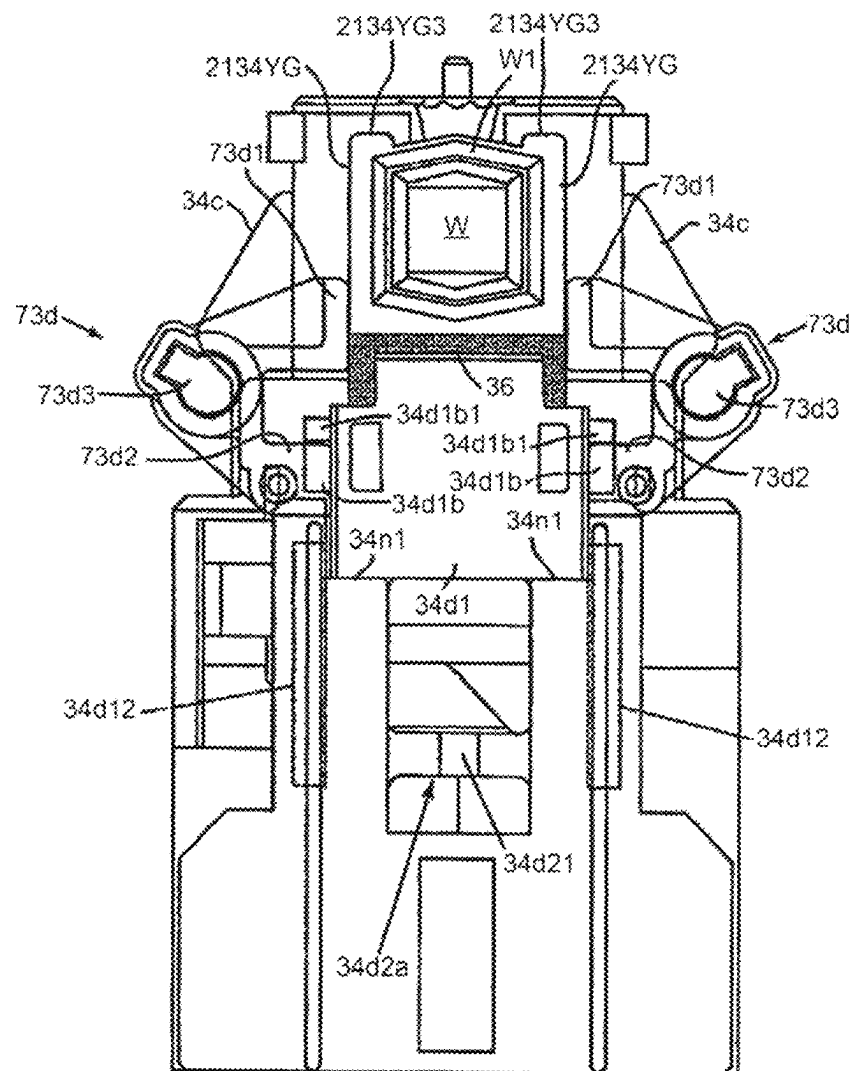


FIG.87

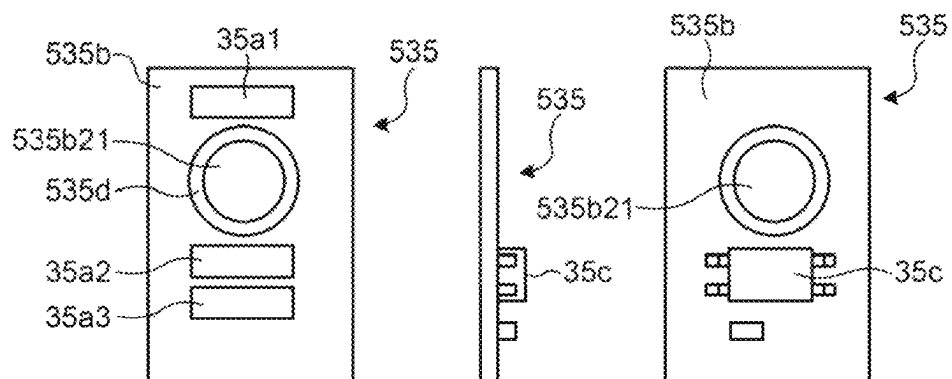


FIG.88A

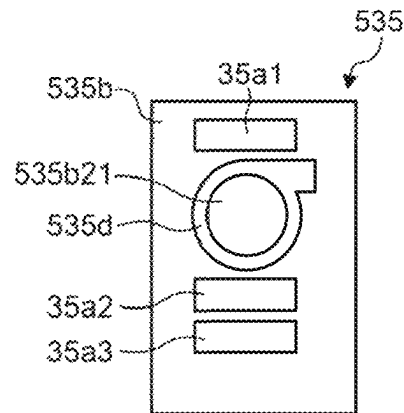


FIG.88B

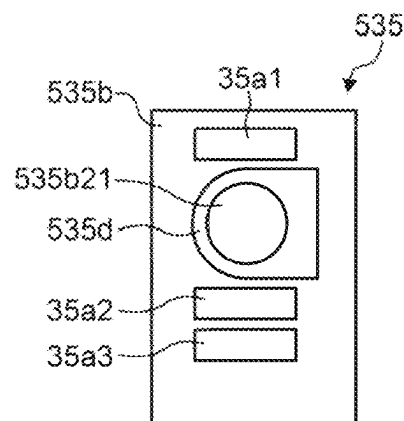
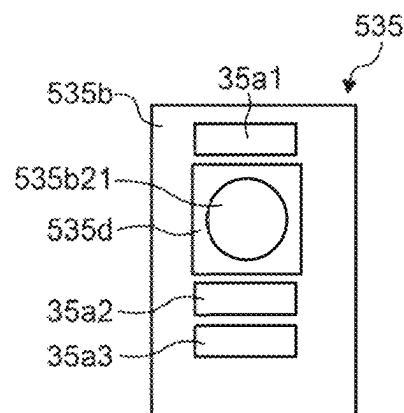


FIG.88C



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INFORMATION STORAGE SYSTEM INCLUDING A PLURALITY OF TERMINALS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/869,795, filed Jul. 21, 2022, which is a continuation of U.S. application Ser. No. 17/110,348, filed Dec. 3, 2020 (now U.S. Pat. No. 11,429,036), which is a continuation of U.S. application Ser. No. 16/929,446, filed Jul. 15, 2020 (now U.S. Pat. No. 11,188,007), which is a continuation of Ser. No. 16/584,982, filed Sep. 27, 2019 (now U.S. Pat. No. 10,725,398), which is a continuation of U.S. application Ser. No. 15/970,427, filed May 3, 2018 (now U.S. Pat. No. 10,754,275), which is a continuation of U.S. application Ser. No. 15/430,261, filed Feb. 10, 2017 (now U.S. Pat. No. 9,989,887), which is a continuation of U.S. application Ser. No. 14/750,679, filed Jun. 25, 2015 (now U.S. Pat. No. 9,599,927), which is a continuation of U.S. application Ser. No. 14/047,755, filed Oct. 7, 2013 (now U.S. Pat. No. 9,110,402), which is divisional of U.S. application Ser. No. 13/691,023, filed Nov. 30, 2012 (now U.S. Pat. No. 8,660,441), which is a continuation of U.S. application Ser. No. 13/448,987, filed on Apr. 17, 2012 (now U.S. Pat. No. 8,346,105), which is a continuation of PCT International Application No. PCT/JP2011/063993, filed Jun. 13, 2011 which designates the United States, and which claims the benefit of priority from Japanese Patent Application No. 2010-134560, filed Jun. 11, 2010, Japanese Patent Application No. 2011-062216, filed Mar. 22, 2011, Japanese Patent Application No. 2011-062283, filed Mar. 22, 2011, Japanese Patent Application No. 2011-084820, filed Apr. 6, 2011, and Japanese Patent Application No. 2011-087786, filed Apr. 11, 2011; the entire contents of each of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to an image forming apparatus such as a copying machine, a printer, a facsimile, or a multi-function peripheral (MFP), a removable device and a developer container that are removably installed therein, and an information storage device installed therein.

2. Description of the Related Art

Conventionally, in an image forming apparatus such as a copying machine, a technique of removably installing a removable device such as a developer container (a toner bottle, a toner storage container, or an ink cartridge) or a process cartridge on an image forming apparatus body has been usually used (for example, Patent Literature 1: Japanese Patent Application Laid-open No. 2009-69417, Patent Literature 2: Japanese Patent Application Laid-open No. 2006-209060, and Patent Literature 3: Japanese Patent Application Laid-open No. 2002-196629).

In the removable device, an information storage device (an information recording unit or a non-volatile memory) such as an ID chip storing information to be exchanged with the image forming apparatus body is installed. In a state in which the removable device is set to the image forming apparatus body, information (for example, information such as manufacturing year, month, and date of the removable device, a manufacturing lot number, or a color of toner, or

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a kind of toner) stored in the information storage device is transmitted to a control unit of the image forming apparatus body, or information (information such as a use history of the image forming apparatus) is transmitted from the image forming apparatus body to the information storage device, so that fulfilling quality control of the image forming apparatus body and the removable device is performed.

Patent Literature 1 discloses a contact-type information storage device (an information recording unit). Specifically, in the contact-type information storage device (an ID chip), when the removable device (a toner storage container) is set to the image forming apparatus body, a metal pad (a terminal) comes in contact with a body side terminal of a connector installed in the image forming apparatus body. As a result, information can be exchanged between the information storage device of the removable device and the control unit (the body side information recording unit) of the image forming apparatus.

Further, a feeding opening for allowing the stored toner to flow out to the outside is installed in the developer container. The opening needs remain closed until it is loaded onto a developing device so as to prevent the toner from being scattered or leaking.

As a configuration for achieving the above desire, there has been suggested a configuration in which a shutter for opening/closing the opening installed in the developer container is installed. Further, as a configuration of the shutter, there has been suggested a configuration in which a flat plate-like shutter that is movable in a direction traversing the toner and an outlet is installed (For example, Patent Literature 4: Japanese Patent Application Laid-open No. 2010-066638).

However, the conventional techniques described above have the following problems.

As a first problem, the conventional contact-type information storage device may electrically get damaged since an electric circuit of the information storage device is not sufficiently earthed and so becomes an electrically floating state when the removable device is attached to or removed from the device body.

There is a need to solve the first problem described above and provides an information storage device, a removable device, a developer container, and an image forming apparatus in which electrical damage is difficult to occur in the information storage device even when the contact-type information storage device is installed in the removable device removably installed in the image forming apparatus body.

As a second problem, in the conventional contact-type information storage device, there may occur a problem in that contact sections thereof are misaligned (a contact failure) due to wrong positioning of the terminal (metal pad) installed in the information storage device and the terminal of the image forming apparatus body. Particularly, when the terminal of the information storage device is small, the problem becomes important.

There is a need to solve the second problem described above and provides a removable device, a developer container, and an image forming apparatus in which a contact failure caused by a positioning failure with the body side terminal of the connector of the image forming apparatus body is difficult to occur even when the contact-type information storage device is installed in the removable device removably installed in the image forming apparatus body.

A third problem is as follows. In recent years, toner having a small particle diameter has been used so as to improve the resolution. Improving a filter function so as to

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cope with using the toner may increase the material or processing cost. That is, when a foamable material is used, it is necessary to prescribe mesh fineness that does not let the toner through, a so-call foaming degree, but as mesh fineness increases, flexibility tends to decrease. This tendency may be difficult to go along with movement of the shutter, and a sealing characteristic may get worse.

There is a need to improve a shutter mechanism of the conventional toner feeding device to provide a developer storage container and an image forming apparatus which have a configuration capable of reliably preventing the toner from leaking from the developer storage container that is replaced by an attaching/detaching operation at a low cost.

SUMMARY OF THE INVENTION

It is an object of the present invention to at least partially solve the problems in the conventional technology.

An information storage device installed in a removable device configured to be removably installed in an image forming apparatus body, includes: an information storage unit that stores information communicated between the image forming apparatus body and the removable device; a terminal that comes in contact with a body side terminal installed in the image forming apparatus body and communicates the information with the image forming apparatus body; and a substrate that holds the information storage unit and the terminal and includes a hole configured to be engaged with a protruding section installed in the image forming apparatus body. The terminal includes a plurality of terminals each including one of a plurality of metallic plates arranged in a transverse direction thereof with a clearance therebetween. An earth terminal, which comes in contact with a body side earth terminal formed in the protruding section of the image forming apparatus body, is formed in the hole in the substrate. The hole in the substrate is disposed at a position sandwiched between two metallic plates among the plurality of metallic plates.

A removable device that is installed removably in an image forming apparatus body and is any one of a toner cartridge inside which toner is contained, a process cartridge inside which toner is contained, and an ink cartridge inside which ink is contained, includes the foregoing information storage device.

A developer container that is installed removably in an image forming apparatus body in a state in which a longitudinal direction of the developer container is horizontal, and that stores a developer thereinside, includes: a cylindrical container body that includes an opening formed at one end in the longitudinal direction and is configured such that the developer stored thereinside is conveyed toward the opening; a cap in which the opening of the container body is inserted and that includes a discharge opening used to discharge the developer, which has been discharged from the opening of the container body, outside the developer container; and a shutter that is held in the cap and moves to open/close the toner discharge opening in conjunction movement of the developer container when the developer container is attached to/detached from the image forming apparatus body. The cap includes the foregoing information storage device installed in an end section in the longitudinal direction and a positioning hole engaged with a positioning pin installed in the image forming apparatus body.

The above and other objects, features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed descrip-

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tion of presently preferred embodiments of the invention, when considered in connection with the accompanying drawings.

In the present invention, a "process cartridge" is defined as a removable device that is configured such that at least one of a charging unit for charging an image carrier, a developing unit (a developing device) for developing a latent image formed on the image carrier, and a cleaning unit for cleaning the surface of the image carrier is integrally formed with the image carrier and that is installed removably on the image forming apparatus body.

Further, in the present invention, a "nearly rectangular metallic plate" is defined to include a nearly rectangular one as well as a rectangular one. Thus, one in which all or part of an angular section of the rectangular metallic plate is chamfered and an R-shaped one are also included in the "nearly rectangular metallic plate."

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an overall configuration view illustrating an image forming apparatus according to an embodiment;

FIG. 2 is a cross sectional view illustrating an image forming unit;

FIG. 3 is a schematic view illustrating a state in which a toner container is installed in a toner feeding device;

FIG. 4 is a schematic perspective view illustrating a state in which four toner containers are installed in a toner container storage unit;

FIG. 5 is a schematic perspective view illustrating a state in which one toner container is installed in a toner container storage unit;

FIG. 6 is a side view illustrating a state in which a toner container is installed in a toner container storage unit;

FIG. 7 is a cross sectional view illustrating a state in which a cap section is installed in a cap receiving section;

FIG. 8 is a perspective view illustrating a cap receiving section of a toner container storage unit;

FIG. 9 is an enlarged perspective view illustrating a neighborhood of a leading end section of a bottle receiving section;

FIG. 10 is a back view illustrating a state in which a cap section is set to a bottle receiving section in a regular toner container;

FIG. 11 is a back view illustrating a state in which a cap section is set to a bottle receiving section in a non-regular toner container;

FIG. 12 is a perspective view illustrating a cap receiving section to which a cap section is set;

FIG. 13 is a front view illustrating a cap receiving section in a state in which a cap section is set;

FIG. 14A is a back view illustrating a cap receiving section, and FIG. 14B is a partial enlarged view illustrating a neighborhood of a contacted groove encircled by a dotted line in a cap receiving section of FIG. 14A;

FIG. 15 is a perspective view illustrating a cap receiving section from obliquely below;

FIG. 16 is a perspective view illustrating a connector;

FIG. 17 is a schematic view illustrating a state in which an information storage device of a cap section is set to a connector of a cap receiving section;

FIG. 18 is a perspective view illustrating a toner container from obliquely below;

FIG. 19 is a side view illustrating a toner container;

FIG. 20 is a perspective view illustrating a cap section side of a toner container from obliquely below;

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FIG. 21 is a front view illustrating a toner container from a cap section side;

FIG. 22 is a perspective view illustrating a state in which a shutter member of a toner container closes a toner discharge opening;

FIG. 23 is a perspective view illustrating a state in which a shutter member of a toner container opens a toner discharge opening;

FIGS. 24A to 24C are schematic views illustrating an opening operation of a shutter member that is in conjunction with a mounting operation of a toner container on a toner container storage unit;

FIG. 25 is a perspective view illustrating a shutter member;

FIG. 26 is another perspective view illustrating a shutter member;

FIG. 27 is a perspective view illustrating a state in which an information storage device is extracted;

FIG. 28 is a six-plane view illustrating a holding member of an information storage device;

FIG. 29 is a three-plane view illustrating an information storage device;

FIG. 30 is a cross sectional view illustrating a neighborhood of a cap section of a toner container;

FIG. 31 is a schematic cross sectional view illustrating a toner container according to a second embodiment;

FIG. 32 is a back view illustrating a cap section in the toner container of FIG. 31;

FIG. 33 is a perspective view illustrating a holding cover engaged with a holding member;

FIG. 34 is a schematic view illustrating a state in which an information storage device of a toner container according to a third embodiment is set to a connector of a cap receiving section;

FIG. 35 is a three-plane view illustrating a substrate of an information storage device according to a fourth embodiment;

FIG. 36 is a three-plane view illustrating a substrate of an information storage device according to a fifth embodiment;

FIG. 37 is a perspective view illustrating an information storage device, a holding member, and a connector;

FIG. 38 is a perspective view illustrating a state in which an information storage device is engaged with a connector;

FIGS. 39A and 39B are schematic views illustrating an electric circuit of an information storage device and an electric circuit of a connector;

FIGS. 40A and 40B are front views illustrating an information storage device;

FIG. 41 is a view illustrating an information storage device in an inspection process;

FIGS. 42A and 42B are perspective views illustrating a toner container according to a sixth embodiment;

FIG. 43 is a front view illustrating a toner container in which a face plate is not installed;

FIG. 44 is a cross sectional view illustrating a toner container in which an information storage device and a face plate are installed;

FIG. 45 is a view illustrating a state in which an information storage device is being inserted into a connector;

FIGS. 46A and 46B are perspective views illustrating a toner container of another form;

FIGS. 47A to 47C are views illustrating a toner container of another form;

FIG. 48 is an exploded perspective view illustrating a toner container according to a seventh embodiment;

FIG. 49 is a cross sectional view illustrating the toner container of FIG. 48;

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FIG. 50 is a perspective view illustrating an image forming apparatus according to an eighth embodiment;

FIGS. 51A and 51B illustrate toner cartridges installed in the image forming apparatus of FIG. 50, FIG. 51A is a cross sectional view, and FIG. 51B is a bottom view;

FIG. 52 is a perspective view illustrating an image forming apparatus according to a ninth embodiment;

FIG. 53 is a schematic view illustrating a state in which a connector is connected to an information storage device in the image forming apparatus of FIG. 52;

FIG. 54 is a perspective view illustrating an ink cartridge according to a tenth embodiment;

FIG. 55 is a top view illustrating an image forming apparatus in which the ink cartridge of FIG. 54 is installed;

FIG. 56 is a perspective view illustrating a connector of an image forming apparatus according to the tenth embodiment;

FIG. 57 is a three-plan view illustrating an information storage device that comes in contact with the connector of FIG. 56;

FIG. 58 is a three-plane view illustrating an information storage device of another form;

FIG. 59 is a perspective view illustrating a toner container according to a twelfth embodiment;

FIG. 60 is an enlarged perspective view illustrating configurations of an information storage device and a holding member according to the twelfth embodiment;

FIG. 61 is an exploded perspective view illustrating the configurations of the information storage device and the holding member according to the twelfth embodiment;

FIG. 62 is an enlarged perspective view illustrating a fixing state between the information storage device and the holding member according to the twelfth embodiment;

FIG. 63 is an enlarged perspective view illustrating a fixing state between an information storage device and a holding member according to a thirteenth embodiment;

FIG. 64 is an enlarged perspective view illustrating configurations of the information storage device and the holding member according to the thirteenth embodiment;

FIG. 65 is an enlarged perspective view illustrating a fixing state between an information storage device and a holding member according to a fourteenth embodiment;

FIG. 66 is an enlarged perspective view illustrating configurations of the information storage device and the holding member according to the fourteenth embodiment;

FIG. 67 is a cross sectional view illustrating a cap section illustrated in FIG. 18;

FIG. 68 is a perspective view, viewed from a bottom surface of a shutter, for explaining a configuration of a shutter used in a cap section illustrated in FIG. 18;

FIGS. 69A and 69B are views, corresponding to FIG. 18, for explaining an opening/closing state of a shutter illustrated in FIGS. 69A and 69B;

FIGS. 70A to 70C are views for explaining a configuration of the shutter illustrated in FIGS. 69A and 69B;

FIGS. 71A to 71C are views illustrating an opening state of the shutter illustrated in FIGS. 70A to 70C and a cross section of the state;

FIG. 72 is a plane view for explaining a relation between a body side shutter closing mechanism and a shutter;

FIG. 73 is a plane view illustrating a state of the body side shutter closing mechanism illustrated in FIG. 72;

FIG. 74 is a plane view illustrating a state of the body side shutter closing mechanism that has changed from the state illustrated in FIG. 73;

FIGS. 75A to 75D are views for explaining a positional relation between a toner discharge opening and a shutter and a sealing state of a seal material;

FIGS. 76A and 76B are views illustrating a configuration of an information storage device used in a sixteenth embodiment;

FIG. 77 is a perspective view of a cap receiving side that becomes part of an electrical connection section with an information storage device;

FIG. 78 is a perspective view illustrating a common electronic substrate including a shutter connected with an information storage device;

FIG. 79 is a view for explaining a connection state between the information storage device used in the sixteenth embodiment and a connector at a cap receiving section side;

FIG. 80 is a perspective view, viewed from a front right side in an insertion direction of a cap in the state in which a shutter is closed, for explaining a modification related to a configuration of a cap section according to a seventeenth embodiment;

FIG. 81 is a perspective view viewed from a front left side in an insertion direction of the cap illustrated in FIG. 80;

FIG. 82 is an exploded perspective view of the cap section illustrated in FIG. 80;

FIG. 83 is a perspective view illustrating a modification of a main part of the cap section illustrated in FIG. 80;

FIG. 84 is a plane view for explaining an aspect of a body side shutter closing mechanism targeting on the cap section illustrated in FIG. 80;

FIG. 85 is a plane view illustrating the body side shutter closing mechanism illustrated in FIG. 84;

FIG. 86 is a plane view illustrating a state of the body side shutter closing mechanism that has changed from the state illustrated in FIG. 85;

FIG. 87 is a three-plane view illustrating an alternative of the substrate illustrated in FIG. 36; and

FIGS. 88A to 88C are plane views illustrating further alternatives of the substrate illustrated in FIG. 36.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings. In the drawings, the same or corresponding parts are denoted by the same reference numerals, and thus a duplicated description thereof will be appropriately simplified or omitted.

First Embodiment

A first embodiment of the present invention will be described in detail with reference to FIGS. 1 to 30.

First, a configuration and operation of the entire image forming apparatus will be described.

As illustrated in FIG. 1, in a toner container storage unit 70 above an image forming apparatus body 100, toner containers 32Y, 32M, 32C, and 32K (developer containers) are removably (replaceably) installed as four removable devices corresponding to respective colors (yellow, magenta, cyan, and black) (also see FIGS. 3 to 5).

An intermediate transfer unit 15 is disposed below the toner container storage unit 70. Image forming units 6Y, 6M, 6C, and 6K corresponding to respective colors (yellow, magenta, cyan, and black) are disposed in line to face an intermediate transfer belt 8 of the intermediate transfer unit 15.

Toner feeding devices 60Y, 60M, 60C, and 60K are disposed below the toner containers 32Y, 32M, 32C, and 32K as the removable devices (developer containers), respectively. The toners stored in the toner containers 32Y, 32M, 32C, and 32K are supplied (fed) to the inside of the developing devices of the image forming units 6Y, 6M, 6C, and 6K by the toner feeding devices 60Y, 60M, 60C, and 60K, respectively.

Referring to FIG. 2, the image forming unit 6Y corresponding to yellow includes a photosensitive drum 1Y, a charging unit 4Y disposed around the photosensitive drum 1Y, a developing device 5Y (a developing section), a cleaning unit 2Y, a neutralizing unit (not shown), or the like. An image forming process (a charging process, an exposure process, a developing process, a transfer process, and a cleaning process) is performed on the photosensitive drum 1Y, and so a yellow image is formed on the photosensitive drum 1Y.

The remaining three image forming units 6M, 6C, and 6K have almost the same configuration as the image forming unit 6Y corresponding to yellow except that colors of used toner are different. Hereinafter, a description of the remaining three image forming units 6M, 6C, and 6K will be appropriately omitted, and a description will be made in connection with the image forming unit 6Y corresponding to yellow.

Referring to FIG. 2, the photosensitive drum 1Y is rotationally driven clockwise in FIG. 2 by a driving motor (not shown). The surface of the photosensitive drum 1Y is uniformly charged at the position of the charging unit 4Y (the charging process).

Thereafter, the surface of the photosensitive drum 1Y reaches an irradiation position of laser light L emitted from an exposure unit 7 (see FIG. 1), and an electrostatic latent image corresponding to yellow is formed by exposure scanning at this position (the exposure process).

Then, the surface of the photosensitive drum 1Y reaches the position facing the developing device 5Y, and the electrostatic latent image is developed at this position, so that a yellow toner image is formed (the developing process).

Next, the surface of the photosensitive drum 1Y reaches the position facing the intermediate transfer belt 8 and a primary transfer bias roller 9Y, and the toner image on the photosensitive drum 1Y is transferred onto the intermediate transfer belt 8 at this position (a primary transfer process). At this time, a slight amount of non-transfer toner remains on the photosensitive drum 1Y.

Thereafter, the surface of the photosensitive drum 1Y reaches the position facing the cleaning unit 2Y, and the non-transfer toner remaining on the photosensitive drum 1Y is mechanically collected by a cleaning blade 2a at this position (the cleaning process).

Finally, the surface of the photosensitive drum 1Y reaches the position facing the neutralizing unit (not shown), and residual potential on the photosensitive drum 1Y is removed at this position.

Thus, a series of image forming processes performed on the photosensitive drum 1Y are finished.

The above described image forming process is performed even in the other image forming units 6M, 6C, and 6K in the same manner as in the yellow image forming unit 6Y. That is, the laser light L based on image information is irradiated from the exposure unit 7 disposed below the image forming units onto the photosensitive drums of the image forming units 6M, 6C, and 6K. Specifically, the exposure unit 7 emits the laser light L from a light source and irradiates the laser

light L onto the photosensitive drum through a plurality of optical elements while scanning the laser light L by a polygon mirror that is rotationally driven.

Thereafter, toner images of respective colors formed on the respective photosensitive drums through the developing process are transferred onto the intermediate transfer belt 8 in a superimposed manner. As a result, a color image is formed on the intermediate transfer belt 8.

Referring to FIG. 1, the intermediate transfer unit 15 includes the intermediate transfer belt 8, four primary transfer bias rollers 9Y, 9M, 9C, and 9K, a secondary transfer bias roller 12, a plurality of tension rollers, an intermediate transfer cleaning unit, and the like. The intermediate transfer belt 8 is stretched over and supported by a plurality of roller members and endlessly moves in a direction of an arrow in FIG. 1 as one roller member 12 is rotationally driven.

The four primary transfer bias rollers 9Y, 9M, 9C, and 9K sandwich the intermediate transfer belt 8 together with the photosensitive drums 1Y, 1M, 1C, and 1K, respectively, to form primary transfer nips. A transfer bias reverse to a polarity of the toner is applied to the primary transfer bias rollers 9Y, 9M, 9C, and 9K.

The intermediate transfer belt 8 moves in a direction of an arrow and sequentially passes through the primary transfer nips of the primary transfer bias rollers 9Y, 9M, 9C, and 9K. The toner images of respective colors on the photosensitive drums 1Y, 1M, 1C, and 1K are primary-transferred onto the intermediate transfer belt 8 in a superimposed manner.

Thereafter, the intermediate transfer belt 8 onto which the toner images of respective colors are transferred in a superimposed manner reaches the position facing a secondary transfer roller 19. At this position, the secondary transfer bias roller 12 sandwiches the intermediate transfer belt 8 together with the secondary transfer roller 19 to form a secondary transfer nip. The toner images of four colors formed on the intermediate transfer belt 8 are transferred onto a recording medium P such as a transfer sheet conveyed to the position of the secondary transfer nip. At this time, the non-transfer toner that has not been transferred onto the recording medium P remains on the intermediate transfer belt 8.

Thereafter, the intermediate transfer belt 8 reaches the position of the intermediate transfer cleaning unit (not shown). At this position, the non-transfer toner on the intermediate transfer belt 8 is collected.

As a result, a series of transfer processes performed on the intermediate transfer belt 8 are finished.

The recording medium P conveyed to the position of the secondary transfer nip is conveyed through a paper feeding roller 27, a pair of resist rollers 28, and the like from a paper feeding unit 26 disposed below the apparatus body 100.

Specifically, a plurality of recording media P such as transfer sheets are stored in a superimposed manner in the paper feeding unit 26. If the paper feeding roller 27 is rotationally driven counterclockwise in FIG. 1, the top recording medium P is fed toward between the rollers of the pair of resist rollers 28.

The recording medium P fed to the pair of resist rollers 28 stops at the position of a roller nip of the pair of resist rollers 28 that has stopped rotational driving. In synchronization with timing of the color image on the intermediate transfer belt 8, the pair of resist rollers 28 is rotationally driven, and the recording medium P is conveyed toward the secondary transfer nip. Thus, a desired color image is transferred onto the recording medium P.

Thereafter, the recording medium P onto which the color image has been transferred at the position of the secondary

transfer nip is conveyed to the position of a fixing device 20. At this position, the color image transferred onto the surface is fixed to the recording medium P by heat and pressure by a fixing belt and a pressing roller.

Thereafter, the recording medium P passes through between rollers of a pair of ejecting rollers 29 and then is ejected to the outside of the apparatus. A recording medium P ejected to the outside of the apparatus by the pair of ejecting rollers 29 is sequentially stacked on a stack unit 30 as an output image.

Thus, in the image forming apparatus, a series of image forming processes are finished.

Next, a configuration and operation of the developing device in the image forming unit will be described in further detail with reference to FIG. 2.

The developing device 5Y includes a developing roller 51Y facing the photosensitive drum 1Y, a doctor blade 52Y facing the developing roller 51Y, two conveying screws disposed in developer storage units 53Y and 54Y, a density detecting sensor 56Y for detecting the density of the toner contained in the developer, and the like. The developing roller 51Y is configured with a magnet fixedly disposed to the inside thereof, a sleeve rotating around the magnet, and the like. A two-component developer G composed of a carrier and a toner is stored in the developer storage units 53Y and 54Y. The developer storage unit 54Y is communicated with a toner falling conveying path 64Y through an opening formed thereabove.

The developing device 5Y having the above described configuration operates as follows.

The sleeve of the developing roller 51Y rotates in a direction of an arrow in FIG. 2. The developer G supported on the developing roller 51Y by a magnetic field formed by the magnet moves on the developing roller 51Y as the sleeve rotates.

The developer G inside the developing device 5Y is adjusted so that a ratio of toner (toner density) contained in the developer can be within a predetermined range. Specifically, as the toner inside the developing device 5Y is consumed, the toner stored in the toner container 32Y is fed to the inside of the developer storage unit 54Y through the toner feeding device 60Y (for example, see FIG. 3). A configuration and operation of the toner feeding device will be described later in detail.

Thereafter, the toner fed to the inside of the developer storage unit 54Y circulates through the two developer storage units 53Y and 54Y while being mixed and agitated together with the developer G by the two conveying screws 55Y (movement in a direction vertical to a paper plane of FIG. 2). The toner in the developer G is absorbed into the carrier by frictional electrification with the carrier and supported on the developing roller 51Y together with the carrier by magnetic force formed on the developing roller 51Y.

The developer G supported on the developing roller 51Y is conveyed in a direction of an arrow in FIG. 2 and then reaches the position of the doctor blade 52Y. The developer G on the developing roller 51Y is adjusted to an appropriate developer amount at this position and then conveyed up to the position (a developing area) facing the photosensitive drum 1Y. The toner is absorbed into a latent image formed on the photosensitive drum 1Y by a magnetic field formed on the developing area. Thereafter, as the sleeve rotates, the developer G remaining on the developing roller 51Y reaches above the developer storage unit 53Y and leaves the developing roller 51Y at this position.

Next, the toner feeding devices **60Y**, **60M**, **60C**, and **60K** will be described in detail with reference to FIGS. **3** to **5**.

Referring to FIG. **3**, the toners inside the toner containers **32Y**, **32M**, **32C**, and **32K** installed in the toner container storage unit **70** of the apparatus body **100** are appropriately fed to the inside of the developing devices by the toner feeding devices **60Y**, **60M**, **60C**, and **60K** respectively installed for toner colors as the toners inside the developing devices of respective colors are consumed.

The four toner feeding devices **60Y**, **60M**, **60C**, and **60K** and the toner containers **32Y**, **32M**, **32C**, and **32K** (the developer containers) have almost the same configuration except that the toner colors used in the image forming process are different. Thus, a description will be made focusing on the toner feeding devices **60Y** and the toner container **32Y** corresponding to yellow, and a description of the toner feeding devices **60M**, **60C**, and **60K** and the toner containers **32M**, **32C**, and **32K** corresponding to the remaining three colors will be appropriately omitted.

Referring to FIG. **1**, if a body cover (not shown) installed on the front side of the apparatus body **100** (the front side in a direction vertical to the paper plane in FIG. **1**) is opened, the toner container storage unit **70** (an insertion opening **71**) is exposed. In the state in which a longitudinal direction of the toner containers **32Y**, **32M**, **32C**, and **32K** (the developer containers) is a horizontal direction, performed is an attaching/detaching operation of the toner containers **32Y**, **32M**, **32C**, and **32K** to/from the front side of the apparatus body **100** (an attaching/detaching operation in which the longitudinal direction of the toner container is an attaching/detaching direction).

As illustrated in FIG. **4**, when the toner containers **32Y**, **32M**, **32C**, and **32K** are mounted on the toner container storage unit **70** of the apparatus body **100** (movement in a direction of an arrow **Q**), in conjunction with the mounting operation, a shutter member **34d** of the toner containers **32Y**, **32M**, **32C**, and **32K** moves, and so a toner discharge opening **W** (a discharging opening) is opened, so that toner feeding openings **73w** (for example, see FIG. **3**) of the toner feeding devices **60Y**, **60M**, **60C**, and **60K** are communicated with the toner discharge opening **W**. The toner stored in the toner containers **32Y**, **32M**, **32C**, and **32K** is discharged from the toner discharge opening **W** and stored in a toner tank unit **61Y** through the toner feeding opening **73w** of the toner feeding devices **60Y**, **60M**, **60C**, and **60K**.

Referring to the schematic view of FIG. **3**, the toner container **32Y** includes a cap section **34Y** that is a nearly cylindrical-shaped toner bottle and is usually non-rotatably held on the toner container storage unit **70** and a container body **33Y** (a bottle body) in which a gear **33c** is integrally formed. The container body **33Y** is relatively rotatably held on the cap section **34Y** and is rotationally driven in a direction of an arrow in FIG. **3** by a driving unit **91** (including a driving motor, a driving gear **81**, and the like). As the container body **33Y** rotates, the toner stored inside the toner container **32Y** (the container body **33Y**) is conveyed in a longitudinal direction (conveyance from the left to the right in FIG. **3**) by a protrusion **33b** formed on an inner peripheral surface of the container body **33Y** in a helical form, and the toner is discharged from the toner discharge opening **W** of the cap section **34Y**. That is, as the container body **33Y** of the toner container **32Y** is appropriately rotationally driven by the driving unit **91**, the toner is appropriately supplied to the toner tank unit **61Y**. Further, when each of the toner containers **32Y**, **32M**, **32C**, and **32K** reaches the end of its life (when the stored toner is almost consumed and becomes empty), it is replaced with a new one.

Referring to FIG. **3**, the toner feeding devices **60Y**, **60M**, **60C**, and **60K** include the toner container storage unit **70**, the toner tank unit **61Y**, a toner conveying screw **62Y**, an agitating member **65Y**, a toner end sensor **66Y**, the driving unit **91**, and the like.

The toner tank unit **61Y** is disposed below the toner discharge opening **W** of the toner container **32Y** and stores the toner discharged from the toner discharge opening **W** of the toner container **32Y**. The bottom of the toner tank unit **61Y** is connected with an upstream section of the toner conveying screw **62Y**.

The toner end sensor **66Y** that detects that the toner stored in the toner tank unit **61Y** has become smaller than a predetermined amount is installed on the wall surface of the toner tank unit **61Y** (at the position of a predetermined height from the bottom). A piezoelectric sensor or the like may be used as the toner end sensor **66Y**. When the control unit **90** detects that the toner stored in the toner tank unit **61Y** has become smaller than a predetermined amount (toner end detection) through the toner end sensor **66Y**, the driving unit **91** rotationally drives the container body **33Y** of the toner container **32Y** during a predetermined time under control of the control unit **90**, so that the toner is fed to the toner tank unit **61Y**. Further, when toner end detection by the toner end sensor **66Y** is not released even if such control is repeated, it is recognized that there is no toner in the toner container **32Y**, and a message for encouraging the replacement of the toner container **32Y** is displayed on a display unit (not shown) of the apparatus body **100**.

Further, the agitating member **65Y** that prevents the toner stored in the toner tank unit **61Y** from being agglomerated is installed on the central section of the toner tank unit **61Y** (near the toner end sensor **66Y**). The agitating member **65Y** has a flexible member installed on a shaft section and rotates clockwise in FIG. **3** to agitate the toner inside the toner tank unit **61Y**. Further, the leading end of the flexible member of the agitating member **65Y** comes in sliding contact with the detection surface of the toner end sensor **66Y** at a rotation period, thereby preventing a problem in that the toner is fixed to the detection surface of the toner end sensor **66Y** and so a degree of detection accuracy decreases.

Even though not shown, the toner conveying screw **62Y** conveys the toner stored in the toner tank unit **61Y** obliquely upward. Specifically, the toner conveying screw **62Y** linearly conveys the toner from the bottom of the toner tank unit **61Y** (the lowest point) toward the top of the developing device **5Y**. The toner conveyed by the toner conveying screw **62Y** falls through the toner falling conveying path **64Y** (for example, see FIG. **2**) by its own weight and is fed to the inside of the developing device **5Y** (the developer storage unit **54Y**).

Referring to FIG. **4**, the toner container storage unit **70** mainly includes a cap receiving section **73** for holding the cap section **34Y** of the toner container **32Y**, a bottle receiving section **72** (a container body bearing) for holding the container body **33Y** of the toner container **32Y**, and an insertion opening **71** that functions as an insertion opening at the time of the mounting operation of the toner container **32Y**.

Next, the toner container storage unit **70** (the bottle receiving section **72** and the cap receiving section **73**) will be described in detail with reference to FIGS. **6** to **17**.

First, as described above with reference to FIGS. **4** and **5**, the bottle receiving section **72**, the cap receiving section **73**, and the insertion opening **71** (that is not shown in FIG. **5**) are formed in the toner container storage unit **70**. The toner container **32Y** is mounted on the toner container storage unit

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70 through the insertion opening 71 by a user gripping a gripping section 33d in a state in which the longitudinal direction is the horizontal direction and the longitudinal direction is the mounting direction in which the cap section 34Y is the head of the container body 33Y. The toner container 32Y inserted through the insertion opening 71 is pushed toward the cap receiving section 73 by the user while sliding on a bottle receiving surface 72a of the bottle receiving section 72 (for example, see FIGS. 5, 6, and 9). Referring to FIG. 6, in the bottle receiving section 72, the bottle receiving surface 72a is formed for each color, and the toner containers 32Y, 32M, 32C, and 32K corresponding to respective colors are inserted (inserted in a direction of a white arrow). Further, referring to FIG. 8, even in the cap receiving section 73, bottle receiving sections 73Y, 73M, 73C, and 73K are formed for respective colors, and the toner containers 32Y, 32M, 32C, and 32K corresponding to respective colors are inserted (inserted in a direction of a white arrow). At this position, the cap receiving section is non-rotatably held.

Referring to FIGS. 5 and 24(A), the bottle receiving surface 72a, a stopper release urging section 72b, and the like are formed in the bottle receiving section 72 of the toner container storage unit 70.

The bottle receiving surface 72a functions as a sliding surface of the toner container 32Y at the time of the attaching/detaching operation of the toner container 32Y and functions as a holding unit of the rotationally driven container body 33Y after the toner container 32Y has been completely set.

Referring to FIG. 5, the stopper release urging section 72b is a trapezoidal rib formed above the bottle receiving surface 72a (at the downstream side of the toner container 32Y in the mounting direction). Referring to FIG. 24, the stopper release urging section 72b pushes a stopper release section 34d/21 of the shutter member 34d up and releases a contact state between a stopper section 34d/22 and a contact section 34d/5 in conjunction with the mounting operation of the toner container 32Y (allows an opening operation of the shutter member 34d).

Referring to FIGS. 14A, 14B, and 15, in the cap receiving section 73 of the toner container storage unit 70, a main reference pin 73a, a sub reference pin 73b, a contacted groove 73m, a lateral groove 73h, a wall section 73g, a through hole 73f, and the like are disposed.

The main reference pin 73a and the sub reference pin 73b as positioning pins are fitted into a first positioning hole 34a and a second positioning hole 34b of the cap section 34Y of the toner container 32Y illustrated in FIGS. 20 and 21, respectively. Positioning of the cap section 34Y is performed in the cap receiving section 73.

Referring to FIG. 7, the main reference pin 73a is formed to be longer than the sub reference pin 73b in the longitudinal direction (the position of the reference surface that is the base section is formed on the same plane surface). Further, the main reference pin 73a has a tapered leading end portion. Thus, in the attaching operation of the toner container 32Y to the cap receiving section 73 in the longitudinal direction, the toner container 32Y can be smoothly mounted on the cap receiving section 73.

Further, referring to FIGS. 14A, 14B, and 15, the contacted groove 73m is the inner wall of the cap receiving section 73 and is also a concave section that is formed, above the main reference pin 73a, in a groove shape at an upstream side in the mounting direction further than the leading end section of the main reference pin 73a. A guide rail section 34e that is formed to extend in the longitudinal direction in

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an upper outer circumference of the cap section 34Y of the toner container 32Y which will be described later is fitted into the contacted groove 73m before the main reference pin 73a is inserted into the positioning hole 34a.

Referring to FIGS. 12 and 15, the lateral groove 73h that is formed to extend in the longitudinal direction and is penetrated toward the outer circumferential side of the cap receiving section 73 is formed on each of both sides of the inner wall of the cap receiving section 73 in a left-right symmetrical relation. Further, referring to FIGS. 12 and 13, cap section sandwiching members 73r that have a nearly pentagonal shape when viewed from the top and a groove section 73r1 (that is formed to be connected with a lateral groove 73h) when viewed in the longitudinal direction are disposed on an outer circumferential side of the cap receiving section 73 in a left-right symmetrical relation.

The cap section sandwiching member 73r is formed of a member different from the cap receiving section 73, fitted into a dent formed on the outer circumferential surface of the cap receiving section 73, urged by a torsion coil spring 93 disposed thereabove centering on a cylindrical axis, and thus pressed against the side of the lateral groove 73h. As a result, the lateral groove 73h is connected with the groove section 73r1 of the cap section sandwiching member 73r, and a pair of deeper left and right groove sections is apparently formed.

In the case of attaching or detaching the toner container 32Y, the lateral protrusion 34c formed in the cap section 34Y pushes and passes through the cap section sandwiching member 73r urged by the torsion coil spring 93 inside the above described deeper groove section (one in which the groove section 73r1 is formed integrally with the lateral groove 73h). Thus, the user who performs the attaching/detaching operation of the toner container 32Y to/from the image forming apparatus body 100 (the cap receiving section 73) can feel a click feeling synchronized with the attaching/detaching operation and perform the attaching/detaching operation of the toner container 32Y at an optimum speed (acceleration) other than a half-hearted speed.

Referring to FIGS. 14A and 15, on the inner side wall surface of the cap receiving section 73 (the wall surface rising in a vertical direction at an apparatus direction inner side), the through hole 73f having a shape obtained by connecting and overlapping edge lines of an elliptical hole and a quadrate hole extending in the vertical direction is formed. A connector 73e (for example, see FIG. 16) which will be described later is installed to be exposed in the inner wall side of the cap receiving section 73 through the through hole 73f (for example, see FIG. 17). When the toner container 32Y is mounted on the cap receiving section 73 (the apparatus body 100), the connector 73e comes in face contact with an ID chip 35 disposed at the leading end of the cap section 34Y, and so information communication can be performed between the ID chip 35 and the apparatus body 100 (the control unit 90).

An installation form of the connector 73e on the cap receiving section 73 of the toner container storage unit 70 will be described below.

The four connectors 73e are disposed in the cap receiving sections 73, corresponding to the toner containers 32Y, 32M, 32C, and 32K of respective colors of yellow, magenta, cyan, and black. Referring to FIG. 8, the four connectors 73e are disposed in line on a single rectangular common electronic substrate 95. Specifically, by fitting a snap fit 73e4 formed on the bottom of the connector 73e into a hole (not shown) formed in the common electronic substrate 95, the connector 73e is fixed onto the common electronic substrate 95.

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Further, referring to FIGS. 8 and 17, the common electronic substrate 95 to which the four connectors 73e are fixed are installed and fixed along the arrangement direction of the four cap receiving sections 73K, 73C, 73M, and 73Y in the state in which the four connectors 73e are inserted into the inside of the cap receiving section 73 through the through holes 73f, respectively. Specifically, four screws 99 are screwed into female screw sections 73x formed below the outer wall sections of the four cap receiving sections 73K, 73C, 73M, and 73Y through holes formed in the common electronic substrate 95, and the common electronic substrate 95 is screw-coupled with the cap receiving section 73 from the outside.

A configuration and operation of the connector 73e will be described below.

Referring to FIG. 16, the connector 73e includes a connector body 73e1, four body side terminals 73e2, two positioning pins 73e3 (positioning protruding sections), and the like. The four body side terminals 73e2 of the connector 73e are flat (or linear) metallic members, respectively, have one terminal side as a fixed terminal and the other terminal side as a free end, and are fixed to the connector body 73e1. The four body side terminals 73e2 have curved sections (sections that become contact points with a metallic pad 35a as a metallic plate) that curve toward the side of the ID chip 35 at the other end side thereof. By the mounting operation of the cap section 34Y to the cap receiving section 73, the curved section of the body side terminal 73e2 is displaced in a -X direction of FIG. 16 and slides toward the left in FIG. 29 (near a first virtual line S1) from a longitudinal direction central section while gradually increasing contact pressure on the metallic pad 35a (the metallic plate) of the ID chip 35 disposed in the cap section 34Y.

As illustrated in FIG. 16, the leading end section of the positioning pin 73e3 has a tapered shape so that engagement with a notch 35b1 of the ID chip 35 can be smoothly performed.

Referring to FIGS. 14A, 14B, 15, and 17, a wall section 73g is installed to surround the lower section and the side section of the through hole 73f in which the connector 73e is installed. By forming the wall section 73g, even if the toner is scattered from the vicinity of the toner discharge opening W of the toner container 32Y to the outside, since the scattered toner is blocked by the wall section 73g, the scattered toner is difficult to stick directly to the connector 73e and the ID chip 35. Thus, a contact failure (a communication failure) between the connector 73e (the body side terminal 73e2) and the ID chip 35 (the metallic pad 35a) caused by the scattered toner can be suppressed.

Necessary information is exchanged between the ID chip 35 (the information storage device) of the toner containers 32Y, 32M, 32C, and 32K and the connector 73e of the apparatus body 100. Information communicated between both sides includes information such as a manufacturing number, a manufacturing date, and the number of recycling times of the toner container or the ID chip, information such as capacity, a lot number, and color of a toner, and information such as a use history of the image forming apparatus body 100. In the ID chip 35 (the information storage device), the electronic information is stored in advance before it is installed in the image forming apparatus body 100 (or information received from the apparatus body 100 after it is installed is stored). The ID chip 35 (the information storage device) will be described later in further detail.

Next, the toner containers 32Y, 32M, 32C, and 32K will be described in detail with reference to FIGS. 18 to 30.

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Referring to FIGS. 18 to 20, the toner container 32Y mainly includes the container body 33Y (a bottle body) and the cap section 34Y (a bottle cap) disposed at the head thereof. Further, the ID chip 35 as the information storage device or the like is detachably installed in the cap section 34Y of the toner container 32Y.

On the head of the container body 33Y, the gear 33c that integrally rotates together with the container body 33Y and an opening A are disposed on one end side in the longitudinal direction (the left-right direction in FIG. 30) (for example, see FIGS. 19 and 30). The opening A is disposed on the head of the container body 33Y (the position that becomes the front side in the mounting operation) and discharges the toner stored in the container body 33Y toward the space inside the cap section 34Y (a hollow space, for example, see FIG. 30).

Further, as the toner is consumed at the image forming apparatus body side, toner conveyance from the inside of the container body 33Y to the hollow space inside the cap section 34Y (rotational driving of the container body 33Y) is appropriately performed.

The gear 33c meshes with a driving gear 81 disposed in the toner container storage unit 70 of the apparatus body 100 and rotationally drives the container body 33Y centering on a rotational shaft. Specifically, the gear 33c is exposed through a notch hole 34x (for example, see FIG. 18) formed on the outer circumferential surface of the cap section 34Y which will be described later and meshes with the driving gear 81 of the apparatus body 100 at the obliquely downward meshing position in FIGS. 3 and 21. Further, driving force is transferred from the driving gear 81 to the gear 33c, and the container body 33Y rotates clockwise in FIG. 21. In the present first embodiment, the driving gear 81 and the gear 33c are spur gears.

Referring to FIG. 18, on the other end side of the container body 33Y in the longitudinal direction (the rear end section in the mounting direction), the gripping section 33d gripped by the user when performing the attaching/detaching work of the toner container 32Y to the image forming apparatus body 100 while gripping the gripping section 33d (movement of the toner container 32Y in a direction of an arrow in FIG. 18).

Further, on the outer circumferential surface of the container body 33Y, the helical protrusion 33b (a helical groove when viewed from the outer circumference surface side) is disposed. The helical protrusion 33b rotationally drives the container body 33Y in a predetermined direction and discharges the toner through the opening A. The container body 33Y having the above described configuration may be fabricated by blow molding together with the gear 33c disposed on the circumferential surface thereof and the gripping section 33d.

Referring to FIG. 30, in the toner container 32Y according to the present first embodiment, an agitating member 33f that rotates together with the container body 33Y is fitted into a bottle mouth section 33a (the opening A) illustrated in FIG. 19. The agitating member 33f is a rod-like member that extends from the hollow space inside the cap section 34Y to the inside of the container body 33Y. Since the agitating member 33f rotates together with the opening A of the container body 33Y, discharging efficiency of the toner from the opening A is improved.

Referring to FIGS. 19 and 30, an engaged section 33j (a flange section), which is engaged with a claw section 34j (for example, see FIG. 21) of the cap section 34Y to connect both members 33Y and 34Y with each other, is formed

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around the opening A of the container body 33Y to make one round around the outer circumference. As described above, the container body 33Y is rotatably fitted into the cap section 34Y.

Further, referring to FIGS. 19 and 30, a head section 33Yc of the container body 33Y (near the position at which the gear 33c is formed) is formed to have the inner diameter smaller than the inner diameter of a storage section 33Ya storing the toner (the position at which the helical protrusion 33b is formed). In the container body 33Y, a pumping section 33Yb (a section encircled by an alternate long and short dash line in FIG. 20) formed to protrude from the inner circumferential surface thereof toward the inside is disposed between the head section 33Yc and the storage section 33Ya. As the container body 33Y rotates, the toner conveyed toward the opening A by the helical protrusion 33b is pumped to the small diameter section of the head section 33Yc by the pumping section 33Yb. Thereafter, the toner pumped to the small diameter of the head section 33Yc is discharged toward the hollow space of the cap section 34Y from the opening A while being agitated by the agitating member 33f.

Referring to FIGS. 20 to 23, the ID chip 35 (the information storage device), the shutter member 34d, a shutter seal 36, and the like are installed in the cap section 34Y of the toner container 32Y.

Referring to FIG. 22, the cap section 34Y has a structure in which roughly a cylindrical section (a larger diameter cylindrical section 34Y1, a medium diameter cylindrical section 34Y2, and a small diameter cylindrical section 34Y3) in which the outer diameter and the inner diameter decreases from the container body 33Y side toward the shutter member 34d side in three stages is combined with a box section (a wide width box section 34Y11 and a narrow width box section 34Y12), disposed at the bottom, in which the width in the horizontal direction decreases in two stages are combined.

An insertion section 34z (for example, see FIG. 30) including the larger diameter cylindrical section 34Y1, the medium diameter cylindrical section 34Y2, the wide width box section 34Y11, and part of the narrow width box section 34Y12 is formed in the cap section 34Y. The head section 33Yc of the container body 33Y and part of the pumping section 33Yb are inserted into the insertion section 34z. Referring to FIG. 30, in the insertion section 34z, the medium diameter cylindrical section 34Y2 is formed to have the inner diameter D smaller than the tip diameter of the gear 33c and larger than the outer diameter of the opening A of the container body 33Y. Further, the small diameter cylindrical section 34Y3 is formed to have the inner diameter B smaller than the inner diameter D of the medium diameter cylindrical section 34Y2 and smaller than the outer diameter of the opening A.

An annular cap seal 37 (an elastic seal) in which the opening diameter becomes nearly the same as the inner diameter B is attached to an annular vertical wall surface (the surface facing the circumference of the opening A of the container body 33Y), which connects the medium diameter cylindrical section 34Y2 with the small diameter cylindrical section 34Y3, by a double-sided tape. The head section 33Yc and part of the pumping section 33Yb are inserted into the insertion section 34z such that an edge of the opening A of the head section 33Yc of the container body 33Y comes in contact with and bites into the cap seal 37. By the above described configuration, a functional section such as part of the gear 33c (a section excluding a section exposed from the notch hole 34x) and a connection section between the cap

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section 34Y and the container body 33Y are covered with the larger diameter cylindrical section 34Y1. For this reason, even when the toner container 32 is solely held by the user, the user can be prevented from touching the functional portion, and even if unexpected external force (for example, careless hitting) is applied to the toner container 32Y, toner leak from the connection section or damage of the tooth surface of the gear 33c can be alleviated. Further, since the annular cap seal 37 is excellent in sliding property and elasticity of the surface, even if the container body 33Y rotates while biting into the annular cap seal 37, there does not occur toner leak caused by a gap generated between the container body 33Y and the cap section 34Y. As a material of the cap seal 37, a high-density microcell urethane sheet having a structure that is high in density, fine, and uniform unlike general soft polyurethane foam (PUR) is used. As a result, compared to the case of using the general PUR, settling of the cap seal 37 is small, and the sealing property between the container body 33Y and the cap section 34Y can be maintained for a long time.

Referring to FIGS. 23 and 30, inside the narrow width box section 34Y12 positioned below the small diameter cylindrical section 34Y3 of the cap section 34Y, disposed is a toner falling path C having a hole of a hexagonal cylindrical shape for discharging the toner discharged from the opening A of the container body 33Y to the container outside downward in the vertical direction (falling by its own weight). The toner falling path C has a predetermined flow passage area of a hexagonal cross section and communicates the lower side circumferential surface inside the small diameter cylindrical section 34Y3 with the toner discharge opening W (discharge opening). The toner discharged to the inside of the small diameter cylindrical section 34Y3 of the cap section 34Y from the opening area A of the container body 33Y falls through the toner falling path C of the hexagonal cylindrical shape by its own weight and then is smoothly discharged from the toner discharge opening W to the container outside (the toner tank section 61Y).

On the bottom of the narrow width box section 34Y12, part of the shutter member 34d (a main shutter section 34d1) for performing opening/closing of the toner discharge opening W in conjunction with the attaching/detaching operation of the toner container 32Y to/from the toner container storage unit 70 is held to be slidably movable.

FIGS. 22 and 23 illustrate an operation in which the shutter member 34d starts opening of the toner discharge opening W and then completes opening. FIGS. 24(A) to 24(C) are schematic views illustrating the opening operation of the shutter member 34d (a shutter deforming section 34d2) at that time. Further, FIGS. 25 and 26 are perspective views illustrating the shutter member 34d. In FIGS. 24(B) and 24(C), the cap section 34Y, the cap receiving section 73, and the bottle receiving section 72 which are illustrated in FIG. 24(A) are partially omitted.

Referring to FIGS. 22 to 26, the shutter member 34d is formed of a resin material such as polystyrene and mainly includes a plate-like main shutter section 34d1 and a shutter deforming section 34d2 that protrudes the main shutter section 34d1, is thinner in thickness than the main shutter section 34d1, and has elasticity.

Referring to FIGS. 25 and 26, in the main shutter section 34d1 of the shutter member 34d, vertical wall 34d13 standing at both side end sections (vertical walls extending in parallel to the mounting direction of the toner container 32Y) and a shutter slider 34d12 having a plurality of protruding objects protruding from the vertical walls 34d13 are formed on both side end sections, respectively. The

shutter slider **34d12** includes a slide protruding section **34d12a** protruding from the inner side surface of the vertical wall **34d13**, an L-shaped engaged protruding section **34d12b** protruding from the outer side surface of the vertical wall **34d13**, and a pair of prismatic sections **34d12c** that is disposed to protrude from the same outer side surface as the engaged protruding section **34d12b** and extends from the body of the main shutter section **34d1** to the wide width box section **34Y11**. Meanwhile, in the narrow width box section **34Y12** of the cap section **34Y**, a pair of slide grooves **34t** (for example, see FIG. 23) extending in both side walls in the longitudinal direction is formed by a rib. The slide protruding section **34d12a** is fitted into the slide groove **34t**, and thus the main shutter section **34d1** of the shutter member **34d** is slide-movably supported on the cap section **34Y**.

Further, a shutter seal **36** adheres to the upper surface of the main shutter section **34d1** (the surface facing the toner discharge opening **W**) as a seal member. The shutter seal **36** is a thin rectangular parallelepiped-like elastic seal, and similarly to the cap seal **37**, a high density microcell urethane sheet is used as a material in view of sliding property and elasticity of the surface. For this reason, even if the opening/closing operation of the shutter member **34d** is repeated, a sealing characteristic in the toner discharge opening **W** can be maintained in the state in which the shutter member **34d** closes the toner discharge opening **W**.

The slide protruding section **34d12a** of the shutter slider **34d12** is fitted into the slide groove **34t** of the narrow width box section **34Y12** (the cap section **34Y**). Further, in this state, the shutter seal **36** is sandwiched between a protrusion **34r** (for example, see FIG. 23) of a hexagonal ring shape protruding downward along an edge of the hexagonal toner discharge opening **W** of the narrow width box section **34Y12** and a main shutter section **34d1**, and the shutter seal **36** becomes a slightly compressed state. In this state, the shutter member **34d** moves along the slide groove **34t**, and thus the main shutter section **34d1** opens or closes the toner discharge opening **W** while suppressing toner leak. Further, in the state in which the main shutter section **34d1** (the shutter member **34d**) has closed the toner discharge opening **W**, the toner leak from between the main shutter section **34d1** and the toner discharge opening **W** is prevented.

Specifically, the shutter member **34d** relatively moves in the longitudinal direction from the cap section **34Y** side to the container body **33Y** side (moves to the left in FIG. 30) to open the toner discharge opening **W** and relatively moves in the longitudinal direction from the container body **33Y** side to the cap section **34Y** side (moves to the right in FIG. 30) to close the toner discharge opening **W**. The opening/closing operation of the shutter member **34d** (the opening/closing operation of the toner discharge opening **W**) is performed in conjunction with the attaching/detaching operation of the toner container **32Y** to/from the toner container storage unit **70** (the apparatus body **100**) in the longitudinal direction.

Referring to FIGS. 25 and 26, the shutter deforming section **34d2** of the shutter member **34d** is formed integrally with the main shutter section **34d1** and formed at the board thickness thinner than the board thickness of the main shutter section **34d1** as described above. The shutter deforming section **34d2** mainly includes two spindly flat plate sections **34d23** extending from the end surface of the main shutter section **34d1** at the container body **33Y** and a plate-like member **34d24** extending in a direction orthogonal to the longitudinal direction to connect the two flat plate sections **34d23** with each other near the leading end sections (the free ends). The shutter deforming section **34d2** is

formed to be elastically deformed in the vertical direction from a fixed end (a connection portion) with the main shutter section **34d1** as a reference point. On the leading end sections (the free ends) of the two flat plate sections **34d23**, formed are stopper sections **34d22** for fixing the shutter member **34d** so as to prevent careless opening of the toner discharge opening **W** as will be described later. On the bottom side of the plate-like member **34d24**, formed is the stopper release section **34d21** that is an inclined protrusion (having a triangular cross section) protruding, in a mountain shape, downward in the vertical direction and that releases fixing of the shutter member **34d** in cooperation with the stopper release urging section **72b** of the cap receiving section **73** as will be described later.

Referring to FIGS. 22 and 23, in the wide width box section **34Y11** positioned below the larger diameter cylindrical section **34Y1** of the cap section **34Y**, formed is a shutter storage section **34n** that stores the shutter deforming section **34d2** therein at the time of shutter opening. Among the four side surfaces of the wide width box section **34Y11**, the two side surfaces facing in the longitudinal direction (a direction of an arrow in FIG. 22) are opened. Particularly, in the side surface formed at the toner discharge opening **W** side, part of the wall surface is formed at both side ends of the bottom side, but most of it functions as an opening **34n1** extending in the horizontal direction. The opening **34n1** is formed such that two surfaces including the side surface and the bottom surface that are at the toner discharge opening **W** side of the wide width box section **34Y11** are cut out. Among edge sections of the opening **34n1**, an edge section formed to stand from the bottom surface of the wide width box section **34Y11** in the vertical direction becomes the contact section **34n5**.

The stopper section **34d22** of the shutter deforming section **34d2** is a wall section formed at a farthest end section of the shutter deforming section **34d2** (the leading end of the shutter deforming section **34d2** away from the main shutter section **34d1**) in the opening direction (the left direction in FIG. 24). The stopper section **34d22** of the shutter deforming section **34d2** comes in contact with the contact section **34n5** and thus restricts the shutter member **34d** from moving in the direction of opening the toner discharge opening **W** from the state in which the toner discharge opening **W** is closed. That is, when the toner container **32Y** is in a separated state (the state in which it is not set to the apparatus body **100** yet), since the stopper section **34d22** of the shutter member **34d** comes in contact with the contact section **34n5**, a phenomenon that the shutter member **34d** moves in the opening direction on its own and so opens the toner discharge opening **W** does not happen.

Meanwhile, the stopper release section **34d21** comes in contact with the stopper release urging section **72b** (for example, see FIGS. 5 and 24) formed in the bottle receiving section **72** in conjunction with the mounting operation of the toner container **32Y** to the toner container storage unit **70** and is pushed upward by the stopper release urging section **72b** (as external force is applied upward). Then, the shutter deforming section **34d2** is elastically deformed upward, and the stopper section **34d22** is also displaced upward. As a result, the contact state between the stopper section **34d22** and the contact section **34n5** is released, so that the shutter member **34d** can move in the opening direction.

An operation of the shutter member **34d** in conjunction with the mounting operation of the toner container **32Y** to the toner container storage unit **70** will be described below in detail with reference to FIGS. 24(A) to 24(C). The position of the shutter member **34d** in FIGS. 24(A) and

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24(C) corresponds to the position of the shutter member 34d in FIGS. 22 and 23, respectively.

As illustrated in FIG. 24, the mounting operation of the toner container 32Y to the toner container storage unit 70 (movement in the left direction in FIG. 24) starts, and when the stopper release section 34d21 of the shutter member 34d does not reach the position of the stopper release urging section 72b (for example, also see FIG. 5) formed in the bottle receiving section 72, the stopper section 34d22 of the shutter member 34d comes in contact with the contact section 34n5, and thus movement of the shutter member 34d in the opening direction is restricted. Further, on the upper surface of the bottle receiving section 72 at the cap receiving section 73 side near the stopper release urging section 72b, a bristle brush 72f is disposed to rub the bottom surface of the shutter member 34d and clean contamination. Particularly, the bristle brush 72f is effective in cleaning flying toner sticking to the bottom surface of the shutter member 34d during an operation of attaching/detaching the toner container 32Y.

Thereafter, when the mounting operation of the toner container 32Y proceeds, as illustrated in FIG. 24(B), the stopper release section 34d21 is pushed up by the stopper release urging section 72b, and so the shutter deforming section 34d2 is elastically deformed from the connection position (a section encircled by an alternate long and short dash line) as the reference point. As a result, the contact state between the stopper section 34d22 and the contact section 34n5 is released, and so the shutter member 34d can relatively move in the opening direction.

Thereafter, the shutter member 34d comes in contact with the wall section (a section indicated as "contact position" in the drawings) formed around the toner feeding opening 73w of the cap receiving section 73, and so movement in the toner container storage unit 70 (the cap receiving section 73) is restricted (the shutter member 34d does not absolutely move in the longitudinal direction). However, since movement in the mounting direction of the toner container 32Y is performed, the shutter member 34d moves relative to the toner discharge opening W in the opening direction. That is, as illustrated in FIG. 24(C), the shutter member 34d relatively moves to the container body 33Y side, and so the shutter deforming section 34d2 is stored in the shutter storage section 34n. As a result, opening of the toner discharge opening W is completely finished by movement of the shutter member 34d in the opening direction. Further, the toner discharge opening W matches with the toner feeding opening 73w of the cap receiving section 73 in a superimposed manner, and an integrated toner feeding passage leading from the toner container 32Y to the toner feeding device is formed. At this time, the stopper release section 34d21 of the shutter member 34d is stored in a notch 34n6 (for example, see FIGS. 22 and 23) that is an extension section of the opening 34n1 of the cap storage section 34n. Thus, it is possible to prevent a problem in that the shutter deforming section 34d2 stored in the shutter storage section 34n is held in a greatly elastically deformed state by contact between the stopper release section 34d21 and the shutter storage section 34n.

As described above, in the toner container 32Y according to the present first embodiment, the shutter deforming section 34d2 that is elastically deformed from the connection position with the main shutter section 34d1 as the reference point is disposed in the shutter member 34d, and the stopper section 34d22 that restricts movement of the shutter member 34d in the opening direction and the stopper release section 34d21 that releases the stopper section 34d22 are disposed

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in the shutter deforming section 34d2. Thus, the shutter member 34d does not open the toner discharge opening W as it pleases in the state in which the toner container 32Y is removed, and only when the toner container 32Y is set to the image forming apparatus body 100, the shutter member 34d opens the toner discharge opening W in conjunction with the mounting operation thereof.

An operation of the L-shaped engaged protruding section 34d12b related to the shutter opening/closing operation will be described below.

As illustrated in FIG. 25, the L-shaped engaged protruding sections 34d12b are formed on both side end sections of the main shutter section 34d1, respectively. Meanwhile, even though not shown because it is illustrated in the drawings of Japanese Patent Application No. 2011-9782, on the bottom surface inside the cap receiving section 73, a pair of urging members is installed to face a pair of engaged protruding sections 34d12b. The pair of urging members is an L-shaped lever (a spindle that becomes a rotation center is formed near an L-shaped curved portion) that forms a shape symmetrical to each other, and an arm section at one side thereof is urged by a torsion coil spring. If the toner container 32Y (the cap section 34Y) is mounted on the cap receiving section 73, an arm section of the other end side of the urging member is engaged with the engaged protruding section 34d12b and applies urging force in a direction resisting the direction in which the shutter member 34d is opened. The user pushes the toner container 32Y by force overcoming urging force by the above described urging member, and opening of the shutter member 34d is rapidly performed. As a result, a state in which the toner discharge opening W does not match with the toner feeding opening 73w of the cap receiving section 73 occurs only instantaneously, and toner leak from between the toner discharge opening W and the toner feeding opening 73w can be suppressed.

On the other hand, when the toner container 32Y (the cap section 34Y) is removed from the cap receiving section 73, the arm section of the urging member becomes a state urged to push the engaged protruding section 34d12b in the mounting direction. The user attempts to pull out the toner container 32Y by force overcoming urging force by the urging member, and so closing of the shutter member 34d is rapidly performed. As a result, similarly to the time of the opening operation, toner leak from between the toner discharge opening W and the toner feeding opening 73w can be suppressed.

Referring to FIGS. 20, 21, and 30, in the upper section (the ceiling portion) of the cap section 34Y, the first positioning hole 34a extending in the longitudinal direction from the end surface of the cap section 34Y orthogonal to the longitudinal direction is formed. The first positioning hole 34a becomes a main positioning reference of the cap section 34Y in the image forming apparatus body 100. Specifically, the main reference pin 73a (for example, see FIGS. 14A, 14B and 17) as the positioning pin of the cap receiving section 73 is fitted into the first positioning hole 34a of the cap section 34Y in conjunction with the mounting operation of the toner container 32Y to the toner container storage unit 70 in the longitudinal direction.

In the lower section (the bottom portion) of the cap section 34Y, the second positioning hole 34b extending in the longitudinal direction from the end surface of the cap section 34Y orthogonal to the longitudinal direction is formed not to reach the position of the toner discharge opening W. The second positioning hole 34b becomes a sub positioning reference of the cap section 34Y in the image

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forming apparatus body 100. Specifically, the sub reference pin 73b (for example, see FIGS. 14A, 14B and 17) as the positioning pin of the cap receiving section 73 is fitted into the second positioning hole 34b of the cap section 34Y in conjunction with the mounting operation of the toner container 32Y to the toner container storage unit 70 in the longitudinal direction. Further, the second positioning hole 34b as illustrated in FIG. 21 is a oblong hole in which a vertical direction is a longitudinal direction (this “longitudinal direction” has a different meaning from the “longitudinal direction” of the toner container 32Y described in the other sections).

Positioning of the cap section 34Y in the toner container storage unit 70 is performed by the two positioning holes 34a and 34b having the above described configuration.

Referring to FIG. 30, the hole depth of the first positioning hole 34a is set to be larger than the hole depth of the second positioning hole 34b. The length of the main reference pin 73a in the longitudinal direction is set to be larger than the length of the sub reference pin 73b in the longitudinal direction. In the mounting operation of the toner container 32Y to the toner container storage unit 70 (the cap receiving section 73) in the longitudinal direction, the main reference pin 73a starts to be fitted into the first positioning hole 34a that is the main positioning reference, and then the sub reference pin 73b starts to be fitted into the second positioning hole 34b that is the sub positioning reference, so that the toner container 32Y can be smoothly mounted on the toner container storage unit 70 (the cap receiving section 73). Further, since the first positioning hole 34a that is long in the longitudinal direction is disposed in the ceiling section of the cap section 34Y (a section that is not buried in the toner), there is no influence on a conveying property (a flow property) of the toner inside the cap section 34Y. The second positioning hole 34b that is short in the longitudinal direction is formed on the bottom of the cap section 34Y but sufficiently performs a function as the sub positioning reference since it can be disposed using a small space from the end surface of the cap section 34Y to the position of the toner discharge opening W.

Referring to FIG. 20, on the outer circumferential surface of the cap section 34Y above the first positioning hole 34a of the cap section 34Y, a guide rail section 34e extending in an axial direction of the first positioning hole 34a is formed. The guide rail section 34e protrudes upward in the vertical direction from the outer circumferential surface of the cap section 34Y to be line-symmetrical to a virtual vertical line passing through the hole center of the first positioning hole 34a when viewed in a cross section orthogonal to the longitudinal direction (a cross section parallel to a front view of FIG. 21) and extends in the longitudinal direction (a direction vertical to a paper plane of FIG. 21). Before the main reference pin 73a is inserted into the positioning hole 34a, the guide rail section 34e is fitted into the contacted groove 73m (a concave section), which is formed in a groove shape in the inner wall of the cap receiving section 73 above the main reference pin 73a, from the upstream side in the mounting direction further than the leading end section of the main reference pin 73a and restricts a posture of the cap section 34Y in the horizontal direction orthogonal to the longitudinal direction at the time of mounting movement to the image forming apparatus body 100 (the cap receiving section 73). Further, in the leading end of the guide rail section 34e, a protruding section 34e1 slightly protruding in the longitudinal direction from the end surface of the first positioning hole 34a is formed. The protruding section 34e1 is formed in a tapered shape as illustrated in FIG. 20. The

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guide rail section 34e enters the contacted grooves 73m formed on the cap receiving section 73, and so the cap section 34Y is guided to the inside of the cap receiving section 73. Thus, when the cap section 34Y is mounted on the cap receiving section 73, in the first positioning hole 34a, the tapered protruding section 34e1 is fitted into the contacted groove 73m before the first positioning hole 34a is fitted into the main reference pin 73a, and thus the cap section 34Y is smoothly mounted on the cap receiving section 73.

Referring to FIGS. 20 and 21, lateral protrusions 34c for restricting a posture of the cap section 34Y in the rotation direction in the image forming apparatus body 100 (the cap receiving section 73) are formed on both side sections of the cap section 34Y, respectively. The lateral protrusion 34c protrudes to both sides in the horizontal direction from the outer circumferential surface of the cap section 34Y to be disposed on a virtual horizontal line passing through the center of a virtual line segment connecting the hole center of the first positioning hole 34a with the hole center of the second positioning hole 34b when viewed in a cross section orthogonal to the longitudinal direction and extends in the longitudinal direction (a direction vertical to a paper plane of FIG. 21). The two lateral protrusions 34c are engaged with the lateral groove 73h and the groove section 73r1, for example, illustrated in FIG. 12 while being pressed to be pushed back in a direction opposite to pushing by the cap section sandwiching member 73r (for example, see FIG. 12) in the cap receiving section 73. Thus, the cap section 34Y is attached to or detached from the cap receiving section 73 while the posture of the cap section 34Y in the rotation direction is being restricted, and the posture of the cap section 34Y in the rotation direction is restricted in the state in which the cap section 34Y is mounted on the cap receiving section 73.

In further detail, in the lateral protrusion 34c, the leading end in the longitudinal direction (the mounting direction) is formed in a tapered shape as illustrated in FIG. 20. Here, when the cap section 34Y is mounted on the cap receiving section 73, the guide rail section 34e is first fitted into the contacted groove 73m, and then the two lateral protrusions 34c having the tapered leading end are fitted into the lateral grooves 73h and the groove sections 73r1. Thus, the cap section 34Y is smoothly mounted on the cap receiving section 73 in the state in which the posture of the cap section 34Y is restricted with a high degree of certainty.

Referring to FIGS. 20 and 21, on both ends of the bottom of the cap section 34Y, disposed are convex sections 34g and 34h for securing incompatibility of the toner container 32Y (the developer container). In detail, on the upper surface side of a flat plate-like blade member extending laterally from the bottom of the cap section 34Y, the convex sections 34g and 34h are disposed to protrude upward. The convex sections 34g and 34h are configured to be fitted into a fitting section 72m (that is formed in the bottle receiving section 72 of the toner container storage unit 70) illustrated in FIG. 9 when the mounting operation of the toner container 32Y to the toner container storage unit 70 is correct (when the toner container 32Y is mounted to the true position of the toner container storage unit 70).

Specifically, even though not shown, the convex sections 34g and 34h are disposed at the different positions depending on color of the toner stored in the toner container (the container body). In detail, if it is assumed that the leading end in the mounting direction when the toner container 32Y is mounted on the image forming apparatus body 100 is the front, the convex sections 34g and 34h are disposed so that

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the protruding positions are not superimposed when viewed from the front and are disposed at the different positions depending on the color. The convex sections **34g** and **34h** of the toner container corresponding to cyan are formed at the position to be fitted only into the cyan fitting section **72m** of the toner container storage unit **70**, the convex sections **34g** and **34h** of the toner container corresponding to magenta are formed at the position to be fitted only into the magenta fitting section **72m** of the toner container storage unit **70**, the convex sections **34g** and **34h** of the toner container corresponding to yellow are formed at the position to be fitted only into the yellow fitting section **72m** of the toner container storage unit **70**, and the convex sections **34g** and **34h** of the toner container corresponding to black are formed at the position to be fitted only into the black fitting section **72m** of the toner container storage unit **70**.

The above described configuration prevents a problem in that the toner container of a different color (for example, the toner container of yellow) is set to the toner container storage unit of a predetermined color (for example, the toner container storage unit of cyan) and thus a predetermined color image cannot be formed. That is, the toner container is prevented from being erroneously set to the toner container storage unit.

FIG. **10** illustrates the state when the mounting operation of the toner container **32Y** to the toner container storage unit **70** is correct. The convex sections **34g** and **34h** of the cap section **34Y** do not interfere with the fitting section **72m** of the bottle receiving section **72**. On the other hand, FIG. **11** illustrates the state when the mounting operation of the toner container **32Y** to the toner container storage unit **70** is not correct. The convex sections **34g** and **34h** of the cap section **34Y** interfere with the fitting section **72m** of the bottle receiving section **72**.

The ID chip **35** (the information storage device) that is characteristic in the toner container **32Y** (the removable device) according to the present first embodiment will be described below in detail.

Referring to FIG. **20**, on the end surface of the cap section **34Y**, the ID chip **35** as the information storage device in which various electronic information is stored is installed at the position of a holding member **34k** installed between the first positioning hole **34a** and the second positioning hole **34b**. The ID chip **35** is configured to be connected to the connector **73e** of the cap receiving section **73** in the state in which the cap section **34Y** is mounted to the toner container storage unit **70** (the cap receiving section **73**) (for example, see FIGS. **3** and **17**). Specifically, in the state in which the cap section **34Y** is mounted on the toner container storage unit **70** (the cap receiving section **73**), a plurality of metallic pads **35a** (metallic plates) of the ID chip **35** come in contact with a plurality of body side terminals **73e2** of the connector **73e**. The ID chip **35** performs communication (wire line communication) with the control unit **90** through the connector **73e** as illustrated in FIG. **3** in the state in which the cap section **34Y** is held on the cap receiving section **73**.

Referring to FIGS. **27** to **29**, in the present first embodiment, a holding mechanism installed in the toner container **32Y** (the removable device) removably installed to the image forming apparatus body **100** includes the ID chip **35** as the information storage device, the holding member **34k** as the holding section, and the like. The ID chip **35** as the information storage device held on the holding mechanism includes a substrate **35b**, an information storage unit **35c**, metallic pads **35a** as a plurality of terminals (metallic plates), and the like.

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Referring to FIG. **29**, the information storage unit **35c** is an electronic circuit in which various information exchanged between the control unit **90** of the image forming apparatus body **100** and the toner container **32Y** is stored. In FIG. **29**, the information storage unit **35c** is illustrated as a box-like one by hatched lines for simplicity but corresponds to an assembly of a memory IC, a condenser for noise reduction, a resistor, and the like. The information storage unit **35c** is disposed on the back surface side of the substrate **35b** (the side facing the end surface of the cap section **34Y**) and electrically connected to all or some of the metallic pads **35a** as a plurality of metallic plates.

The metallic pads **35a** as the plurality of terminals come in contact with the plurality of body side terminals **73e2** of the connector **73e** installed in the cap receiving section **73** (the apparatus body **100**), respectively, and exchange an electrical signal related to information with the image forming apparatus body **100** (the control unit **90**). The plurality of metallic pads **35a** are disposed at the front surface side of the substrate **35b** (the side facing the cap receiving section **73**). Further, the plurality of metallic pads **35a** are formed in a nearly rectangular shape and arrayed in a transverse direction thereof with a clearance therebetween (a Z direction (vertical direction) in FIG. **29(A)**).

On the substrate **35b** on which the information storage unit **35c** and the metallic pads **35a** are disposed, positioning notches **35b1** (that has a shape of a half when an elliptical circumference is divided into two by a straight line) are formed on both ends in the vertical direction, respectively. The positioning notch **35b1** is fitted into the positioning pin **73e3** (for example, see FIGS. **16** and **17**) as a cylindrical positioning protruding section installed in the connector **73e** (the image forming apparatus body **100**) and is used for positioning the plurality of metallic pads **35a** on the plurality of body side terminals **73e2**.

The ID chip **35** (the information storage device) having the above described configuration is held on the holding member **34k** (the holding section) that is configured removably from the cap section **34Y**.

The holding member **34k** (the holding section) holds the contact type ID chip **35** (the information storage device) to be able to move on a virtual plane (a virtual plane substantially orthogonal to) intersecting with the movement direction (a direction of an arrow in FIG. **17**) in which the metallic pads **35a** (terminals) approach and come in contact with the body side terminals **73e2**.

Specifically, in the present first embodiment, the holding member **34k** holds the ID chip **35** (the substrate **35b**) to be able to move on a virtual plane (an XZ plane in FIG. **20**) orthogonal to the attaching/detaching direction of the toner container **32Y** to/from the image forming apparatus body **100**. That is, the ID chip **35** (the substrate **35b**) is configured to be able to move (about 1 mm) on the XZ plane in FIG. **20** freely to some extent in a state held by the holding member **34k** (the cap section **34Y**) as illustrated in FIG. **20**. Specifically, the ID chip **35** (the substrate **35b**) is held loosely to some extent inside the box-like holding member **34k** (the holding mechanism). That is, the ID chip **35** is held with a predetermined gap between the ID chip **35** and the side wall inside the holding member **34k**. Referring to FIGS. **28** and **29**, the ID chip **35** is held such that a small clearance Qt (for example, " $QtQt$ " is about 0.85 to 1.05 mm) is formed in a DY direction on the thickness t (about 0.8 mm) of the substrate **35b** inside the holding member **34k**. For this reason, it is possible to make the substrate **35b** stand to orthogonally cross in the insertion direction of the positioning pin **73e3** to some extent. Thus, prevented is a problem

in that the substrate **35b** is excessively laid down in the insertion direction of the positioning pin **73e3**, the positioning pin **73e3** is seized by the notch **35b1**, and thus the positioning pin **73e3** fails to be fitted into the notch **35b1**.

Through the above described configuration, even when the size of the image forming apparatus body **100** or the toner container **32Y** decreases and so the plurality of metallic pads **35a** (terminals) on the substrate **35b** are densely arranged so as to reduce the size of the ID chip **35** installed thereon, a contact failure that is caused by a positioning failure between the plurality of metallic pads **35a** and the body side terminal **73e2** of the connector **73e** is difficult to occur regardless of whether or not a degree of dimension accuracy or a degree of assembly accuracy of an associated component is high or low.

Specifically, referring to FIG. 17, if the mounting operation of the cap section **34Y** of the toner container **32Y** on the cap receiving section **73** proceeds, the positioning pins **73a** and **73b** are inserted into the positioning holes **34a** and **34b**, and so the cap section **34Y** is positioned in the cap receiving section **73**. If the mounting operation of the cap section **34Y** further proceeds, the positioning pin **73e3** of the connector **73e** (for example, see FIGS. 16 and 17) is inserted into the notch **35b1** of the substrate **35b** of the ID chip **35**, and the substrate **35b** (the plurality of metallic pads **35a**) is positioned with respect to the connector **73e** (the plurality of body side terminal **73e2**). In further detail, the positioning pin **73e3** comes in contact with the edge section (or the inner surface) of the notch **35b1**, and so movement of the substrate **35b** is restricted. At this time, since the ID chip **35** (the substrate **35b**) having the notch **35b1** formed therein is configured to be movable on the XZ plane in the holding member **34k**, an allowable tolerance of a degree of dimension accuracy or a degree of assembly accuracy of an associated component according to mass production has to be large. Thus, when positioning between the toner container **32Y** and the image forming apparatus body **100** is performed, even if the position of the notch **35b1** is misaligned with the positioning pin **73e3** from the beginning, the ID chip **35** (the substrate **35b**) is restricted by the tapered leading end of the positioning pin **73e3** and moves on the XZ plane, and thus the cylindrical section of the positioning pin **73e3** can be fitted into the notch **35b1**. That is, independently of positioning between the toner container **32Y** and the image forming apparatus body **100**, the cylindrical section of the positioning pin **73e3** can be fitted into the notch **35b1**. Thus, the contact failure that is caused by the positioning failure between the plurality of metallic pads **35a** and the body side terminals **73e2** of the connector **73e** is difficult to occur.

Referring to FIG. 27, the holding member **34k** as the holding section is removably configured on the cap section **34Y** and is a box-like member having an insertion opening **34k1**, formed in the upper side thereof, through which the ID chip **35** is inserted or separated.

In detail, when assembling the holding mechanism to the cap section **34Y**, the ID chip **35** (the information storage device) is first inserted into the holding member **34k** through the insertion opening **34k1** (movement in a direction of an arrow in FIG. 27). Thereafter, the holding member **34k** (the holding section) having the ID chip **35** mounted thereon moves in the arrow direction in FIG. 27 and is press-fitted into the convex section of the cap section **34Y**. At this time, the holding member **34k** is fixed and held at the position contacting a pedestal section **34q** (disposed at the position not contacting the substrate **35b**) disposed in the convex section of the cap section **34Y**. Further, when ejecting the ID

chip **35** from the cap section **34Y**, an operation is performed in a procedure reverse to the above described procedure. The pedestal section **34q** is a rib standing in the mounting direction of the toner container **32Y** (or toward the holding member **34k**) inside the concave section of the cap section **34Y** and disposed at the position other than a place into which the positioning pin **73e3** of the connector **73e** is to be inserted.

In the present first embodiment, the holding member **34k** is press-fitted into and fixed to the concave section of the cap section **34Y**, but the holding member **34k** may be mounted on and screw-fixed to the concave section of the cap section **34Y**. Specifically, a flaky rib having a hole in the side wall of the holding member **34k** protrudes to form a female screw section in the end surface of the cap section **34Y**. The holding member **34k** is mounted on the concave section of the cap section **34Y**, and in the state in which the flaky rib of the holding member **34k** comes in contact with the end surface of the cap section **34Y**, a screw is screwed into the female screw section of the cap section **34Y** through the hole of the rib related to the holding member **34k**. Even in this configuration, the holding member **34k** can be comparatively easily attached to or detached from the cap section **34Y**.

As described above, since the ID chip **35** (the substrate **35b**) is configured to be attached to or detached from the cap section **34Y**, efficiency of assembling the ID chip **35** (the substrate **35b**) to the toner container **32Y** as the removable device increases, and at the same time, efficiency of a disassembly operation of the ID chip **35** (the substrate **35b**) when recycling the toner container **32Y** increases. Particularly, in the present first embodiment, the substrate **35b** of the ID chip **35** is a small substrate having the size of about 12 mm □ 8 mm to 15 mm □ 10 mm. If data input/output (data input/output when a probe terminal comes in contact with the metallic pad **35a**) is about to be performed in the state in which the substrate **35b** is mounted on the cap section **34Y** in the manufacturing process, it may be difficult to do a work on the cap section **34Y** of a complicated shape, and a process time may increase. Thus, in the present first embodiment, the ID chip **35** (the substrate **35b**) is removably configured, and thus there is a great effect since a data input/output operation can be performed on the ID chip **35** alone (or for each holding member **34k**) if necessary.

Referring to FIGS. 17 and 27, in the cap section **34Y**, formed is a standing member **34f** that blocks the insertion opening **34k1** in the state in which the holding member **34k** is mounted on the concave section.

Thus, prevented is a problem in that the ID chip **35** falls off from the insertion opening **34k1** of the holding member **34k** after the ID chip **35** (the holding member **34k**) is mounted on the cap section **34Y**.

Referring to FIG. 28, inside the holding member **34k** (the box-like member), formed are a first facing section **34k4** and a second facing section **34k5**. The first facing section **34k4** faces a first plane of the substrate **35b** (referring to FIG. 29, a surface where the plurality of metallic pads **35a** are disposed) and is formed to come in contact only with and slide on an outer peripheral area of the first plane where the metallic pads **35a** are not disposed so as not to interfere with the plurality of metallic pads **35a**. The second facing section **34k5** faces a second plane of the substrate **35b** (referring to FIG. 29, a surface where the information storage unit **35c** is disposed) and is formed to slide on part of the second plane so as not to interfere with the information storage unit **35c**. Thus, inside the holding member **34k**, the ID chip **35** can freely move on the XZ plane (can move to slide on the facing

sections 34k4 and 34k5) without falling off from the holding member 34k, and without the metallic pads 35a or the information storage unit 35c getting damaged.

Further, openings 34k2 and 34k3 are formed in the front and back surfaces of the holding member 34k, respectively. The first opening 34k2 is formed to allow the plurality of metallic pads 35a and the positioning notch 35b1 to be exposed at the side facing the connector 73e even when the substrate 35b moves in the XZ plane to some extent. Thus, in accompany with movement of the substrate 35b in the XZ plane, the positioning pin 73e3 and the notch 35b1 can be engaged with each other, and the metallic pads 35a and the body side terminals 73e2 can be connected with (come in contact with) each other. Further, the second opening 34k3 to allow the information storage unit 35c to be exposed at the side facing the concave section of the cap section 34Y even when the substrate 35b moves on the XZ plane to some extent.

Referring to FIG. 28, the opening 34k2 formed in the front surface of the holding member 34k is formed such that the left side has a convex shape and the right side has a concave shape. Thus, an area encircled by a dotted line in FIG. 28 functions as a hook (a stopper) for preventing the ID chip 35 from falling off from the opening 34k2.

FIG. 29 is a three-plane view illustrating the ID chip 35.

As illustrated in FIG. 29, in the ID chip 35, the metallic pads 35a as four metallic plates are disposed in line in the Z direction on the first plane of the substrate 35b. The metallic pad 35a has a multi-layer structure having three layers of a copper layer, a nickel layer, and a metallic layer which are disposed in order from the substrate 35b side, and the metallic layer as the surface layer is comparatively expensive but disposed to prevent oxidation. The metallic pad 35a is formed by electric field evaporation on the substrate 35b that is masked in advance.

The positioning notches 35b1 are formed on both end sections of the four metallic pads 35a in the arrangement direction (the Z direction) to sandwich the four metallic pads 35a. In the present first embodiment, a first virtual straight line S1 that passes through the centers of the notches 35b1 and is parallel to the arrangement direction of the plurality of metallic pads 35a is configured to be positioned not to overlap a second virtual straight line S2 that connects the centers of the plurality of metallic pads 35a in the longitudinal direction. Specifically, the first virtual straight line S1 that connects the two positioning notches 35b1 (a virtual line that connects the centers of original oblong holes in the notches 35b1 each of which has a shape of a half when the oblong hole is divided into two in the longitudinal direction) is configured to be positioned not to overlap the second virtual straight line S2 that connects the centers of the plurality of metallic pads 35a in the longitudinal direction. That is, in the notches 35b1, the virtual straight line S1 that connects sections that are most bitten toward the inside of the substrate 35b is positioned not to overlap the virtual straight line S2. Further, the virtual straight line S1 is configured to be nearly parallel to the virtual straight line S2.

In the present first embodiment, dimensions a to f in FIG. 29 have been set to 6.2 mm, 5.2 mm, 1.5 mm, 2 mm, 6 mm, and 11.7 mm, respectively. The substrate 35b having the small area size is small in deflection amount even if external force is applied and comparatively large in resistance characteristic (stiffness) against shearing force. In the present first embodiment, the ID chip 35 is held to be movable inside the holding member 34k, and employed is a positioning method in which the positioning pin 73e3 is inserted into the notch 35b1 and is likely to be "seized" (a state in which the

positioning pin 73e3 enters the notch 35b1 obliquely rather than vertically, a sliding load between the notch 35b1 and the positioning pin 73e3 increases, and so the substrate 35b is deflected and does not move) in the case of the ID chip 35 having the large area size. However, since the substrate 35b has the small area size, stiffness increases, and so the positioning method that is difficult to cause deflection that causes "seizure" is implemented. Further, an interval between the metallic pads 35a in the substrate 35b is narrow, but the contact failure between the metallic pads 35a and the body side terminals 73e2 can be prevented by high accuracy positioning in accompany with movement of the substrate 35b in the XZ plane, and thus the area size of the expensive metallic pads 35a having the metal layer can be suppressed to a minimal area size.

FIG. 16 is a schematic perspective view illustrating the connector 73e at the apparatus body 100 side.

Referring to FIG. 16, the four body side terminals 73e2 in the connector 73e are plat-like (linear) metallic members, have one end as the fixed end and the other end as the free end, and are fixed to and supported on the connector body 73e1. On the other end sides of the four body side terminals 73e2, formed are curved sections that curve toward the ID chip 35 (toner container 32Y) side. That is, the body side terminals 73e2 curve toward the ID chip 35 like a knee (or a boomerang). The curved sections of the body side terminals 73e2 are sections that functions as contact points with the metallic pads 35a.

In accompany with the mounting operation of the cap section 34Y (the toner container 32Y) on the cap receiving section 73, the curved section of the body side terminal 73e2 comes in contact with nearly the central section of the metallic pad 35a in the longitudinal direction. Then, when the mounting operation of the cap section 34Y further proceeds, the ID chip 35 (the substrate 35b) approaches to the connector 73e side, the body side terminal 73e2 is deformed so that the curved section of the body side terminal 73e2 get close to the first virtual straight line S1 while being pressed by the metallic pad 35a and elastically deformed (a state in which a curved knee stretches). That is, in accompany with the mounting operation of the cap section 34Y, the curved section of the body side terminal 73e2 slides toward the left in FIG. 29 from the central section in the longitudinal direction (approaches the first virtual straight line S1) while gradually increasing contact pressure on the metallic pad 35a.

Through the above described configuration, even if the position of the cap section 34Y (the metallic pad 35a) in the longitudinal direction (the Y direction) is misaligned with the cap receiving section 73 (the body side terminal 73e2) depending on whether or not a degree of dimensions accuracy or a degree of assembly accuracy of an associated component is high or low (a dimension variation), the contact failure between the body side terminal 73e2 and the metallic pad 35a can be prevented with a high degree. Finally, since the body side terminal 73e2 and the metallic pad 35a come in contact with each other near the position where the positioning pin 73e3 is engaged with the notch 35b1 (near the first virtual straight line S1), the distance between the positioning section and the connect section can be reduced. As a result, a degree of accuracy of the contact position between the body side terminal 73e2 and the metallic pad 35a increases.

Further, in the present first embodiment, the plurality of metallic pads 35a are disposed in line in the arrangement direction that is the vertical direction.

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As a result, the positioning direction of the cap section 34Y in the cap receiving section 73 (the arrangement direction of the positioning pins 73a and 73b and the positioning holes 34a and 34b) is the same as the positioning direction of the connector 73e and the substrate 35b (the arrangement direction of the positioning pin 73e3 and the notch 35b1), and thus the contact failure between the body side terminal 73e2 and the metallic pad 35a is difficult to occur.

Referring to FIG. 21, the positioning holes 34a and 34b of the cap section 34Y are formed at positions apart from each other in the vertical direction to sandwich the ID chip 35 (the information storage device). A third virtual straight line S3 that connects the centers of the two positioning holes 34a and 34b is configured to be parallel to the first virtual straight line S1 that connects the two positioning notches 35b1.

As described above, since the ID chip 35 is disposed be fixed between the first positioning hole 34a (the main reference) and the second positioning hole 34b (the sub reference), the position of the ID chip 35 with respect to the connector 73e of the cap receiving section 73 is decided with a high degree of accuracy. Thus, the communication failure caused by position misalignment between the connector 73e and the ID chip 35 can be suppressed. Particularly, since the positioning direction of the cap section 34Y in the cap receiving section 73 (the arrangement direction of the positioning pins 73a and 73b and the positioning holes 34a and 34b) is the same as the positioning direction of the connector 73e and the substrate 35b (the arrangement direction of the positioning pin 73e3 and the notch 35b1), the positioning operation of the toner container 32Y on the image forming apparatus body 100 contributes to making it easier to position of the substrate 35b on the connector 73e. As a result, there is an effect in that the contact failure between the body side terminal 73e2 and the metallic pad 35a is difficult to occur.

Further, a procedure in which components of the bottle receiving section 72 and the cap receiving section 73 are concerned with the cap section 34Y when the mounting operation of the toner container 32Y on the toner container storage unit 70 proceeds is as follows.

First, the cap section 34Y slides on the bottle receiving surface 72a, and thereafter, the guide rail section 34e of the cap section 34Y is fitted into the engaged groove 73m of the cap receiving section 73, the lateral protrusion 34c of the cap section 34Y is fitted into the lateral groove 73h and the groove section 73r1 of the cap receiving section 73, and the posture of the cap section 34Y in the cap receiving section 73 in the vertical and horizontal directions is restricted. At this time, shaking of the cap section 34Y before being inserted into the cap receiving section 73 is prevented by the cap section sandwiching member 73r. The first positioning hole 34a of the cap section 34Y is fitted into the main reference pin 73a of the cap receiving section 73, and so positioning of the main reference is performed. Thereafter, the second positioning hole 34b of the cap section 34Y is fitted into the sub reference pin 73b of the cap receiving section 73, and so main and sub positioning is completed. Further, while the positioning is being completed, the contact state between the stopper section 34d22 of the shutter member 34d of the cap section 34Y and the contact section 34n5 is released by the stopper release urging section 72b, and the postures of the shutter member 34d and the cap section 34Y in the cap receiving section 73 are decided by a body side shutter closing mechanism (not shown). In this state, the opening operation of the shutter member 34d is performed. The toner discharge opening W opened in the cap section 34Y is communicated with the toner feeding

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opening 73w of the cap receiving section 73. The notch 35b1 of the ID chip 35 of the cap section 34Y is engaged with the positioning pin 73e3 of the connector 73e of the apparatus body 100, the position of the ID chip 35 in the cap section 34Y is decided, and the plurality of metallic pads 35a of the ID chip 35 come in contact with the plurality of body side terminals 73e2 of the connector 73e, respectively, with a high degree of certainty. Thus, setting of the cap section 34Y (the toner container 32Y) in the cap receiving section 73 (the toner container storage unit 70) is completed. At this time, the gear 33c of the container body 33Y meshes with the driving gear 81 of the apparatus body 100.

Meanwhile, when extracting (removing) the toner container 32Y from the toner container storage unit 70 (the cap receiving section 73), the procedure reverse to the procedure at the time of mounting is performed.

As described above, in the image forming apparatus according to the present first embodiment, as the user performs one action of moving the toner container 32Y in the longitudinal direction while gripping the gripping section 33d (excluding the opening/closing operation of the body cover), the opening/closing operation of the toner discharge opening W by the shutter member 34d is also performed in conjunction with the operation, and the mounting operation and the removing operation of the toner container 32Y is completed.

Further, in the toner container 32Y according to the present first embodiment, the toner discharge opening W having the comparatively large opening area size is disposed downward in the vertical direction, and so the toner can be discharged such that it falls directly from the toner discharge opening W by its own weight.

Further, since the toner container 32Y is not placed from above the toner container storage unit 70 (the apparatus body 100) but attached to or detached from the front surface of the toner container storage unit 70 (the apparatus body 100), a degree of freedom of the layout above the toner container storage unit 70 increases. For example, even when a scanner (a document reading section) is disposed directly above the toner feeding device, operability or workability at the time of attachment/detachment of the toner container 32Y does not deteriorate.

Further, since the toner container 32Y is disposed in the apparatus body 100 such that the longitudinal direction is the horizontal direction, it does not have any influence on the overall layout of the image forming apparatus body 100 in the height direction. The toner capacity of the toner container 32Y increases, and so the replacement frequency can be reduced.

As described above, in the present first embodiment, the contact-type ID chip 35 (the information storage device) is held on the holding member 34k (the holding section) to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads 35a (terminals) approach and come in contact with the body side terminals 73e2. Thus, even when the contact-type ID chip 35 (the information storage device) is installed in the toner container 32Y (the removable device) installed removably on the image forming apparatus body 100, the contact failure caused by the positioning failure with the body side terminals 73e2 of the connector 73e of the image forming apparatus body 100 is difficult to occur.

Second Embodiment

A second embodiment of the present invention will be described in detail with reference to FIGS. 31 to 33.

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FIG. 31 is a schematic cross sectional view illustrating a toner container 232Y according to the present second embodiment. FIG. 32 is a back view illustrating a cap section 234Y of a toner container 232Y. FIG. 33 is a perspective view illustrating a holding cover 234k8 that is fitted into a holding member 234k.

The toner container 232Y according to the present second embodiment is different from the first embodiment in configuration of the holding member 234k in a holding mechanism for holding the information storage device.

Similarly to the first embodiment, the toner container 232Y according to the present second embodiment includes the container body 33Y and a cap section 234Y. The ID chip 35 as the information storage device is removably installed in the cap section 234Y.

In the cap section 234Y according to the present second embodiment, the holding member 234k in which the opening 34k2 exposing part of the ID chip 35 (the metallic pads 35a and the notch 35b1) is formed is integrally formed.

The ID chip 35 is inserted into from the inner side of the cap section 234Y in a direction of an arrow in FIG. 31 and set at the position of the holding member 234k. In the state in which the ID chip 35 is set at the position of the holding member 234k, the holding cover 234k8 is inserted into from the inner side of the cap section 234Y in the direction of the arrow in FIG. 31 and fitted into the holding member 234k (the state of FIG. 32).

In the holding cover 234k8, a pedestal section 234q that is fitted into the holding member 234k is disposed so as not to come in contact with the substrate 35b.

Further, the holding cover 234k8 is tightly installed without a clearance with the inner wall of the cap section 234Y so as to prevent the toner leaking from the toner container 232Y from sticking to the ID chip 35.

Even in the present second embodiment, the ID chip 35 (the substrate 35b) is held in the holding member 234k (and the holding cover 234k8) to be movable on the XZ plane.

As described above, similarly to the first embodiment, even in the present second embodiment, the contact-type ID chip 35 (the information storage device) is held on the holding member 234k (the holding section) to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads 35a (terminals) approach and come in contact with body side terminals 73e2. Thus, even when the contact-type ID chip 35 (the information storage device) is installed in the toner container 232Y (the removable device) installed removably on the image forming apparatus body 100, the contact failure caused by the positioning failure with the body side terminals 73e2 of the connector 73e of the image forming apparatus body 100 is difficult to occur.

Third Embodiment

A third embodiment of the present invention will be described in detail with reference to FIG. 34.

FIG. 34 is a schematic view illustrating a state in which the information storage device 35 of a toner container 332Y according to the present third embodiment is set to the connector 73e of the cap receiving section 73. FIG. 34 is a view corresponding to FIG. 27 in the first embodiment.

The present third embodiment is different from the first embodiment in that a cushion material 334k10 is installed inside the holding member 34k and a configuration of a wall section 373g of a cap receiving section 373 is different.

Similarly to the above embodiments, a toner container 332Y according to the present third embodiment includes a

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container body 33Y and the cap section 34Y. The ID chip 35 as the information storage device is removably installed in the cap section 34Y. Further, the ID chip 35 is held in the holding member 34k to be movable in the XZ plane.

In the present third embodiment, the cushion material 334k10 is disposed between the inner wall (the second facing section 34k5) of the holding member 34k and the substrate 35b. The cushion material 334k10 is made of an elastic material such as foamed polyurethane, and a low frictional material adheres to a section facing the substrate 35b. Thus, it is possible to alleviate damage occurring in the substrate 35b when the positioning pin 73e3 is engaged with the notch 35b1 without hindering movement of the ID chip (the substrate 35b) in the XZ plane.

In the present third embodiment, the wall section 373g of the cap receiving section 373 is installed to surround four sides of the connector 73e. In order to cope with it, the concave section for avoiding interference with the wall section 373g is formed in the cap section 34Y. By disposing the wall section 373g as described above, even if the toner is scattered from the vicinity of the toner discharge opening W of the toner container 332Y to the outside, the scattered toner is difficult to stick directly to the connector 73e or the ID chip 35. Thus, the contact failure (the communication failure) between the connector 73e (the body side terminal 73e2) and the ID chip 35 (the metallic pad 35a) caused by the scattered toner can be prevented.

As described above, similarly to the above embodiments, even in the present third embodiment, the contact-type ID chip 35 (the information storage device) is held on the holding member 34k (the holding section) to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads 35a (terminals) approach and come in contact with body side terminals 73e2. Thus, even when the contact-type ID chip 35 (the information storage device) is installed in the toner container 332Y (the removable device) installed removably on the image forming apparatus body 100, the contact failure caused by the positioning failure with the body side terminals 73e2 of the connector 73e of the image forming apparatus body 100 is difficult to occur.

Fourth Embodiment

A fourth embodiment of the present invention will be described in detail with reference to FIG. 35.

FIG. 35 is a three-plane view illustrating a substrate 435b of an information storage device 435 according to the present fourth embodiment. FIG. 35 is a view corresponding to FIGS. 14A and 14B in the first embodiment.

Similarly to the above embodiments, a toner container (432Y) according to the present fourth embodiment also includes the container body 33Y and the cap section 34Y. The ID chip 435 as an information storage device is removably installed in the cap section 34Y. Further, the ID chip 435 is held in the holding member 34k to be movable in the XZ plane.

In a substrate 435b of the ID chip 435 according to the present third embodiment, instead of the positioning notch 35b1 in the above embodiments, positioning holes 435b11 and 435b12 are disposed. In accompany with the mounting operation of the toner container (432Y), the substrate 435b freely moves in the XZ plane, and the positioning holes 435b11 and 435b12 are engaged with the positioning pins 73e3 of the connector 73e. Specifically, edges (or inner surfaces) of the holes 435b11 and 435b12 come in contact with the positioning pins 73e3, and movement of the sub-

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strate **435b** is restricted. Thus, the contact failure caused by the positioning failure between the plurality of metallic pads **35a** and the body side terminals **73e2** of the connector **73e** is difficult to occur.

In the present fourth embodiment, in view of the fact that the substrate **435b** is positioned at the lower side of the holding member **34k** due to the weight of the substrate **435b** directly before the positioning holes **435b11** and **435b12** are engaged with the positioning pins **73e3**, the circular hole **435b11** is formed in the lower section of the substrate **435b**, and the elliptical hole **435b12** is formed in the upper section of the substrate **435b**. The hole **435b11** in the lower section is picked up by the positioning pin **73e3**, and so the substrate **435b** is raised, and the positioning pin **73e3** is smoothly inserted into the other elliptical hole (the hole **435b12**). If the hole in the lower section is the elliptical hole and the hole in the upper section is the circular hole, the substrate **435b** may not be raised by the positioning pin **73e3**, and the positioning pin **73e3** may be difficult to be inserted into the circular hole in the upper portion.

In the present fourth embodiment, the two positioning holes **435b11** and **435b12** are formed in the substrate **35b** of the ID chip **35**. On the other hand, in the substrate **435b** of the ID chip **435**, one may be formed of the positioning holes **435b11** (or **435b12**), and the other may be formed of the positioning notch **35b1** (that has been used in the above embodiments). Even in this case, the same effect as in the present fourth embodiment can be obtained.

As described above, similarly to the above embodiments, even in the present fourth embodiment, the contact-type ID chip **435** (the information storage device) is held on the holding member **34k** (the holding section) to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads **35a** (terminals) approach and come in contact with body side terminals **73e2**. Thus, even when the contact-type ID chip **435** (the information storage device) is installed in the toner container (the removable device) installed removably on the image forming apparatus body **100**, the contact failure caused by the positioning failure with the body side terminals **73e2** of the connector **73e** of the image forming apparatus body **100** is difficult to occur.

Fifth Embodiment

A fifth embodiment of the present invention will be described in detail with reference to FIGS. **36** to **41**.

FIG. **36** is a three-plane view illustrating a substrate of an information storage device **535** according to the present fifth embodiment and is a view corresponding to FIG. **29** in the first embodiment. FIG. **37** is a perspective view illustrating the information storage device **535**, a holding member **534k** (**534k25**), and a connector **573e** and is a perspective view illustrating a relative positional relation of the three members **534k** (**534k25**), **535**, and **573e**. FIG. **38** is a perspective view illustrating a condition in which the information storage device **535** is engaged with the connector **573e**. FIGS. **39A** and **39B** are circuit diagrams illustrating an electric circuit of the information storage device **535** and an electric circuit of the connector **573e**. FIG. **40A** is a front view illustrating a condition in which the information storage device **535** is held on the connector **573e**, and FIG. **40B** is a front view illustrating a condition in which the information storage device **535** is rotating on a positioning hole **535b21**. FIG. **41** is a view illustrating the information storage device **535** that comes in contact with a probe **400** in an inspection process when manufactured in a factory.

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The present fifth embodiment is different from the first to fourth embodiments in that only one positioning hole **535b21** is formed in the substrate **535b** of the information storage device **535**, and the positioning hole **535b21** is disposed between a plurality of rectangular metallic pads **35a1**, **35a2**, and **35a3** (metallic pads).

Referring to FIG. **36**, the ID chip **535** as the information storage device according to the present fifth embodiment has the positioning hole **535b21** that is formed at the upper position in the vertical direction further than the gravity center of the substrate **535b**. An earth metallic terminal **535d** is disposed on an inner surface of the hole **535b21** and around the hole **535b21**. In the present fifth embodiment, the metallic terminal **535d** formed on the surface of the substrate **535b** includes two protruding sections **535d1** that are formed to extend from a rink-like section in the horizontal direction.

Further, the rectangular metallic pad **35a1** is installed above the positioning hole **535b21** in the vertical direction, and the two rectangular metallic pads **35a2** and **35a3** are installed below the positioning hole **535b21** in the vertical direction.

Further, on the back side of the substrate **535b** (the side facing the cap section **34Y**), a protection member **535e** that is made of a resin material having a substantially hemispherical shape or a shape of an inverted pan such as epoxy and covers the information storage unit is disposed. In the present fifth embodiment, although it depends on the shape of the substrate **535b** or the configuration/arrangement of the back surface such as the protection member **535e**, by disposing the hole **535b21** above the protection member **535e** that may include the information storage unit such as an integrated circuit (IC) therein and is a heaviest component, a positional relation in which the hole **535b21** is installed vertically above the gravity center of the ID chip **535** is implemented. Specifically, referring to FIG. **40A**, the ID chip **535** (the information storage device) according to the present fifth embodiment is formed so that the center position of the positioning hole **535b21** is at the distance **Za** above the gravity center of the ID chip **535**.

Referring to FIG. **37**, the connector **573e** includes a connector body **573e21** that is made of resin and is a hollow box, and a positioning pin **573e23** (a positioning protruding section) that is a hollow cylinder and having a tapered tip is disposed on the connector body **573e21** to stand in the horizontal direction. A body side earth terminal **573e25** (an earth terminal) is installed in the positioning pin **573e23**. The body side earth terminal **573e25** is a plat-like (or linear) metallic member, partially stored in the hollow section of the positioning pin **573e23** formed integrally with a connector **573e21**, and has a curved section that is exposed from a slit-like opening formed in part of the circumferential surface of the hollow cylinder and protrudes from the cylindrical outer circumferential surface. Further, one body side terminal **573e2** is installed vertically above the positioning pin **573e23** (the body side earth terminal **573e25**), and two body side terminals **573e2** are installed vertically below the positioning pin **573e23** (the body side earth terminal **573e25**). The body side terminals **573e2** are plat-like (or linear) metallic members and are formed almost in the same manner as in the above embodiments except that the installation position is different.

Further, swing preventing members **573e24** as a pair of guiding members protrude from the right and left sides of the positioning pin **573e23**. The guiding members include a pair of plates whose tips have inner tapered surfaces facing each other and the guide the both sides of the ID chip **535** to be upright.

Similarly to the above embodiments, the holding member **534k** (the holding section) is fixed to a toner container (**532Y**) and positioned between the connector **573e** and the ID chip **535**. The holding member **534k** has almost the same function (a function for movably holding the ID chip **35**) as in the above embodiments. Referring to FIG. 37, the holding member **534k** according to the present fifth embodiment has a first facing section **534k24** that is configured to be linearly symmetrical on a center axis in the vertical direction and is formed to cover an area from two upper corners of the ID chip **535** to both sides of the hole **535b21**. The holding member **534k** is formed to cover the lower section of the substrate **535b** further than the lowest metallic pad **35a3**, and through the above described configuration, the ID chip **535** is prevented from falling from the holding member **534k**.

Further, in the holding member **534k**, the first facing section **534k24** including an area facing the four body side terminals **573e2** and **573e25** of the connector **573e** is formed of an opening. Particularly, in the holding member **534k**, formed is an opening **534k22** that is opened up to a section corresponding to a pair of swing preventing members **573e24**. At the time of mounting of the toner container **532Y**, the positioning pin **573e23** is inserted into the opening **534k22**, and thereafter the pair of swing preventing members **573e24** (the pair of guiding members) is also inserted into the holding member **534k** through the opening **534k22**.

A second facing section **534k25** of a flat plate form facing the back surface of the ID chip **535** (the side of the holding member **535e**) is fixed to the holding member **534k** by adhesion or a snap fit (not shown). The second facing section **534k25** includes an opening **534k26** similarly to the opening **534k22** and so can avoid interference with the holding members **535e** or the inserted swing preventing member **573e24**. Meanwhile, when the positioning pin **573e23** is inserted into the hole **535b21** of the ID chip **535**, the ID chip **535** is pushed, but since the second facing section **534k25** supports the substrate **535b** from the rear side, the contact condition between the terminals can be maintained.

FIG. 38 is a schematic perspective view illustrating the condition in which positioning of the connector **573e** and the ID chip **535** at the apparatus body **100** side is completed. The condition is that the toner container **532Y** according to the fifth embodiment is mounted on the apparatus body **100**, and the body side terminals **573e2** and **573e25** are connected with the abovementioned metallic pads **35a1** to **35a3** and the earth terminal **535d**. In FIG. 38, for the sake of easy understanding, the holding member **534k** (**534k25**) between the connector **573e** and the ID chip **535** and the metallic pads **35a1** to **35a3** are omitted.

When mounting the toner container **532Y** on the image forming apparatus body, the main and subordinate positioning holes **34a** and **34b** of the cap section **534Y** are fitted into the main and subordinate positioning pins **73a** and **73b** of the cap receiving section **73**, and positioning of the cap section **534Y** is performed. This mounting operation is the same as the mounting operation of the first embodiment. After the position of the cap section **534Y** is decided, the hole **535b21** of the ID chip **535** is fitted into the positioning pin **573e23** to be picked up by the tapered tip of the positioning pin **573e23** of the connector **573e**. As a result, the position of the ID chip **535** in the horizontal direction and the vertical direction is decided at the same time. Further, as illustrated in FIG. 40A, the swing preventing members **573e24** (the pair of guiding members) of the connector **573e** that includes the pair of plates are inserted into the lower edge portions that are at both right and left sides of the substrate **535b** and at the lower area further than the center of the hole

535b21. At this time, it is a possible case that the posture of the ID chip is misaligned as illustrated in FIG. 40B. Even in this case, if the inner tapered surfaces of the plates comes in contact with the lower edge portions, it causes the substrate **535b** to rotate in a direction for having the posture to be vertical by the action of the gravity center, and the misalignment of the posture in the rotation direction (rotation in a direction of an arrow illustrated in FIG. 40B) is corrected (it becomes a condition of FIG. 40A). Thus, positioning of the ID chip **535** is completed. At this time, part of the earth terminal **535d** of the ID chip **535** (a section corresponding to the inner surface of the hole **535b21**) comes in contact with the body side earth terminal **573e25** of the positioning pin **573e23** illustrated in FIG. 38, and the ID chip **535** is earthed (conduction). Further, after the earth is connected the three metallic pads **35a** (**35a1**, **35a2**, and **35a3**) of the ID chip **535** also come in contact with the three body side terminals **573e2** of the connector **573e**, respectively, and so information can be transmitted between the ID chip **535** and the body side connector **573e** (the apparatus body **100**).

As described above, in the present fifth embodiment, a high-accuracy positioning mechanism is implemented by an inexpensive configuration by adding various ideas such as the following ideas (1) to (5).

- (1) It is only one positioning hole **535b21** for ID chip **535** to need to be positioned. Thus, the drill processing time of the substrate **535b** in the manufacturing become shorter than two positioning holes type's ID chip, and the manufacturing cost can be reduced.
- (2) The body side earth terminal **573e25** is integrally installed on the side circumferential surface of the positioning pin **573e23**. Thus, the distance between the positioning pin **573e23** and the body side earth terminal **573e25** can become real zero (0), and a degree of position accuracy of the earth terminal **535d** with respect to the body side terminal **573e25** can increase.
- (3) In the mounting completion condition of FIG. 38, the positional relation between the positioning hole **535b21** and the curved sections of the body side terminals **573e2** is adjusted so as to match the hole center of the hole **535b21** with the line that connects the curved sections (connection sections) of the three body side terminals **573e2** at the connector **573e** side. Thus, the distance in the horizontal direction from the hole **535b21** as the positioning section to the contact sections can be reduced to nearly 0 mm. As a result, when the three metallic pads **35a1**, **35a2**, and **35a3** come in contact with the body side terminals **573e2**, a degree of position accuracy is improved.
- (4) A plurality of metallic pads **35a1**, **35a2**, and **35a3** are lined, and the positioning hole **535b21** is arranged at either of two spaces formed between two of the three lined pads. Thus, compared to another arrangement type's ID chip in which the hole is disposed at the lower side or the upper side outside a row of the plurality of metallic pads **35a1**, **35a2**, and **35a3**, the distance (that corresponds to the arm length of the pendulum) from the center of the positioning hole **535b21** to the farthest metallic pad **35a** can be reduced. Specifically, on the another type, the arm length becomes the distance corresponding to the three metallic pads from the hole center. However, in the present fifth embodiment, the arm length can be the distance corresponding to the two metallic pads. Since the arm length of the pendulum is short, even if the parallelism of the farthest metallic pad **35a** on the body side terminal **73e2** is misaligned due

to, for example, variability in the mass production, the misalignment can be suppressed to a minimum.

- (5) When a user stores the toner container in some space out of the image forming apparatus, an alien substance may enter the holding member **534k** and so the nipping position between the ID chip **535** and the facing sections **534k24** and **534k25** may remain misaligned. On this problem, in the present fifth embodiment, the positional relation between the hole **535b21** and the gravity center of the ID chip **535** is improved. The hole **535b21** is arranged vertically above the gravity center of the ID chip **535**. Thus, when the pair of the swing preventing members **573e24** is inserted below the hole **535b21** that is the rotation center, the substrate **535b** contacts with the tapered tips of the swing preventing member **573e24**. Then, the posture of the ID chip **535** is urged to rotate in the vertical direction by the force of gravity and is corrected to be upright. As a result, even if there is one positioning hole **535b21**, a degree of position accuracy of the plurality of metallic pads **35a1**, **35a2**, and **35a3** on the plurality of body side terminals **573e2** can be increased at the same time.

Depending on a number of produced planned to be made, there is a possibility that it is determined that the conduction inspection device has a margin in durability. In this case, inspection can be conducted using a inspection device having a configuration similar to the connector of the image forming apparatus body **100**, i.e. by inserting a conduction probe into the hole in which the earth terminal **535d** is formed. In this case, as in a modification of the ID chip illustrated in FIG. **87**, a circular earth terminal that does not include the protruding sections **535d1** may be employed. The same also applies to a case in which a manufacturing method in which the conduction inspection is omitted based on improvement in production process is employed. With this modification, the area of the metallic members can be reduced and the manufacturing cost can be reduced. In FIG. **87**, the ID chip is illustrated assumed as being of a type in which no protection member is provided on the IC circuit on the back surface and, therefore, the IC circuit is exposed. If the manufacturing environment, etc. allows absence of the protection member, such an ID chip is useful for cost reduction. Although any layout is available so long as the IC circuit keeps out from both the right-and-left areas for smooth sliding, it is preferable to arrange a relatively large IC in a lower section so that the center of gravity is located at a lower position.

Moreover, if the earth conduction probe is of one pin, it is allowable to provide, as illustrated in FIG. **88A**, one protruding section with which the earth probe **401** comes in contact. For easy earth inspection, the modification illustrated in FIG. **88B** that has a size-increased probe contacting section may be employed. This is useful in particular in a case of manual inspection. The modification illustrated in FIG. **88C** that has not a circular frame but a square frame may also be employed. Any of the modifications of FIGS. **88A**, **88B**, and **88C** are designed freely so long as the outer circumferential area of the terminal formed in the hole keeps out from both the right-and-left sliding areas. The back surfaces of the ID chips of FIGS. **88A**, **88B**, and **88C** can be either covered by the protection member or uncovered.

As described in the above described ideas (1) to (5), each of the five ideas can provide each function effect, and even if an inexpensive configuration in which the area size of the metallic pad **35a** becomes minimal is employed, it is possible to highly increase a degree of positioning accuracy between the plurality of terminals **35a1**, **35a2**, and **35a3** and

535d including the earth terminal at the ID chip **535** and the plurality of body side terminals **573e2** and **573e25**.

Further, in the present fifth embodiment, the ideas and function effects different from ones described above will be described.

Firstly, each of the metallic pads **35a1**, **35a2**, and **35a3** are described. The metallic pad **35a1**, which is at the highest level, receives a clock signal for communication control. While a serial communication method that is low-speed but low-cost because of sequential data transfer is employed and an I2C (Inter-Integrated Circuit) is employed as a serial bus, The metallic pad **35a1** forms a signal line to which a serial clock (SCL) is input when the signal line is connected to the body side connector. The metallic pad **35a1** corresponds to a terminal to which a clock-signal is input. Because a clock signal flows in one way, it is expected that the possibility that the ID chip **535** breaks down if to a short circuit occurs between the metallic pad **35a1** and a later-described Vcc (the power supply or the metallic pad **35a3**) than between the other terminals and the Vcc. Therefore, to prevent break down of the ID chip **535**, the metallic pad **35a1** is arranged more distant from the Vcc. This is because the possibility of bread down is lower if a short circuit occurs between the metallic pad **35a1** and the GND (the earth terminal **535d**).

The metallic pad **35a2** also employs a serial communication method, employing an I2C as a serial bus, and forms a signal line to which serial data (SDA) is input/output when the signal line is connected to the body side connector. Because this pad has a bidirectional input/output mechanism, the possibility that the ID chip **535** breaks down due to a short circuit is lower than the possibility due to the one-way input metallic pad **35a1**.

The metallic pad **35a3** forms a power input portion (Vcc) to which a 5V voltage or a 3.3V voltage is input when it is connected to the body side connector. To decrease the risk of break down of the entire device due to a short circuit between the power supply and the GND, the serial-data input terminal (the metallic pad **35a2**) is arranged between the GND (the earth terminal **535d**) and the serial-clock input terminal (the metallic pad **35a1**). As illustrated in FIG. **36**, the Vcc or the metallic pad **35a3** overlaps with the protection member **535e** that is on the back side of the ID chip in such a manner that the substrate **535b** is between them; therefore, the metallic pad **35a3** is close to an IC driving circuit included in the protection member **535e**. With this arrangement, a short and thick line can be used as a power-supply line, which enables stable power-supply operations (=suppression of malfunction due to noises).

Secondly, ideas for earth are described. In the mounting operation of the toner container **532Y**, the earth terminal **535d** of the ID chip **535** comes in contact with the body side earth terminal **573e25** of the positioning pin **573e23** (the connector **573e**), and then the three metallic pads **35a1**, **35a2**, and **35a3** of the ID chip **535** start to come in contact with the three body side terminals **573e2** of the connector **573e**. That is, in the detaching operation of the toner container **532Y**, contact between the three metallic pads **35a1**, **35a2**, and **35a3** of the ID chip **535** and the three body side terminals **573e2** of the connector **73e** is released, and then the earth terminal **535d** of the ID chip **535** is released from the contact condition with (separated from) the body side earth terminal **573e25** of the positioning pin **573e23** (the connector **573e**). Specifically, referring to FIG. **39A**, the body side earth terminal **573e25** in the connector **573e** has the contact start position closer to the ID chip **535** side than the three body side terminals **573e2**.

Through the above described configuration, in the mounting operation of the toner container 532Y, the metallic pads 35a1, 35a2, and 35a3 always start to be connected with the body side terminals 573e2 in the condition in which the ID chip 535 is earthed, and in the detaching operation of the toner container 532Y, the metallic pads 35a1, 35a2, and 35a3 always start to be separated from (released from the contact condition with) the body side terminals 573e2 in the condition in which the ID chip 535 is earthed. Thus, an electric circuit at the ID chip 535 is prevented from being not earthed and so becoming an electrically floating condition, and so the ID chip 535 is difficult to be electrically damaged.

In detail, when the electric circuit at the ID chip 535 is not earthed and becomes an electrically floating condition, the electrical circuit becomes a condition that is earthed with very large impedance. If static electricity, which is generated when the metallic pads 535a come in contact with or are separated from the body side terminals 573e2, slightly flows to the electric circuit, a high voltage that is the same as impedance is applied to the current is generated. The high voltage causes insulation breakdown inside the IC in the ID chip 535, and thus the IC is broken. This problem easily occurs when the contact start positions of the three body side terminals 573e2 and the body side earth terminal 573e25 on the ID chip 535 are formed at the same position, with respect to the connector 573e, as illustrated in FIG. 39B.

On the other hand, in the present fifth embodiment, the curved section of the body side earth terminal 573e25 exposed from the slit-like opening of the positioning pin 573e23 is disposed to be closer to the ID chip 535 than the curved section of the body side terminal 73e2 that most protrudes to the ID chip 535 side. Thus, since the earth is first connected, and at the time of separation and the earth is lastly disconnected at the time of contact, the impedance is always theoretically zero, and even if static electricity flows to the inside of the electric circuit, insulation breakdown inside the IC is prevented.

Further, in the ID chip 535 (the information storage device) according to the present fifth embodiment, the two protruding sections 535d1 are disposed on part of the outer circumference of the earth terminal 535d as described above with reference to FIG. 36.

Since the protruding sections 535d1 are disposed on the front surface of the substrate 535b of the ID chip 535 as described above, in the inspection process (a process of inspecting whether or not the ID chip 535 is defective) at the time of manufacturing in a factory, an operation of contacting a conduction inspection probe can be easily performed. In detail, as illustrated in FIG. 41, leading ends of a plurality of probes 401 of a conduction inspection device 400 are pressed downward against the metallic pads 35a and the earth terminal 535d of the ID chip 535 placed on an inspection table. At this time, since the protruding section 535d1 of the earth terminal 535d has an area that can sufficiently come in contact with the leading ends of the probes 401, a conduction inspection failure caused by a contact failure of the probes 401 can be prevented. Further, since conduction inspection is performed by pressing the leading end sections of the probes 401 downward against the earth terminal 535d (the protruding section 535d1), compared to when conduction inspection is performed by inserting the probes 401 into the hole 535b21, a resistance characteristic of the probes 401 that are repetitively used for inspection can be improved, and a problem in that the hole 535b21 of the ID chip 535 wears by conduction inspection can be prevented.

In a surplus space broadening in a wedge form between the annular earth terminal 535d and the rectangular metallic pad 35a1, 35a2, the protruding section 535d1 has a horizontal direction boundary (boundary line) that comes in contact with the annular outer circumference and is disposed to be parallel to the metallic pads 35a1, 35a2, and 35a3. Thus, the protruding section 535d1 does not protrude in the vertical direction, the protruding section 535d1 can be prevented from protruding to the left and sliding areas of the substrate 535b that slides against the first facing section 534k24 (protruding in the horizontal direction). As a result, the size of the substrate 535b does not increase, and at the time of manufacturing, it is possible to obtain as many substrates 535b as possible from a substrate material having the standard size. Further, the cost of the ID chip 535 can be suppressed from increasing.

As described above, similarly to the above embodiments, even in the present fifth embodiment, the contact-type ID chip 535 (the information storage device) is held on the holding member 534k (the holding section) to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads 35a1, 35a2, and 35a3 (terminals) approach and come in contact with body side terminals 573e2. Thus, even when the contact-type ID chip 535 (the information storage device) is installed in the toner container 532Y (the removable device) installed removably on the image forming apparatus body 100, the contact failure caused by the positioning failure with the body side terminals 573e2 of the connector 573e is difficult to occur.

Further, in the present fifth embodiment, even when the contact-type ID chip 535 (the information storage device) is installed in the toner container 532Y (the removable device) installed removably on the image forming apparatus body 100, since the earth terminal 535d that is engaged with the body side earth terminal 573e25 formed in the positioning pin 573e23 (the protruding section) of the connector 573e (the image forming apparatus body 100) is formed in one hole 535b21 formed in the substrate 535b of the ID chip 535, the ID chip 535 is difficult to be electrically damaged.

Sixth Embodiment

A sixth embodiment of the present invention will be described in detail with reference to FIGS. 42A to 47C.

FIGS. 42A and 42B are perspective views illustrating a toner container 632Y according to a sixth embodiment. Particularly, FIG. 42A is an exploded view illustrating a state in which the ID chip 535 described in the fifth embodiment is not mounted, and FIG. 42B is a view illustrating a state in which the ID chip 535 is mounted. FIG. 43 is a front view illustrating the toner container 632Y in which a face plate 634p is not installed and is a view corresponding to FIG. 21 in the first embodiment. FIG. 44 is a cross-sectional view illustrating a main part of the toner container 632Y in which the ID chip 535 and the face plate 634p are installed. FIG. 45 is a view illustrating a state in which the ID chip 535 is inserted into the connector 573e.

In the present sixth embodiment, the ID chip 535 as the information storage device is the same as in the fifth embodiment. The present sixth embodiment is different from the fifth embodiment in that the ID chip 535 is loosely held in a concave section disposed in a cap section 634Y and movably covered by the face plate 634p, and the remaining configuration is the same as in the fifth embodiment.

Similarly to the above embodiments, the toner container 632Y according to the present sixth embodiment also

includes the container body 33Y and the cap section 634Y. The ID chip 535 as the information storage device is removably installed in the cap section 634Y.

Referring to FIGS. 42A and 42B, in the present sixth embodiment, the ID chip 535 is not installed in the cap section 534Y in a state in which it is loosely inserted into the holding member 534k, and the falling prevention face plate 634p is screw-coupled to the cap section 634Y in a state in which the ID chip 535 is loosely held in the concave section (in which a pedestal section 634q is formed) formed in the cap section 634Y.

In detail, referring to FIG. 42A and FIG. 43, the concave section for holding the ID chip 535 to be movable in the XZ plane is formed on the end surface of the cap section 634Y. In the concave section, formed is the pedestal section 634q that comes in surface contact with part of the ID chip 535. In the state in which the ID chip 535 is loosely held in the concave section of the cap section 634Y, the face plate 634p for preventing the ID chip 535 from falling from the concave section is attached. Referring to FIG. 42B, FIG. 43, and FIG. 44, the face plate 634p is screw-coupled to come in contact with part of the substrate 35b of the ID chip 35 in the state in which the metallic pads 35a1, 35a2, and 35a3, the positioning hole 535b21 (the earth terminal 535d), and the like of the ID chip 535 formed similarly to the fifth embodiment are exposed.

In further detail, in the cap section 634Y, a positioning pin 634s1 for positioning the face plate 634p is formed on the right side of the concave section, and a screw hole 634s2 for screw-fixing the face plate 634p is formed on the left of the concave section with the concave section interposed therebetween. Meanwhile, in the face plate 634p, a positioning hole 634p1 is formed at the position corresponding to the positioning pin 634s1, and a hole 634p2 through which a screw 680 passes is formed at the position corresponding to the screw hole 634s2. In the lower section of the face plate 634p, a contact section 634p3 that comes in contact with the outer circumferential edge of the second positioning hole 34b and functions as a rotation stopper is formed. The position of the face plate 634p with respect to the cap section 634Y is decided by the positioning hole 634p1 and the contact section 634p3 for rotation stopping. The screw 680 is screwed into the screw hole 634s2 formed in the cap section 634Y through the hole 634p2 formed in the face plate 634p, and so the face plate 634p is fixed to the cap section 634Y. Thus, the ID chip 535 does not fall from the cap section 634Y and is held on the cap section 634Y to be movable in the XZ plane. Referring to FIG. 45, similarly to the fifth embodiment, in accompany with the mounting operation of the toner container 632Y, the positioning hole 535b21 (the earth terminal 535d) of the ID chip 535 is engaged with the positioning pin 573e23 (the body side earth terminal 573e25) of the connector 573e of the apparatus body 100, thereafter the body side terminal 573e2 of the connector 573e comes in contact with the metallic pads 35a1, 35a2, and 35a3 of the ID chip 535, and so electrical contact between the connector 573e and the ID chip 535 is completed. In this case, since the ID chip 535 in the cap section 634Y of the toner container 632Y is held to be movable in the XZ plane, similarly to the above embodiments, the contact failure caused by the positioning failure with the body side terminals 73e2 and 573e25 of the connector 573e of the apparatus body 100 is difficult to occur. In the normal state, the ID chip 535 remains down to the lower side of the concave section of the cap section 534Y due to gravity, and the center position of the hole 535b21 of the ID chip 535 is misaligned downward on the axial center

position of the positioning pin 573e23 like the most left one among the three ID chips 535 illustrated in FIG. 45.

Then, when the mounting operation of the toner container 632Y starts and the ID chip 535 comes in contact with the positioning pin 573e23, the ID chip 535 moves upward (in the Z direction) (is scooped up) such that the hole 535b21 follows the tapered leading end section of the positioning pin 573e23, the hole 535b21 is fitted into the positioning pin 573e23, and finally the body side terminals 573e2 comes in contact with the metallic pads 35a1, 35a2, and 35a3.

In the present sixth embodiment, the face plate 634p has been fixed (screw-coupled) to the cap section 34Y by the screw 680.

On the other hand, as illustrated in FIGS. 46A and 46B, a face plate 734p may be fixed to a cap section 734Y by snap fit fixing. In detail, as illustrated in FIG. 46A, a plurality of snap fit fixing engaging section 734p2 are formed on the outer circumferential section of the face plate 734p, and snap fit fixing engaged sections 734s2 are formed at the positions of the cap section 734Y corresponding thereto. As illustrated in FIG. 46B, in the state in which the ID chip 535 is loosely inserted into the concave section of the cap section 734Y, the face plate 734p is snap fit-fixed to the cap section 734Y. In further detail, while aligning a hole 734p3 formed in the face plate 734p with a positioning boss section 734s3 formed in the cap section 734Y, the engaging section 734p2 of the face plate 734p is engaged with the engaged section 734s2 of the cap section 734Y, and the face plate 734p is positioned and fixed to the cap section 734Y. Even in the case of this configuration, the same effect as in the sixth embodiment can be obtained.

Further, in the present sixth embodiment, since replacement can be made again even after the face plate 634p (or the face plate 734p illustrated in FIGS. 46A and 46B) is assembled, the toner container can be manufactured by a procedure in which the toner container and the face plate manufactured by a foreign partner company are first imported, and then, within the country, after or before a process of filling the toner container with the toner, the ID chip 535 purchased from another company is assembled, and toner information is input to the ID chip 535. Thus, the manufacturing process of the toner container can be effectively performed.

Further, a recycling process of collecting the used toner container from the market and filling the toner again after cleaning it may be performed by a procedure of replacing the ID chip 535 or removing the ID chip 535 from the toner container, rewriting information, and mounting the ID chip 535 on the cap section again. Thus, the reusing process of the toner container can be effectively performed.

However, referring to FIG. 47C (that is a cross-sectional view illustrating a cap section 834Y on which the ID chip 535 is mounted), when it is desired to increase a assembly strength between a face plate 834p and a cap section 834Y without needing to remove the ID chip 535, only the positioning boss 734s3 (for example, see FIGS. 46A and 46B) may be disposed in the cap section without disposing the shape for screw coupling or the shape for snap fitting. Then, after the ID chip 535 and the face plate 834p are assembled in the cap section 834Y, the leading end of the positioning boss 734s3 may be thermally molten to fix the face plate 834p to the cap section 834Y, or an adhesive may be coated between the face plate 834p and the cap section 834Y to fix the face plate 834p to the cap section 834Y.

As described above, in the present sixth embodiment, the contact-type ID chip 535 (the information storage device) is held on the cap section 634Y, 734Y, or 834Y to be movable

on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads **35a1**, **35a2**, and **35a3** (terminals) approach and come in contact with body side terminals **573e2**. Thus, even when the contact-type ID chip **535** (the information storage device) is installed in the toner container **632Y**, **732Y**, or **832Y** (the removable device) installed removably on the image forming apparatus body **100**, the contact failure caused by the positioning failure with the body side terminals **573e2** of the connector **573e** of the image forming apparatus body **100** is difficult to occur.

Further, even in the present sixth embodiment, similarly to the fifth embodiment, even when the contact-type ID chip **535** (the information storage device) is installed in the toner container **632Y** (the removable device) installed removably on the image forming apparatus body **100**, since the earth terminal **535d** engaged with the body side earth terminal **573e25** formed in the positioning pin **573e23** (the protruding section) of the connector **573e** (the image forming apparatus body **100**) is formed in one hole **535b21** formed in the substrate **535b** of the ID chip **535**, the ID chip **535** is difficult to be electrically damaged.

FIGS. **47A** and **47B** are views illustrating a toner container **932Y** of another embodiment. Particularly, FIG. **47A** is a front view illustrating a cap section **934Y** on which the ID chip **535** is mounted, and FIG. **47B** is a front view illustrating the cap section **934Y** and the ID chip **535** before the ID chip **535** is mounted. In the toner container **932Y** illustrated in FIGS. **47A** and **47B**, unlike the above embodiments, the ID chip **535** (the information storage device) is fixed to and held on the cap section **934Y** (held to be immovable in the XZ direction). Specifically, a concave section of the cap section **934Y** (a section encircled by a dotted line in FIG. **47B**) is formed in a shape according to an outer circumferential shape of the ID chip **535** so that the ID chip **535** can be fitted therein (fitted therein within a dimension variation range by a fitting tolerance of about 0.3 mm at maximum even if shaken). In this case, unlike the above embodiments, the ID chip **535** cannot move in the XZ plane, but the function effects of the ideas (1) to (4) among the five ideas (1) to (5) described in the fifth embodiment can be obtained. That is, since one positioning hole **535b21** in which the earth terminal **535d** is formed is installed in the ID chip **535**, the effects described in the fifth embodiment can be obtained.

Seventh Embodiment

A seventh embodiment of the present invention will be described in detail with reference to FIGS. **48** and **49**.

FIG. **48** is an exploded perspective view illustrating a toner container **1032Y** according to the seventh embodiment. FIG. **49** is a cross-sectional view illustrating the toner container **1032Y**.

The toner container according to the present seventh embodiment is different from the above embodiments in which the container body **33Y** is rotatably held on the toner container storage unit **70** in that a container body **1033Y** is combined with any one of the cap sections **634Y**, **734Y**, **834Y**, and **934Y** illustrated in the sixth embodiment and non-rotatably held on the toner container storage unit **70** together with the cap section.

Referring to FIGS. **48** and **49**, similarly to the above embodiments, the toner container **1032Y** according to the present seventh embodiment mainly includes the container body **1033Y** (the bottle body) and the cap section **634Y** installed on the head section thereof (or the cap section

734Y, **834Y**, or **934Y** of another form). Hereinafter, the cap section according to the present seventh embodiment will be described using the cap section **634Y** described with reference to FIGS. **42A** and **42B** in the sixth embodiment.

Unlike the above embodiments, in the toner container **1032Y** according to the present seventh embodiment, the container body **1033Y** (the bottle body) is fixed to the cap section **634Y** by a fixing method, for example, it adheres to (fuses with) or is engaged with the cap section **634Y** (the bottle cap). That is, the container body **1033Y** is non-rotatably fixed to the cap section **634Y**.

Unlike the above embodiments, in the container body **1033Y** according to the present seventh embodiment, a helical protrusion is not formed on the circumferential surface thereof. Further, the gear **33c** in the above embodiments is not integrally formed with the container body **1033Y**, and a gear member **1042Y** is installed rotatably on the container body **1033Y** and the cap section **634Y** together with the agitating member **33f**. Inside the container body **1033Y**, unlike the above embodiments, a conveying member **1041Y** for conveying the toner stored in the container body **1033Y** toward the opening **A** is formed such that one end thereof is fixed to the gear **1042Y**, and the other end thereof is rotatably supported on a bearing **1033d1** of the container body **1033Y** which will be described later.

The cap section **634Y** has almost the same configuration as in the fifth embodiment except that it non-rotatably adheres to or is fixed to the container body **1033Y**.

The agitating member **33f** has almost the same configuration, form, and function as in the above embodiments except that it is not fixed to the container body **1033Y** but held only on the gear **1042Y**.

A further detailed description will be made with reference to FIGS. **48** and **49**.

Referring to FIG. **48**, even in the seventh embodiment, on the other end side of the container body **1033Y** in the longitudinal direction (the side opposite to one end side, at which the cap section **634Y** is installed, in the longitudinal direction and an end section at the rear side in the mounting direction on the apparatus body **100**), disposed is a gripping section **1033d** gripped by the user when the attaching/detaching operation of the toner container **1032Y** is performed. In the gripping section **1033d**, a through hole communicating with the inside and outside of the container body **1033Y** is formed, and a cover member **1049Y** that is formed of deformable flexible resin such as polypropylene or polyethylene is removably installed in the through hole. The cover member **1049Y** is used when filling the inside of the toner container **1032Y** (the container body **1033Y**) with the toner (or cleaning), for example, at the time of manufacturing or recycling. The cover member **1049Y** is removed from the container body **1033Y** when filling the toner (cleaning) and mounted to the container body **1033Y** after filling of the toner is completed.

Referring to FIG. **49**, the conveying member **1041Y** installed inside the container body **1033Y** is formed such that a thin flexible agitating member **1041Yb** formed of a material such as mylar (a trade name: a polyester film) adheres to a shaft section **1041Ya**, and an agitator member **1041Yc** is formed at the opposite side. In the shaft section **1041Ya** of the conveying member **1041Y**, an end section at one end side in the longitudinal direction is engaged with and fixed to a connection section **1033/20** installed at the position of the rotation center of the agitating member **33f**. An end section at the other side in the longitudinal direction is rotatably supported on the bearing section **1033d1** (which is a base section of the gripping section **1033d** and formed

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in a section stuck into the container body 1033Y). In the state in which the container body 1033Y and the cap section 634Y are non-rotatably held on the toner container storage unit 70, the agitating member 33f receives driving force from the driving unit 91 and rotates together with the gear member 1042Y, and so the conveying member 1041Y connected with the agitating member 33f at the position of the connection section 1033/20 also rotates. Thus, the toner stored in a container body 1044Y is agitated by agitating force of the agitator member 1041Yc installed in the conveying member 1041Y, and the toner stored in the container body 1033Y is conveyed toward the cap section 1034Y by conveying force of the flexible agitating member 1041Yb installed in the conveying member 1041Y in the shaft direction.

The flexible agitating member 1041Yb of the conveying member 1041Y includes cutouts 1041Yb1 formed at a plurality of positions (in the present seventh embodiment, six positions) in the longitudinal direction. Thus, in accompany with rotation of the conveying member 1041Y, the leading end of the flexible agitating member 1041Yb (the free end side that is not supported on the shaft section 1041Ya) comes in sliding contact with the inner circumferential surface of the container body 1033Y, and the flexible agitating member 1041Yb rotates in the appropriately twisted and bent state, so that the toner stored in the container body 1033Y is agitated and conveyed toward the right side in FIG. 49 in the shaft direction.

As described above, even in the toner container 1032Y according to the present seventh embodiment, similarly to the above embodiments, the toner is discharged from the toner discharge opening W of the cap section 1034Y.

Here, the gear member 1042Y is rotatably attached to the container body 1033Y.

In detail, a gear engaging section (a claw section snap fitted into) (not shown) formed in the gear member 1042Y is caught in a flange section (in which a protrusion 1033e which will be described later is formed) formed to make one round around the outer circumferential surface of a bottle mouth section 1033a of the container body 1033Y, and so the gear member 1042Y is rotatably held on the container body 1033Y. Further, a gear section (a spur gear) is formed on the outer circumferential surface of the gear member 1042Y, and when the toner container 1032Y is set to the apparatus body 100, the gear section meshes with the driving gear 81 of the apparatus body 100.

A seal material is disposed between the gear member 1042Y and the end surface of the bottle mouth section 1033a so as to prevent the toner from leaking to the outside of the toner container 1032Y. The seal material is made of a foamed elastic material such as foamed polyurethane, formed in an annular shape to be bitten into the end surface of the bottle mouth section 1033a, and adheres to the gear member 1042Y. When the gear member 1042Y is set to the container body 1033Y, the seal material is pressed against the opening end surface of the bottle mouth section 1033a, and so a sealing characteristic between both members 1033Y and 1042Y is secured.

The gear member 1042Y is not fixed to the cap section 1034Y but rotatably held on the claw section 34j of the cap section 634Y. A method of holding the gear member 1042Y on the cap section 634Y is similar to the method of holding the cap section 34Y on the bottle mouth section 33a of the container body 33Y described in the above embodiments. That is, the claw section 34j of the cap section 634Y is engaged with a flange-like engaged protruding section 1033j disposed to make one round around the outer circumference

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of the gear member 1042Y, and the gear member 1042Y is rotatably supported on the cap section 1034Y. Through the above described configuration, the container body 1033Y is connected with the cap section 634Y via the gear 1042Y. Further, in order to prevent the container body 1033Y from rotating on the cap section 634Y, the protrusion 1033e formed near the bottle mouth section 1033a of the container body 1033Y is fitted into a notch groove 1034i formed on the side surface of the cap section 634Y to play a role of a rotation stopper.

Further, in the cap section 634Y, a cap seal made of a foamed elastic material adheres to a section where the end surface of the gear member 1042Y (the end surface at the side opposite to the container body 1033Y side) is pressed. Thus, the toner leak from between the gear member 1042Y and the cap section 634Y can be prevented.

The agitating member 33f is attached to the inner surface of the gear member 1042Y. Further, the shaft section 1041Ya (the end section at one end side) of the conveying member 1041Y is connected to the connection section 1033/20 of the agitating member 33f as described above.

As described above, even in the present seventh embodiment, similarly to the above embodiments, the contact-type ID chip 535 (the information storage device) is held on the cap section 634Y to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads 35a1, 35a2, and 35a3 (terminals) approach and come in contact with body side terminals 573e2. Thus, even when the contact-type ID chip 535 (the information storage device) is installed in the toner container 1032Y (the removable device) installed removably on the image forming apparatus body 100, the contact failure caused by the positioning failure with the body side terminals 573e2 of the connector 573e of the image forming apparatus body 100 is difficult to occur.

Further, in the present seventh embodiment, the cap section 634Y described, for example, with reference to FIGS. 42A and 42B in the sixth embodiment has been used as the cap section of the toner container, but as the cap section of the toner container in the present seventh embodiment, the cap section 734Y described with reference to FIGS. 46A and 46B in the sixth embodiment may be used, the cap section 834Y described with reference to FIG. 47C in the sixth embodiment may be used, or the cap section 934Y described with reference to FIGS. 47A and 47B in the sixth embodiment may be used.

Further, even in the present seventh embodiment, similarly to the fifth and sixth embodiments, even when the contact-type ID chip 535 (the information storage device) is installed in the toner container (the removable device) installed removably on the image forming apparatus body 100, since the earth terminal 535d engaged with the body side earth terminal 573e25 formed in the positioning pin 573e23 (the protruding section) of the connector 573e (the image forming apparatus body 100) is formed in one hole 535b21 formed in the substrate 535b of the ID chip 535, the ID chip 535 is difficult to be electrically damaged.

Eighth Embodiment

An eighth embodiment of the present invention will be described in detail with reference to FIGS. 50 to 51B.

FIG. 50 is an exploded perspective view illustrating an image forming apparatus 1100 according to the eighth embodiment. FIG. 51A is a cross-sectional view illustrating part of a toner cartridge 1106Y installed in the image forming apparatus, and FIG. 51B is a bottom view illustrat-

ing part of the toner cartridge **1106Y**. In FIGS. **50** to **51B**, a toner discharge mechanism and a positioning mechanism for having the toner cartridge to operate are omitted.

The image forming apparatus **1100** according to the present eighth embodiment is different from those according to the above embodiments in which the toner container **532Y**, **632Y**, **732Y**, **832Y**, **932Y**, or **1032Y** in which the ID chip **535** is installed is mounted on the apparatus body **100** in the horizontal direction, where the longitudinal direction is the mounting direction in that the toner cartridge **1106Y** in which the ID chip **535** is installed is mounted on the apparatus body **1100** from above.

Referring to FIG. **50**, the image forming apparatus **1100** according to the present eighth embodiment is configured so that toner cartridges **1106Y**, **1106M**, **1106C**, and **1106K** as four removable devices are attached or detached from above. FIG. **50** illustrates the state in which the three toner cartridges **1106M**, **1106C**, and **1106K** except the yellow toner cartridge **1106Y** have been mounted on the apparatus body **1100**.

The toner cartridges **1106Y**, **1106M**, **1106C**, and **1106K** are attached to or detached from an installation section of the apparatus body **1100** in the state in which a body cover **1110** (a body door) is opened as illustrated in FIG. **50**.

Meanwhile, the toner cartridges **1106Y**, **1106M**, **1106C**, and **1106K** include an opening with a shutter that is disposed at the position of the lower side facing the developing device and store toner of corresponding color (one-component developer) therein. Referring to FIGS. **51A** and **51B**, on the lower surfaces of the end sections of the toner cartridges **1106Y**, **1106M**, **1106C**, and **1106K** in the longitudinal direction, the ID chip **535** (the information storage device) is movably held by a holding member **1134k** in the horizontal plane direction (the paper surface direction of FIG. **51B**).

The holding member **1134k** is screw-coupled to the toner cartridge **1106Y** to come in contact with part of the substrate **535b** of the ID chip **535** in the state in which the metallic pads **35a1**, **35a2**, and **35a3**, the positioning hole **535b21** (the earth terminal **535d**), and the like of the ID chip **535** that is the same as that in the fifth embodiment are exposed. In detail, the hole of the holding member **1134k** is combined with a boss section **1181** formed in the end section of the toner cartridge **1106Y**, a screw **1180** is screwed into a screw hole formed at the opposite side with the ID chip **535** interposed between the boss section **1181** of the toner cartridge **1106Y** and the hole formed in the holding member **1134k**, and the holding member **1134k** is fixed to the toner cartridge **1106Y**. Thus, the ID chip **535** does not fall from the toner cartridge **1106Y** and is held to be movable in the horizontal plane. Referring to FIG. **50**, in accompany with the mounting operation of the toner cartridge **1106Y** from above on the apparatus body **1100**, the positioning pin **573e23** (the body side earth terminal **573e25**) of the connector **573e** installed in the installation section of the apparatus body **1100** is fitted into the positioning hole **535b21** (the earth terminal **535d**) of the ID chip **535**. Thereafter, the body side terminal **73e2** of the connector **573e** comes in contact with the metallic pads **35a1**, **35a2**, and **35a3** of the ID chip **535**, and electrical contact between the connector **573e** and the ID chip **535** is completed. In this case, since the ID chip **535** in the toner cartridge **1106Y** is held to be movable in the horizontal plane, similarly to the above embodiments, the contact failure caused by the positioning failure with the body side terminals **73e2** and **573e25** of the connector **573e** of the apparatus body **1100** is difficult to occur.

As described above, in the present eighth embodiment, the contact-type ID chip **535** (the information storage device) is held on the toner cartridge **1106Y** to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads **35a1**, **35a2**, and **35a3** (terminals) approach and come in contact with body side terminals **573e2**. Thus, even when the contact-type ID chip **535** (the information storage device) is installed in the toner cartridge **1106Y** (the removable device) installed removably on the image forming apparatus body **1100**, the contact failure caused by the positioning failure with the body side terminals **73e2** of the connector **573e** of the image forming apparatus body **1100** is difficult to occur.

Further, even in the present eighth embodiment, similarly to the fifth to seventh embodiments, even when the contact-type ID chip **535** (the information storage device) is installed in the toner cartridge **1106Y** (the removable device) installed removably on the image forming apparatus body **1100**, since the earth terminal **535d** engaged with the body side earth terminal **573e25** formed in the positioning pin **573e23** (the protruding section) of the connector **573e** (the image forming apparatus body **1100**) is formed in one hole **535b21** formed in the substrate **535b** of the ID chip **535**, the ID chip **535** is difficult to be electrically damaged.

Ninth Embodiment

A ninth embodiment of the present invention will be described in detail with reference to FIGS. **52** and **53**.

FIG. **52** is a perspective view illustrating an image forming apparatus according to the ninth embodiment and is a view corresponding to FIG. **50** in the eighth embodiment. FIG. **53** is a schematic view illustrating a state in which the connector **573e** is connected to the ID chip **535** in accompany with a closing operation of a body cover **1210** of an apparatus body **1200**.

The image forming apparatus **1200** according to the present ninth embodiment is different from those according to the eighth embodiment in that the ID chip **535** is installed on an upper surface of a process cartridge **1206Y** rather than the toner cartridge, and the connector **573e** is installed in a body cover **1210** of the apparatus body **1200**.

Referring to FIG. **52**, the image forming apparatus **1200** according to the present ninth embodiment is configured so that process cartridges **1206Y**, **1206M**, **1206C**, and **1206K** as four removable devices are attached or detached from above. FIG. **52** illustrates the state in which the three process cartridges **1206M**, **1206C**, and **1206K** except the yellow process cartridge **1206Y** have been mounted on the apparatus body **1200**.

The process cartridges **1206Y**, **1206M**, **1206C**, and **1206K** are attached to or detached from an installation section of the apparatus body **1200** in the state in which the body cover **1210** (the body door) is opened as illustrated in FIG. **52**. Here, in the present ninth embodiment, in the body cover **1210**, LED units **1207Y**, **1207M**, **1207C**, and **1207K** for performing an exposure process are installed at the positions corresponding to the four process cartridges **1206Y**, **1206M**, **1206C**, and **1206K**, respectively (in FIG. **52**, the two LED units **1207Y** and **1207M** are omitted). Referring to FIG. **53**, when the body cover **1210** is closed, the LED unit **1207Y** moves to face the positioning of the photosensitive drum **1201Y** for an electrostatic latent image in the process cartridge **1206Y**.

Meanwhile, in each of the process cartridges **1206Y**, **1206M**, **1206C**, and **1206K**, the photosensitive drum, the charging unit, the developing unit, and the cleaning unit are

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integrally formed, and toner of corresponding color (one component developer) is stored inside the developing unit. Referring to FIG. 52, on the upper surfaces of the end sections of the process cartridges 1206Y, 1206M, 1206C, and 1206K in the longitudinal direction, the ID chip 535 (the information storage device) is held by the holding member (not shown) (or the face plate) to be movable in the horizontal plane direction (the vertical direction and the left-right direction in the paper plane of FIG. 53).

The holding member is screw-coupled to an outer cover of the process cartridge 1206Y to come in contact with part of the substrate 535b of the ID chip 535 in the state in which the metallic pads 35a1, 35a2, and 35a3, the positioning hole 535b21 (the earth terminal 535d), and the like of the ID chip 535 that is formed similarly to that in the fifth embodiment are exposed. Thus, the ID chip 535 does not fall from the process cartridge 1206Y and is held to be movable in the horizontal plane. Referring to FIG. 52, in accompany with the mounting operation of the process cartridge 1206Y from above on the apparatus body 1200 (the mounting operation in accompany with the closing operation of the body cover 1210), the positioning pin 573e23 (the body side earth terminal 573e25) of the connector 573e installed in the body cover 1210 is fitted into the positioning hole 535b21 (the earth terminal 535d) of the ID chip 535. Thereafter, the body side terminal 73e2 of the connector 573e comes in contact with the metallic pads 35a1, 35a2, and 35a3 of the ID chip 535, and electrical contact between the connector 573e and the ID chip 535 is completed. In this case, since the ID chip 535 in the process cartridge 1206Y is held to be movable in the horizontal plane, similarly to the above embodiments, the contact failure caused by the positioning failure with the body side terminals 573e2 and 573e25 of the connector 573e of the apparatus body 1200 is difficult to occur.

Further, in the present ninth embodiment, referring to FIG. 53, the LED unit 1207Y (having an end section in which the connector 573e is installed) is installed to be rotatable (swingable) on the body cover 1210 clockwise or counterclockwise in FIG. 53 via a support arm 1211. The LED unit 1207Y is urged by a compression spring 1212 installed inside the support arm 1211. When the body cover 1210 is closed to mount the four LED units on the process cartridges by the swing function and the urging force against the process cartridge side, as illustrated in FIG. 53, the LED unit 1207Y shakes the neck along the wall surface of the process cartridge 1206Y and is guided to a predetermined position. At the same time, the connector 573e also moves to approach the ID chip 535 and is positioned similarly to the fifth to eighth embodiments. Thus, due to the urging force of the compression spring 1212, the connector 573e comes in contact with the ID chip 535 of the process cartridge 1206Y mounted on the installation section of the apparatus body 1200 with appropriate force.

As described above, in the present ninth embodiment, the contact-type ID chip 535 (the information storage device) is held on the process cartridge 1206Y to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads 35a1, 35a2, and 35a3 (terminals) approach and come in contact with body side terminals 573e2. Thus, even when the contact-type ID chip 535 (the information storage device) is installed in the process cartridge 1206Y (the removable device) installed removably on the image forming apparatus body 1200, the contact failure caused by the positioning failure with the body side terminals 573e2 of the connector 573e of the image forming apparatus body 1200 is difficult to occur.

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Further, in the present ninth embodiment, even when the contact-type ID chip 535 (the information storage device) is installed in the process cartridge 1206Y (the removable device) installed removably on the image forming apparatus body 1200, since the earth terminal 535d engaged with the body side earth terminal 573e25 formed in the positioning pin 573e23 (the protruding section) of the connector 573e (the image forming apparatus body 1200) is formed in one hole 535b21 formed in the substrate 535b of the ID chip 535, the ID chip 535 is difficult to be electrically damaged.

Tenth Embodiment

A tenth embodiment of the present invention will be described in detail with reference to FIGS. 54 and 55.

FIG. 54 is a perspective view illustrating an ink cartridge 1306Y (a developer container) according to the tenth embodiment. FIG. 55 is a front view illustrating an inkjet printer 1300 as an image forming apparatus in which ink cartridges 1306Y, 1306M, 1306C, and 1306K are installed.

The image forming apparatus 1300 according to the present tenth embodiment is different from those according to the above embodiments in that the ink cartridge 1306Y having the side surface on which the ID chip 535 is installed is mounted on the apparatus body 1300 from the side.

Referring to FIG. 55, the image forming apparatus 1300 (the inkjet printer) according to the present tenth embodiment includes a carriage 1301 that includes recording heads 1301a and 1301b and moves in a direction of a double-headed arrow, a guide lock 1302, a supply tube 1303 that supplies ink from the ink cartridges 1306Y, 1306M, 1306C, and 1306K of respective colors to a sub tank of the carriage 1301, a conveying belt 1304 for conveying a recording medium P in a direction of an arrow, and the like. The ink cartridges 1306Y, 1306M, 1306C, and 1306K of respective colors (the removable devices) are removably installed on an installation section disposed in the end section of the apparatus body 1300 (installation having the vertical direction in FIG. 55 as the attaching/detaching direction).

Further, a main configuration of an image forming apparatus 300 is the same as stated in, for example, Japanese Patent Application Laid-open No. 2010-234801 and has been well known, and thus a detailed description thereof will not be repeated.

Referring to FIG. 54, in the ink cartridge 1306Y (an ink bag 1307 is stored therein) as the removable device, the ID chip 535 (the information storage device) held on a holding member 1334k to be movable in the XZ direction is installed on a concave section 1308 formed on the side surface thereof.

The holding member 1334k and the ID chip 535 have a configuration similar to those in the fifth embodiment. That is, the holding member 1334k is fitted into the concave section 1308 of the ink cartridge 1306Y in the state in which the metallic pads 35a1, 35a2, and 35a3, the positioning holes 535b21 (the earth terminal 535d), and the like of the ID chip 535 are exposed. Thus, the ID chip 535 does not fall from the ink cartridge 1306Y and is held on the holding member 1334k to be movable in the XZ plane. Referring to FIG. 54, in accompany with the mounting operation on the apparatus body 1300, the positioning pin 573e23 (the body side earth terminal 573e25) of the connector 573e installed in the apparatus body 1300 is fitted into the positioning hole 535b21 (the earth terminal 535d) of the ID chip 535. Thereafter, the body side terminal 73e2 of the connector 573e comes in contact with the metallic pads 35a1, 35a2, and 35a3 of the ID chip 535, and electrical contact between

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the connector 573e and the ID chip 535 is completed. In this case, since the ID chip 535 in the ink cartridge 1306Y is held to be movable in the XZ plane, similarly to the above embodiments, the contact failure caused by the positioning failure with the body side terminals 573e2 and 573e25 of the connector 573e of the apparatus body 1300 is difficult to occur.

As described above, in the present tenth embodiment, the contact-type ID chip 535 (the information storage device) is held on the ink cartridge 1306Y through the holding member 1334k to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads 35a1, 35a2, and 35a3 (terminals) approach and come in contact with body side terminals 573e2. Thus, even when the contact-type ID chip 535 (the information storage device) is installed in the ink cartridge 1306Y (the removable device) installed removably on the image forming apparatus body 1300, the contact failure caused by the positioning failure with the body side terminals 573e2 of the connector 573e of the image forming apparatus body 1300 is difficult to occur.

Further, even in the present tenth embodiment, even when the contact-type ID chip 535 (the information storage device) is installed in the ink cartridge 1306Y (the removable device) installed removably on the image forming apparatus body 1300, since the earth terminal 535d engaged with the body side earth terminal 573e25 formed in the positioning pin 573e23 (the protruding section) of the connector 573e (the image forming apparatus body 1300) is formed in one hole 535b21 formed in the substrate 535b of the ID chip 535, the ID chip 535 is difficult to be electrically damaged.

Eleventh Embodiment

An eleventh embodiment of the present invention will be described in detail with reference to FIGS. 56 to 58.

FIG. 56 is a perspective view illustrating a connector 1473e of an image forming apparatus according to the tenth embodiment and is a view corresponding to FIG. 16 according to the first embodiment. FIG. 57 is a three-plane view illustrating an ID chip 1435 as an information storage device that comes in contact with the connector 1473e of FIG. 56 and is a view corresponding to FIG. 29 according to the first embodiment. FIG. 58 is a three-plane view illustrating an ID chip 1535 as an information storage device of another form and is a view corresponding to FIG. 35 according to the fourth embodiment.

The eleventh embodiment is different from the first and fourth embodiments in that a body side earth terminal 1473e5 is installed in a positioning pin 1473e3 of the connector 1473e, and a metallic earth terminal 1435d or 1535d (an earth terminal) that comes in contact with the body side earth terminal 1473e5 is installed in the ID chip 1435 or 1535.

Referring to FIG. 56, in the image forming apparatus according to the present eleventh embodiment, similarly to the first embodiment, installed is the connector 1473e that includes a connector body 1473e1, four body side terminals 1473e2, two positioning pins 1473e3 (positioning protruding sections), a snap fit 1473e4, and the like.

In the connector 1473e according to the present eleventh embodiment, the body side terminal 1473e5 (the earth terminal) is installed inside the positioning pin 1473e3 (a section that comes in contact with a notch 1435b1 of the ID chip 1435 or a hole 1535b11).

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Meanwhile, referring to FIG. 57, in the ID chip 1435 (the substrate 1435b) according to the present eleventh embodiment, the metallic earth terminal 1435d (the earth terminal) is installed on the inner surface of the two notches 1435b1 and around the two notches 1435b1.

Through the above described configuration, in the mounting operation of the toner container, the earth terminal 1435d of the ID chip 1435 comes in contact with the body side earth terminal 1473e5 (for example, see FIG. 56) of the positioning pin 1473e3 (the connector 1473e), and then the fourth metallic pads 35a of the ID chip 1435 start to come in contact with the four body side terminals 1473e2 of the connector 1473e. That is, in the detaching operation of the toner container, contact between the four metallic pads 35a of the ID chip 1435 and the four body side terminals 1473e2 of the connector 1473e is released, and then the earth terminal 1435d of the ID chip 1435 is released from the contact state with (separated from) the body side earth terminal 1473e5 of the positioning pin 1473e3 (the connector 1473e). Specifically, the body side earth terminal 1473e5 in the connector 1473e has the contact start position closer to the ID chip 1435 side than the four body side terminals 1473e2.

Through the above described configuration, in the mounting operation of the toner container, the metallic pads 35a always start to be connected with the body side terminals 1473e2 in the state in which the ID chip 1435 is earthed, and in the detaching operation of the toner container, the metallic pads 35a always start to be separated from (released from the contact state with) the body side terminals 1473e2 in the state in which the ID chip 1435 is earthed. Thus, an electric circuit at the ID chip 1435 is prevented from being not earthed and so becoming an electrically floating state, and so the ID chip 1435 is difficult to be electrically damaged.

Further, similarly to a relation between the ID chip according to the first embodiment and the ID chip according to the fourth embodiment, the ID chip 1435 illustrated in FIG. 57 may be replaced with the ID chip 1535 illustrated in FIG. 58.

In detail, referring to FIG. 58, in the ID chip 1535, a metallic earth terminal 1535d (an earth terminal) is installed on the inner surface of one positioning hole 1535b11 and around the positioning hole 1535b11.

Through the above described configuration, in the mounting operation of the toner container, the earth terminal 1535d of the ID chip 1535 comes in contact with the body side earth terminal 1473e5 (for example, see FIG. 56) of the positioning pin 1473e3 (the connector 1473e), and then the four metallic pads 35a of the ID chip 1535 start to come in contact with the four body side terminals 1473e2 of the connector 1473e. That is, in the detaching operation of the toner container, contact between the four metallic pads 35a of the ID chip 1535 and the four body side terminals 1473e2 of the connector 1473e is released, and then the earth terminal 1535d of the ID chip 1535 is released from the contact state with (separated from) the body side earth terminal 1473e5 of the positioning pin 1473e3 (the connector 1473e). Specifically, the body side earth terminal 1473e5 in the connector 1473e has the contact start position closer to the ID chip 1535 side than the four body side terminals 1473e2.

Through the above described configuration, in the mounting operation of the toner container, the metallic pads 35a always start to be connected with the body side terminals 1473e2 in the state in which the ID chip 1535 is earthed, and in the detaching operation of the toner container, the metallic pads 35a always start to be separated from (released from

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the contact state with) the body side terminals **1473e2** in the state in which the ID chip **1535** is earthed. Thus, an electric circuit at the ID chip **1535** is prevented from being not earthed and so becoming an electrically floating state, and so the ID chip **1535** is difficult to be electrically damaged.

As described above, even in the present eleventh embodiment, the contact-type ID chip **1435** or **1535** (the information storage device) is held on the holding member **34k** (the holding section) to be movable on the virtual plane that is substantially orthogonal to the movement direction in which the metallic pads **35a** (terminals) approach and come in contact with the body side terminals **1473e2**.

Thus, even when the contact-type ID chip **1435** or **1535** (the information storage device) is installed in the toner container (the removable device) installed removably on the image forming apparatus body, the contact failure caused by the positioning failure with the body side terminals **1473e2** of the connector **1473e** of the image forming apparatus body is difficult to occur.

As described above, in the present eleventh embodiment, even when the contact-type ID chip **1435** or **1535** (the information storage device) is installed in the toner container (the removable device) installed removably on the image forming apparatus body **100**, since the earth terminal **1435d** or **1535d** engaged with the body side earth terminal **1473e5** formed in the positioning pin **1473e3** (the protruding section) of the connector **1473e** (the image forming apparatus body **100**) is formed in the notch **1435b1** or the hole **1535b11** formed in the substrate **1435b** or **1535b** of the ID chip **1435** or **1535**, the ID chip **1435** or **1535** is difficult to be electrically damaged.

Twelfth Embodiment

A twelfth embodiment of the present invention will be described in detail with reference to FIGS. **59** to **62**.

FIG. **59** is a perspective view illustrating a toner container **1632Y** as a removable device according to the twelfth embodiment. The toner container **1632Y** includes a container body **1633Y** having the same configuration as the container body **33Y**, a cap section **1634Y** that covers a toner discharge opening (not shown) formed in the container body **1633Y** from the outer side, an ID chip as an information storage device attached to the leading end of the cap section **1634Y**, and a holding mechanism **1635** that holds the ID chip. For example, the ID chip **535** described in the fifth embodiment may be used as the ID chip.

The toner container **1632Y** relates to a toner container attachable to and detachable from a toner feeding device of a toner suction conveying type disclosed in Japanese Patent No. 4396946 or U.S. Pat. No. 7,835,675. That is, except for the ID chip, the holding mechanism, and a communication method of the ID chip, the toner container and the toner feeding device disclosed in the relevant patent are employed. The relevant patent is referred to in connection with a positioning configuration, which allows attachment and detachment, disposed in both the toner container and the feeding device, a configuration for driving the container body, and the like. The difference between the toner container of the present embodiment and the toner container of Japanese Patent No. 4396946 or U.S. Pat. No. 7,835,675 will be described later. The toner feeding device of the present embodiment is different from the toner feeding device of Japanese Patent No. 4396946 or U.S. Pat. No. 7,835,675 in that the former employs a contact type communication method, whereas the latter employs a non-contact type communication method (a so-called RFID method). Thus, as

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the body side connector of the former, the connector **573e** of FIGS. **37**, **38**, and **45** described with reference to the fifth embodiment is disposed at a position facing the toner container cap end surface of the toner feeding device of Japanese Patent No. 4396946 or U.S. Pat. No. 7,835,675.

As illustrated in FIG. **60**, the positioning hole **535b21** described above is formed in the ID chip **535**, and, for example, the positioning pin **573e23** of the connector installed in the apparatus body described above is inserted into the positioning hole **535b21**.

The holding mechanism **1635** includes a holding section **1635A** that holds the ID chip **535** in a movable manner in the XZ direction and a holding cover **1635B** as a cover member that is removably fitted into the holding section **1635A**.

As illustrated in FIG. **61**, the holding section **1635A** includes a concave section **1635Aa** formed on an ID chip mounting surface **1634Ya** that is vertically flat and formed at the leading end of the cap section **1634Y**, a pedestal section **1635g**, formed in the concave section **1635Aa**, in which the ID chip **535** is installed, and an ID chip installation wall section **1635Ab** of a substantially frame shape formed to surround the concave section **1635Aa** and the pedestal section **1635g** from the outer side. The ID chip installation wall section **1635Ab** is formed to protrude outward from the ID chip mounting surface **1634Ya** further than the pedestal section **1635g**. The ID chip installation wall section **1635Ab** has a size capable of storing the ID chip **535** having an outward rectangular form and holds the ID chip **535** in a movable manner in the XZ direction when the ID chip **535** is placed. That is, the ID chip **535** is installed in the pedestal section **1635g** but not fixed to the cap section **1634Y**. When installed in the pedestal section **1635g**, the ID chip **535** is installed with a clearance with the ID chip installation wall section **1635Ab** that is formed to surround the ID chip **535** from the outer side.

On the ID chip mounting surfaces **1634Ya**, positioning bosses **1615a** and **1615b** for mounting the holding cover **1635B** are formed to protrude from the ID chip mounting surface **1634Ya**. The positioning bosses **1615a** and **1615b** are integrally formed with the cap section **1634Y** by resin.

The holding cover **1635B** is mounted on and fixed to the holding section **1635A** by a melt-fixing method (for example, heat calking) described below, with the ID chip being disposed in the holding section **1635A**. A central section of the holding cover **1635B** is provided with an opening **1635Bc** that allows a contact point (not shown) and the positioning hole **535b21** of the ID chip **535** to be exposed to the outside and allows the connector terminal (not shown) of the connector and the positioning pin **573e23** to be inserted therethrough. The holding cover **1635B** is configured to sandwich the ID chip **535** set inside the ID chip installation wall section **1635Ab** together with the ID chip installation wall section **1635Ab** so that the ID chip **535** does not separate. Above and below the opening **1635Bc** of the holding cover **1635B**, mounting holes **1635Ba** and **1635Bb** are formed at positions corresponding to the positioning bosses **1615a** and **1615b**.

In this configuration, when mounting the ID chip **535** on the cap section **1634Y**, the back surface of the ID chip **535** comes in contact with the pedestal section **1635g** so that its position in a depth direction is determined. Along this, up, down, left and right positioning is done by the surrounding thanks to the ID chip installation wall section **1635Ab**. The holding cover **1635B** is superimposed on the ID chip installation wall section **1635Ab** in a direction facing the ID chip installation wall section **1635Ab**, and the positioning bosses **1615a** and **1615b** are inserted into the mounting holes

1635Ba and 1635Bb. Thus, the ID chip 535 is positioned in a state covered by the holding cover 1635B, and mounted and held on the cap section 1634Y. In this state, the ID chip 535 is installed on the ID chip mounting surface 1634Ya of the cap section 1634Y but is not fixed directly to the cap section 1634Y. That is, the ID chip 535 is mounted to the cap section 1634Y through the ID chip installation wall section 1635Ab formed on the ID chip mounting surface 1634Ya.

The present embodiment features a fixing method between the holding cover 1635B and the cap section 1634Y. In the present embodiment, a melt-fixing method is employed as a fixing method of the holding cover 1635B and the cap section 1634Y.

Since the holding cover 1635B is held such that the positioning bosses 1615a and 1615b formed at the cap section 1634Y side are inserted into the mounting holes 1635Ba and 1635Bb as described above, in the present embodiment, as illustrated in FIG. 62, the holding cover 1635B is fixed to the cap section 1634Y by heat calking. In FIG. 62, a reference numeral 1640 represents a calking section (a fixing section). For example, the positioning bosses 1615a and 1615b illustrated in FIG. 61 have the size protruding from the mounting holes 1635Ba and 1635Bb. The positioning bosses 1615a and 1615b are heated by a heating member such as a heatable iron, and the calking section 1640 is formed by crushing and thermally deforming the bosses while melting them by heat. Accordingly, the holding cover 1635B can be fastened to and fixed to the cap section 1634Y.

In the present embodiment, as the fixing method between the holding cover 1635B and the cap section 1634Y, fixing by heat calking has been described, but as the fixing method between the holding cover 1635B and the cap section 1634Y, another melt-fixing method such as ultrasonic welding may be used, and a resin melting method not limited to the present embodiment.

Thirteenth Embodiment

A thirteenth embodiment will be described in detail with reference to FIGS. 63 and 64.

In the present embodiment, the holding cover 1635B is not fixed by a process such as heat calking but fixed by a fastening method using a fastening member. The remaining sections of the toner container and the form of the toner feeding device are the same as in the twelfth embodiment. In present embodiment, the holding cover 1635B is fixed to the cap section 1634Y such that fastening members 1650a and 1650b are inserted into the mounting holes 1635Ba and 1635Bb formed in the holding cover 1635B that allows the positioning bosses 1615a and 1615b to be inserted into and screwed into the ID chip mounting surface 1634Ya. For example, when the screw fixing is performed using a self-tap screw that creates a screw groove in an opposing hole at the same time when it is screwed into the opposing hole as the fastening members 1650a and 1650b, all you have to do is to form a tubular pilot hole (corresponding to 1651a and 1651b of FIG. 64) in the ID chip mounting surface 1634Ya.

As another embodiment, for example, there is a case in which the holding cover 1635B is fixed to the cap 1634Y without the ID chip being held therein and then the resultant product is shipped from a toner container manufacturing factory, and then in another factory, the holding cover 1635B is removed, the ID chip is set inside, and the holding cover 1635B is fixed again to the cap 1634Y, or there is a case of recycling the used toner container. In this case, if attachment/detachment of the holding cover 1635B is repeated

within a certain range, the above described tubular pilot hole is preferable. However, if five or six times, or more times of attachments/detachments are expected and the stability of fastening force on each occasion should be considered, it is preferable that screw holes 1651a and 1651b are formed in advance in the ID chip mounting surface 1634Ya, and fixing is performed by screwing screws into the screw holes 1651a and 1651b through the mounting holes 1635Ba and 1635Bb so as to correspond to the pitch of the screw holes 1651a and 1651b as fastening members 1650a and 1650b, as illustrated in FIG. 64. In the present embodiment, the fastening members 1650a and 1650b are fixed at two positions below and above the holding cover 1635B but may be fixed at one position, or more positions than the above. Further, the fastening members 1650a and 1650b may be mounted at the left and right sides rather than the upper and lower sides of the holding cover 1635B, and are not limited in terms of number and position to the present embodiment.

Fourteenth Embodiment

A fourteenth embodiment of the present invention will be described in detail with reference to FIGS. 65 and 66.

In the present embodiment, the holding cover is characterized in that it is not fastened to and fixed to the cap section 1634Y by the process such as heat calking or the fastening member but fixed by a fitting method using a claw member. The remaining sections of the toner container and the form of the toner feeding device are the same as those in the twelfth embodiment.

A holding cover 1635C according to the present embodiment basically has the same function as the holding cover 1635B. Specifically, the mounting holes 1635Ba and 1635Bb are eliminated from the holding cover 1635B, and instead hook sections 1636a and 1636b that pass through up to an opening 1635Cc formed at a central section are formed in an upper section 1635Ca and a lower section 1635Cb. Like the opening 1635Bc, the opening 1635Cc allows the contact point (not shown) and the positioning hole 535b21 of the ID chip 535 to be exposed to the outside and allows the connector terminal (not shown) of the connector and the positioning pin 573e23 to be pass therethrough.

In the present embodiment, the cap section 1634Y, as illustrated in FIG. 66, is provided with claw sections 1637a and 1637b as engaging sections that enter the inside of the hook sections 1636a and 1636b and engage with the hook sections 1636a and 1636b. In the present embodiment, the claw sections 1637a and 1637b are formed to be disposed respectively on the upper section and the lower section of the ID chip installation wall section 1635Ab covered with the holding cover 1635C. The claw sections 1637a and 1637b have inclined surfaces 1637a1 and 1637b1 formed at the insertion side and are configured so as to guide the holding cover 1635C to the tops of the claw sections 1637a when the holding cover 1635C is aligned and mounted.

With this configuration, when the ID chip 535 is set in the ID chip installation wall section 1635Ab and the holding cover 1635C is moved toward the ID chip installation wall section 1635Ab so as to be superimposed on the ID chip installation wall section 1635Ab, the claw section 1637a formed in the ID chip installation wall section 1635Ab that becomes the cap section 1634Y side enters the inside of the hook sections 1636a and 1636b formed in the holding cover 1635C, and the holding cover 1635C can be fixed to the cap section 1634Y by fitting between both sides.

In the present embodiment, the hook sections 1636a and 1636b are fitted into the claw sections 1637a and 1637b at

the two positions of the upper and lower sections of the holding cover **1635C**, but they may be fitted at the left and right sections or at the upper, lower, left, and right sections of the holding cover **1635C** instead of the upper and lower sections. The fitting position and number are not limited to the present embodiment.

In the thirteenth and fourteenth embodiments, the method of fastening or fitting the holding cover **1635C** to be attached to or detached from the cap section **1634Y** has been described. However, as another fixing method, for example, the cover member may be fixed to the ID chip installation wall section **1635Ab** by an adhesive. In this case, the cover member preferably has adhesive force sufficient not to fall off at the time of detachment of the toner container **1632Y** from the apparatus body, and a kind of an adhesive and an adhesion area are not particularly limited.

In the twelfth to fourteenth embodiments, even in any embodiment, even when the ID chip **535** that is the contact-type information storage device is installed in the toner container **1632Y**, the contact failure caused by the unsatisfactory positioning relative the terminal of the connector of the apparatus body is difficult to occur.

Fifteenth Embodiment

In the toner container according to any one of the first to seventh embodiments, part of the inventive (the shutter mechanism) for solving the above described third problem will be described in detail once again as a fifteenth embodiment.

The stopper release urging section **72b** in FIG. **5** will be described with reference to FIG. **18** and the subsequent figures. The stopper release urging section **72b** is a section used to open the toner discharge opening **W** by displacing the shutter **34d** disposed in the cap section **34Y** from the closed state to the open state in conjunction with the insertion (mounting) operation of the developer storage containers **32Y**, **32M**, **32C**, and **32K**. The stopper release urging section **72b** is configured with a trapezoidal rib that protrudes upward from the upper surface of the bottle receiving surface **72a** toward the shutter.

Meanwhile, FIGS. **18** and **20** illustrate the entire configuration of the developer storage containers **32Y**, **32M**, **32C**, and **32K** (see FIG. **18**) and the details of the cap section **34Y** disposed in the container (see FIG. **20**).

In FIG. **18**, the developer storage container **32Y** mainly includes the container body **33Y** (the bottle body) and the cap section **34Y** (the bottle cap) disposed at the head thereof. Further, the ID chip **35** as the information storage device or the like is detachably installed in the cap section **34Y** of the developer storage container **32Y**.

Among the sections described above, the configuration illustrated in FIG. **20** is used at the position where the ID chip **35** is installed so that the ID chip **35** can be mounted.

On the leading end surface of the cap section **34Y**, as illustrated in FIG. **20**, the first and second positioning holes **34a** and **34b** that can be engaged with the first and second positioning pins (not shown) disposed in the cap receiving section **73** are disposed at the two positions in the longitudinal direction (the vertical direction).

Between the first and second positioning holes **34a** and **34b**, formed is a rectangular concave section **34t** that has a shape connectable with the connector disposed at the developer storage container storage unit **70** (see FIG. **5**) and extends in the vertical direction as illustrated in FIGS. **67** and **68**. Inside the concave section, as illustrated in FIG. **67**, the holding member **34k** to which the ID chip is attachable

is mounted. A reference numeral **33f** illustrated in FIG. **67** represents the agitating member having an agitating section positioned inside the cap, and the agitating member rotates in conjunction with the gear **33c** which will be described later.

The mounting position of the holding member **34k** is vertically higher than the position of the toner discharge opening **W** that is opened or closed by the shutter **34d** which will be described later with reference to FIGS. **70A** to **70C** (in FIG. **67**, for convenience, the position having the height **H** between a bottom section **34t1** of the concave section **34t** and the toner discharge opening **W**), and thus the holding member **34k** is separated from the toner discharge opening **W**. Further, a convex wall is disposed at a circumferential edge of the rectangular concave section. Thus, obtained is the state in which part of the concave section **34t** is difficult to be superimposed on part of the toner discharge opening **W** in the transverse direction. That is, the bottom section **34t1** of the concave section **34t** does not get close to the toner discharge opening **W**. Thus, part of the toner discharge opening **W** is prevented from being filled with the bottom section **34t1**, and discharging of the toner is not inhibited. Further, even when the toner leaks and is scattered from the toner discharge opening **W** of the developer storage container **32Y** to the outside, the scattered toner does not reach the connector against its own weight, and the scattered toner is blocked by the convex wall. Thus, the contact failure caused when the toner sticks to the connector can be prevented, and the occurrence of the communication failure can be prevented. The concave section **34t** is disposed at the first positioning hole **34a** side.

Meanwhile, in the head section of the container body **33Y** illustrated in FIG. **20**, as illustrated in FIG. **67**, the gear **33c** integrally rotating together with the container body **33Y** and the opening **A** are disposed at one end side in the longitudinal direction (the left-right direction in FIG. **67**).

The opening **A** is disposed on the head section positioned at the front side when the container body **33Y** is mounted and allows the toner stored in the container body **33Y** to be discharged toward a hollow space section **B** inside the cap section **34Y**.

Further, as the toner is consumed at the image forming apparatus body side, toner conveyance (rotation driving of the container body **33Y**) from the inside of the container body **33Y** to the hollow space **B** inside the cap section **34Y** is appropriately performed.

Next, the configuration of the cap section **34Y** of the developer storage container **32Y** will be described below with reference to FIGS. **20**, **67**, and **68**.

In the cap section **34Y** of the developer storage container **32Y**, installed are the ID chip **35** (the information storage device), the shutter member **34d**, and the shutter seal **36**.

As illustrated in FIG. **68**, the cap section **34Y** has a structure in which roughly a cylindrical body in which the outer diameter and the inner diameter decreases from the container body **33Y** side toward the shutter member **34d** side in three stages (large, medium, and small) is combined with a box section, disposed at the bottom, in which the width in the horizontal direction decreases in two stages (wide width and narrow width). The cap section **34Y** includes an insertion section including the large diameter section and the medium diameter section of the cylindrical section and the wide width box section **34n**.

In the large diameter section of the cap section **34Y**, a cut-out hole **34p0** formed such that part of the outer circumference is removed is disposed, and as illustrated in FIG. **68**, part of the teeth of the gear **33c** is exposed to the outside.

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In the insertion section, in FIG. 68, a circumferential section 34P1 adjacent to the cut-out hole 34P0 in the shaft direction has the outer diameter smaller than a circumferential section 34P2 that is not adjacent to the cut-out hole 34P0 in the circumferential direction. In FIG. 68, for convenience, D1 and D2 representing the outer diameters are attached to the reference numerals of the circumferential sections 34P1 and 34P2, and the relation between the outer diameters is $D1 < D2$.

As described above, the circumferential section adjacent to the cut-out hole 34P0 of the insertion section in the shaft direction has the outer diameter smaller than other sections, and thus the teeth surface of the gear engaged with the gear 33c, which is exposed to the outside through the cut-out hole 34P0, in the shaft direction becomes difficult to interfere with the insertion section outer circumference. As a result, the engagement operation of the gear 33c with the gear moving in the shaft direction can be smoothly performed without being disturbed by part of the insertion section.

Further, in FIG. 68 and FIG. 81 illustrating a seventeenth embodiment, which will be described later, that is a modification in which a main part is shared with the configuration illustrated in FIG. 68, a reference numeral 34YG0 represents a retaining section configured by a step section at the leading end side of a guide rail 34YG. The retaining section 34YG0 is a section that is hit by a slide protruding section 34d1c (see FIGS. 70A to 70C) disposed at the shutter 34d side so that the shutter 34d cannot move forward further, thereby retaining the shutter 34d as will be described later.

As illustrated in detail in FIG. 81, an upper rail rib 34SG that is at a predetermined distance from the guide rail 34YG and parallel to the guide rail 34YG is disposed above the guide rail 34YG. The upper rail rib 34SG prevents a sandwiching section of a body side shutter closing mechanism 73d (see FIG. 72) illustrated in FIG. 72 and drawings subsequent thereto from entering between the cylindrical circumferential surface of the cap 34Y and the guide rail 34YG.

On the upper surface of the guide rail 34YG, a shutter protruding section 34YG2 including a protruding section is disposed at the position that the shutter 34d reaches before hitting the retaining section 34YG0 (see FIG. 81). The shutter protruding section 34YG2 is used as a section for restricting movement of the shutter 34d when the shutter 34d is in the closed state.

An insertion/removal (attachment/detachment) operation of the developer storage container 33Y can be performed by the user gripping the gripping section disposed on the rear side end section of the container body 33Y in the insertion (mounting) direction as indicated by a reference numeral 33d in FIG. 18.

A narrow width box section 34Y1 is formed in the small diameter cylindrical section of the cap section 34Y, and inside the box section 34Y1, as illustrated in FIG. 69B, the toner discharge opening W for discharging (falling by its own weight) the toner, discharged from the opening A of the container body 33Y to the lower side in vertical direction, that is, to the container outside is disposed to communicate with the hollow space section B illustrated in FIG. 67.

As illustrated in FIG. 69B, the toner discharge opening W is formed in a hexagonal shape that is one of polygonal shapes and has a predetermined flow passage area and communicates the lower side circumferential surface of the space B inside the small diameter cylindrical section with the toner discharge opening W (discharge opening). Thus, the toner discharged to the space B inside the small diameter cylindrical section of the cap section 34Y from the opening

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A of the container body 33Y falls from the toner discharge opening W of the hexagonal cylindrical shape by its own weight and then is smoothly discharged to the container outside (the toner tank section 61Y).

In the toner discharge opening W, as illustrated in FIGS. 67 and 69B, a rib W1 protruding toward a seal material 36 of the shutter 34d which will be described later is formed along the opening circumferential edge. The rib W1 has a function of folding and riding up the end section of the seal material 36 which will be described later, a function of improving adhesion of the seal material 36 by coming in press contact with a section other than the end section, and a function of damming the toner that is about to leak from the toner discharge opening W.

In FIGS. 69A and 69B, in the bottom section of the narrow width box section 34Y1 disposed in the lower section of the cap section 34Y, the shutter 34d for performing opening and closing of the toner discharge opening W in conjunction with the attaching/detaching operation of the developer storage container 32Y on the developer storage container storage unit 70 is held to be slidably movable.

The shutter 34d is a feature section of the present invention and has the following configuration which will be described with reference to FIGS. 70A and 70B. In addition, FIG. 70A is a perspective view in which the shutter 34d is viewed from the bottom surface side, and FIG. 70B is a perspective view in which the shutter 34d is viewed from the top surface.

The shutter 34d is made of a resin material such as polystyrene and mainly includes a plate-like main shutter section 34d1 and a shutter deforming section 34d2 that protrudes the main shutter section 34d1, is thinner in thickness than the main shutter section 34d1, and has elasticity.

In the main shutter section 34d1, vertical wall 34d1a standing at both side end sections of a plate section and a pair of shutter sliders 34d12 having a protruding objects protruding from the vertical walls are disposed.

The vertical walls 34d1a includes a pair of slide protruding sections 34d1c that are disposed at the inner side surfaces of the vertical walls to protrude facing each other and L-shaped engaged protruding sections 34d1b that are disposed on the outer side surfaces at the side opposite to the slide protruding sections 34d1c.

The engaged protruding section 34d1b is shaped such that a plate section extending in the shutter moving direction is present on the upper surface, and a protrusion 34d1b1 engaged with a sandwiching section which will be described later extends downward from a section positioned in the front side of the plate section in an insertion direction of the developer storage container.

The shutter slider 34d12 includes a pair of prismatic sections that is disposed to protrude from the surface of the same side as the engaged protruding section 34d1b of the vertical wall 34d1a and extends toward the rear side in the direction of closing the toner discharge opening W of the shutter 34d indicated by an arrow.

In the present embodiment, as illustrated in FIG. 70B, the protrusion 34d1b1 disposed in the engaged protruding section 34d1b is disposed at the position offset from the front end surface of the main shutter section 34d1 (the position where a section corresponding to a distance indicated by a symbol S1 in FIG. 70B is removed). As will be described in FIG. 72 and drawings subsequent thereto, the protrusion 34d1b1 is used as a section for preventing interference when one of sandwiching sections 73d2 (see FIG. 72) disposed in a body side shutter closing mechanism 73d starts to turn.

In the shutter **34d**, the shutter deforming section **34d2** is configured in a cantilever shape, and an inner side angular section of a base station connected to the main shutter section **34d1** is formed of an arc-like curvature-shaped section (a shape indicated by symbol R in FIGS. **70A** and **70B**) and functions to avoid stress concentration when deflectively deformed.

Further, the shutter deforming section **34d2** is formed such that part of the base end positioned at the main shutter section **34d1** side becomes a horizontal surface (a section indicated by symbol S2 in FIG. **70C**), and the remaining section is inclined from the leading end of the horizontal surface as illustrated in FIG. **70C** in which the engaged protruding section **34d1b** is omitted. In this configuration, unlike when the inclined base end of the shutter deforming section **34d2** is directly connected with the main shutter section **34d1**, it is possible to avoid stress from being concentrated at the connection position between the inclined base end of the shutter deforming section **34d2** and the main shutter section **34d1** when the base end side of the shutter deforming section **34d2** oscillates.

The shutter deforming section **34d2** is configured with a cantilever shaped piece section (a section having the length indicated by symbol L in FIG. **70A**) extending to the rear side in the insertion direction of the developer storage container as the base end of the main shutter section **34d1**. The shutter deforming section **34d2** is inclined such that it goes downward from the base end side to the rear side in the insertion direction.

The free ends of the shutter deforming section **34d2** are integrated by a connecting plate section **34d2a** that bridges them laterally. In the central section of the connecting plate section **34d2a** in the bridging direction, a stopper release section **34d21** is disposed to face the stopper release urging section **72b** (see FIG. **5**) that is configured with a trapezoidal rib disposed at the cap receiving section **73** side. On both sides in the bridging direction, as will be described later, disposed are stopper sections **34d22** for fixing the shutter **34d** so as to prevent careless opening of the toner discharge opening W.

The stopper release section **34d21** is formed to have a triangular cross section. By running on the stopper release urging section **72b** (see FIG. **5**) disposed at the cap receiving section **73** side, the stopper release section **34d21** changes the shutter deforming section **34d2** from an inclined state to a horizontal state, so that engagement between the stopper section **34d22** and an engaging end surface **34n1** (see FIGS. **69A** and **69B**) positioned in the wide width box section **34n** that is present on the bottom of the cap section **34Y** can be released. Thus, the shutter **34d** can move in the direction of opening or closing the toner discharge opening W.

The engaging end surface **34n1** positioned in the wide width box section **34n** is disposed as a section for restricting movement of the shutter **34d** in the direction of opening the toner discharge opening W in the state in which the toner discharge opening W is closed.

FIGS. **71B** and **71C** are views for explaining the relation between the engaging end surface **34n1** and the stopper section **34d22** at the shutter deforming section **34d2** side. At the time of closing of the toner discharge opening W illustrated in FIG. **71C**, since the shutter deforming section **34d2** at the shutter **34d** side in the inclined state as the initial state, the stopper section **34d22** positioned at the inclined free end faces the engaging end surface **34n1**, and the shutter **34d** cannot move independently. Thus, the state in which the toner discharge opening W is not carelessly opened is maintained.

Further, as illustrated in FIGS. **69B** and **71B**, when the shutter **34d** has moved in the direction of opening the toner discharge opening W, the front end **34d1d** of the main shutter section **34d1** in the moving direction comes in contact with the engaging end surface **34n1**, and thus the moving position of the main shutter section **34d1** can be specified. FIG. **71C** illustrates the case in which the shutter **34d** has moved in the direction of closing the toner discharge opening W. In this case, the free end section of the shutter deforming section **34d2** becomes inclined, and so the stopper section **34d22** positioned at the free end section comes in face contact with the engaging end surface **34n1**. Thus, movement of the shutter **34d** is stopped unless the stopper release section **34d21** is pushed up.

The seal material **36** is composed of a rectangular parallelepiped body attached to the main shutter section **34d1**. As the seal material **36** hits against the rib W1 illustrated in FIG. **67**, the end section is folded and rides up, and a section other than the end section comes into press contact with the rib W1 and thus is deflectively deformed toward the toner discharge opening W in the fractional contact state. The seal material **36** is an elastic seal made of a flexible material. As the material, a high-density microcell urethane sheet is employed in view of sliding property and elasticity of the surface.

The seal material **36** has the length (the length indicated by symbol L1 in FIG. **70A**) at which the leading end in the direction of closing the toner discharge opening W by the shutter **34d** protrudes to the outer side further than the leading end of the main shutter section **34d1**. The protruding leading end section is a section that easily rides up when hitting against the rib W1 disposed at the circumferential edge of the toner discharge opening W.

The shutter **34d** is stored inside the wide width box section **34n** positioned in the lower portion of the large diameter cylindrical section of the cap section **34Y** and slidably moves.

In the wide width box section **34n**, among four wall surfaces disposed at the side surface, two wall surfaces facing in the longitudinal direction (the shaft direction of the cap section cylinder) are opened. Particularly, since the wall surface is partially left on the corner at the bottom side, an opening extending in the horizontal direction is formed on most of the wall surface at the toner discharge opening W side. The opening is formed such that two surfaces of the side surface and the bottom surface at the toner discharge opening W side in the longitudinal direction of the wide width box section **34n** are cut out.

Meanwhile, in FIGS. **20**, **68**, **69A**, **69B**, and **71A**, lateral protrusions **34c** for restricting the posture of the cap section **34Y** in the rotation direction in the image forming apparatus body **100** (the cap receiving section **73**) are formed on both side sections of the cap section **34Y**, respectively.

The lateral protrusions **34c** are positioned on both sides in a right angle direction in the same plane as a column direction of the positioning holes **34a** and **34b** in the circumferential surface of the medium diameter cylindrical section, has a triangular shape in a planar view, and has the top section at the position away from the circumferential surface of the medium diameter cylindrical section from the head section of the cap section **34Y** to the rear side.

In the inclined surface of the lateral protrusion **34c**, a rising edge angle of an inclined piece positioned behind the top section is larger than a rising edge angle of an inclined piece positioned at the head section side of the cap section **34Y** ahead of the top section.

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The inclined piece at the head section is disposed at the cap receiving section 73 side and can move while contacting the lateral protrusions 34c with a thrusting member (not shown) that is subject to tucking behavior by urging of elastic force. That is, if a section having a small inclined angle, so-called inclined plane, faces the thrusting member when thrusting toward the thrusting member, the inclined plane can enter with respect to the thrusting member without any resistance. If the top section of the inclined plane goes beyond the thrusting member, the inclined surface at the rear side is engaged with the thrusting member, movement resistance from the thrusting member abruptly decreases directly after going beyond the thrusting member, and a feeling of resistance when fitted into the thrusting member, so-called click feeling, is caused.

In the present embodiment, of the inclined pieces of the later protrusion, the angle of the inclined piece at the head section side is set to 30°, and the angle of the inclined piece at the rear side is set to 45°.

In FIGS. 20, 68, 69A, 69B, and 71A to 71C, reference numerals 34g and 34h are convex sections that are disposed on both ends of the bottom section of the cap section 34Y and are for securing incompatibility of the developer storage container 32Y (the developer storage container).

The convex sections 34g and 34h are sections for judging whether or not the mounting operation of the developer storage container 32Y on the developer storage container storage unit 70 is correct. If a fitting state on a fitting section (not shown) disposed at the developer storage container storage unit 70 side is normal, the developer storage container, in which toner of predetermined color is stored, specified at a predetermined position is mounted at that position, and it is judged that it has been correctly mounted. Thus, it is possible to prevent an erroneous operation, so-called erroneous setting, in which color of toner stored in the developer storage container is not mounted in a predetermined mounting section.

Meanwhile, the shutter 34d can be maintained in the state in which the toner discharge opening W is closed by the body side shutter closing mechanism 73d illustrated in FIGS. 72 to 75D. The body side shutter closing mechanism 73d is disposed to solve a problem in that the toner container 32Y is extracted from the apparatus body 100 while the toner discharge opening W is not completely closed, for example, at the time of replacement of the developer storage container.

In FIG. 72, the body side shutter closing mechanism 73d (a shutter sandwiching mechanism) is disposed at the bottom section inside the cap receiving section 73 and at the upstream side of the toner discharge opening W in the mounting direction of the developer storage container 32Y.

In FIG. 72, the body side shutter closing mechanism 73d is a pair of horseshoe shaped members disposed to face each other in the left-right direction of FIG. 72 and is configured to be rotatable on a support shaft 73d3 in which a torsion coil spring is installed.

The body side shutter closing mechanism 73d (the shutter sandwiching mechanism) includes a first sandwiching section 73d1 formed on one end side and a second sandwiching section 73d2 formed on the other end side.

In the sandwiching sections, at the time of the opening/closing operation of the shutter 34d in the developer storage container 32Y, the engaged protruding section 34d1b of the shutter 34d is sandwiched by the second sandwiching members 73d2, and a vertical surface (the surface where an outgoing line leading end section of symbol 34YG in FIG. 73 is positioned) of the guide rail 34YG (see FIGS. 68, 69A,

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69B, and 71A to 71C) of the cap section 34Y is sandwiched by the first sandwiching members 73d1 (the state illustrated in FIG. 73). At the time of the opening/closing operation of the shutter 34d, the postures of the shutter 34d of the cap receiving section 73 and the cap section 34Y are decided, and thus the opening/closing operation can be smoothly performed.

FIGS. 72 to 74 are views illustrating an operation of the body side shutter closing mechanism 73d (the shutter sandwiching mechanism) when opening or closing the shutter 34d.

At the time of the opening operation of the shutter 34d, as illustrated in FIG. 72, in accompany with the mounting operation of the developer storage container 32Y in a white arrow direction, the first sandwiching members 73d1 first come in contact with a leading end 34YG1 (see FIGS. 68, 69A, 69B, and 71A to 71C) of the guide rail 34YG of the shutter 34d, and then, as will be described later, the second sandwiching members 73d2 come in contact with the protrusions 34d1b1 positioned in the engaged protruding sections 34d1b of the shutter 34d.

As illustrated in FIG. 73, when the mounting operation of the developer storage container 32Y proceeds in the white arrow direction, the body side shutter closing mechanism 73d (the shutter sandwiching mechanism) rotates on a support shaft section 73d3.

When the body side shutter closing mechanism 73d rotates, the first sandwiching members 73d1 sandwich the vertical surfaces (the surfaces where an outgoing line leading end section of symbol 34YG in FIG. 73 is positioned) of the guide rails 34YG of the cap section 34Y, and the second sandwiching members 73d2 sandwich the side wall surfaces by coming in face contact with the side wall surfaces of the main shutter section 34d1 where the base ends of the engaged protrusion 34d1b are positioned while being engaged with the protrusions 34d1b1 positioned in the engaged protruding section 34d1b of the shutter 34d.

Thereafter, even though not shown, the shutter 34d comes in contact with the wall section formed around the toner feeding opening at the cap receiving section 73 side and so stops movement in the mounting direction. Then, the vertical surface of the guide rail 34YG is sandwiched by the first sandwiching sections 73d1, and movement of the shutter 34d in the cap receiving section 73 is restricted (the shutter 34d does not absolutely move in the longitudinal direction).

In the state in which movement of the shutter 34d is restricted, when the developer storage container 32Y moves in the mounting direction, the shutter 34d whose movement in the mounting direction is stopped moves in a direction relative to movement of the cap section 34Y in the mounting direction. Further, when the cap section 34Y moves to the front side in the mounting direction further than the shutter 34d whose movement is stopped, the toner discharge opening W is opened as illustrated in FIG. 74.

At this time, as illustrated in FIG. 74, the vertical surfaces of the cap section 34Y are sandwiched by first sandwiching members 73d1, and the protrusions 34d1b1 positioned in the engaged protruding section 34d1b of the shutter 34d are engaged by second sandwiching members 73d2. Since the opening operation of the shutter 34d is performed in the state in which the shutter 34d is sandwiched, the postures of the shutter 34d and the cap section 34Y in the cap receiving section 73 are decided, and thus the opening/closing operation can be smoothly performed.

Meanwhile, when extracting (separating) the developer storage container 32Y from the developer storage container storage unit 70 (the cap receiving section 73), the operation

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is performed in a procedure reverse to the mounting procedure. That is, the operation of the body side shutter closing mechanism **73d** (the shutter sandwiching mechanism) accompanying with the closing operation of the shutter **34d** is performed in order of FIGS. **74**, **73**, and **72**.

The seal state of the seal member **36** on the toner discharge opening **W** at the time of the opening/closing operation of the shutter will be described in connection with the movement position of the shutter **34d** with reference to FIGS. **75A** to **75D**.

FIG. **75A** illustrates the state in which the toner discharge opening **W** of the cap **34** is closed by the shutter **34d**. In this state, since the developer storage container is not loaded on the cap receiving section **73**, the shutter **34d** closes the toner discharge opening **W**. Since the seal material **36** is in press contact with the rib **W1** positioned at the circumferential edge of the toner discharge opening **W**, the state in which the shutter **34d** is in close contact with the toner discharge opening **W** is maintained. A dotted line in FIG. **75A** represents the state in which the stopper release section **34d21** of the shutter **34d** is pushed up by the stopper release urging section **72b** at the cap receiving section **73** side. The shutter deforming section **34d2** is defectively deformed from the inclined state to the horizontal state. The stopper section **34d22** positioned at the free end of the shutter deforming section **34d2** is released from engagement with the engaging end surface **34n1** positioned in the wide width box section **34n** that is at the bottom side of the cap section **34Y** as illustrated in FIGS. **69A** and **69B**.

Thus, as described in FIGS. **72** to **74**, it can move up to the position where the engaged protruding section **34d1b** at the shutter **34d** side is sandwiched by the second sandwiching members **73d2** of the body side shutter closing mechanism **73d**. As described in FIG. **72**, movement of the shutter **34d** in the mounting direction is restricted, whereas the cap section **34Y** can move in the mounting direction, so that the toner discharge opening **W** is opened, and the state of FIG. **75B** is obtained. FIG. **75B** illustrates the state in which the developer storage container is inserted toward the cap receiving section **73**.

FIG. **75C** illustrates a state of a section indicated by symbol **C** in FIG. **75B**, that is, a state directly before the toner container starts an removing operation from the main body, and the shutter **34d** starts to close the toner discharge opening **W** during the operation. In FIG. **75C**, when the shutter **34d** further moves in the direction of closing the toner discharge opening **W**, the corner (a ridgeline section) of the seal material **36** at the leading end side hits against the rib **W1** positioned at the circumferential edge of the toner discharge opening **W** and so gets caught (ride up) between the rib **W1** and the upper seal surface.

FIG. **75D** illustrates the state in which the toner discharge opening is completely closed by the shutter **34d**. At the time of closing completion, the corner of the seal material **36** at the leading end side gets caught in and comes in close contact with the rib **W1** side. The leading end surface of the seal material **36** is pulled by the caught ridgeline section and deformed, and rides up to cover the contact section between the rib **W1** and the seal material **36** when the cap section **34Y** is viewed from the front.

As a result, since the toner discharge opening **W** is sealed by the seal material **36** until the developer storage container is completely mounted, the toner can be prevented from carelessly leaking from the toner discharge opening **W**.

The shutter mechanism according to the present fifteenth embodiment is the invention for solving the above mentioned third problem. In the present embodiment, by the

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configuration in which the rib **W1** is disposed on the circumferential edge for the toner discharge opening **W** used as the existing component and the configuration of the seal material **36** having a section that gets caught by hitting against the rib **W1** for the seal material used as the existing component, adhesion on the toner discharge opening increases without adding any other component, and so leak of the toner can be prevented with a high degree of certainty.

Particularly, since the toner discharge opening **W** has the hexagonal shape, the leading end of the seal material **36** concentratedly receives a load causing turning-up and is easily turned up, and turning-up can be caused while alleviating sliding resistance in the entire end section in continuity with the top section of the hexagon on which the load is concentrated. Thus, adhesion on the entire circumferential edge of the toner discharge opening **W** can be secured.

Sixteenth Embodiment

Next, an embodiment in which an ID chip as another invention is mounted in the developer storage container according to the fifteenth embodiment will be described.

In the present embodiment, a connector, at the cap receiving side, to which an electrical connection relation with an ID chip **2035** illustrated in FIGS. **76A** and **76B** is set increases matching of the connection position, and the contact failure by the toner at the connection position is prevented. A description will be made in connection with this configuration.

FIGS. **76A** and **76B** are front views illustrating a configuration of the ID chip **2035** and a toner container **2032Y** in which the ID chip **2035** is mounted. In the same figure, the ID chip **2035** is configured such that, with respect to on a central section of a rectangular terminal plate body, a terminal **2035a** is provided at the right, and a non-contact type communication area (an antenna section) **2035b** such as a wireless type is provided at the left (see FIG. **76A**). Since both the contact-type and the non-contact type are provided, there are the following merits. For writing on the ID chip in an assembly or toner filling process in a toner container factory, toner information or the like is written in the ID chip by non-contact communication during an assembly line operation. Thus, the manufacturing speed can be remarkably improved, and thus an inexpensive toner container having a low manufacturing cost can be produced. Meanwhile, inside the image forming apparatus, an inexpensive non-contact type electronic substrate can be employed in a body side communication device, thereby contributing to the cost reduction of the image forming apparatus.

Next, a mounting configuration of the ID chip on the toner container according to the sixteenth embodiment will be described. The ID chip **2035** has semicircular notches **2035d** on a central section. As illustrated in FIG. **76B**, the ID chip **2035** is held on the leading end surface of a cap section **2034Y** of the toner container **2032Y** to be slightly movable in the horizontal direction that is the longitudinal direction. As the holding method, the ID chip **2035** is sandwiched and held between two flange pins **2034f**, which are disposed at nearly the center of the relevant leading end surface, at the positions of the notches **2035d**. A gap between the two flange pins **2034f** is larger than the shortest width between the two notches **2035d**, and so the ID chip **2035** is held on the cap section **2034Y** with a backlash.

Meanwhile, in a cap receiving section **2073** of the developer storage container storage unit **70**, as will be described later, a through hole **2073f** that exposes a connector **2073e** (that is not shown in FIG. **77** for convenience) used as an

electrical connection section on the ID chip **2035** and a wall section **2073g** are disposed as illustrated in FIG. 77.

The wall section **2073g** is a section for shielding a surrounding area of the connector **2073e** which will be described later, and when the connector **2073e** which will be described later is exposed through the through hole **2073f**, the wall section **2073g** blocks the toner from entering the connector **2073e**.

The through hole **2073f** is a place that allows the connector **2073e** disposed in a common electric substrate which will be described in FIG. 78 to be exposed and face the ID chip **2035**.

FIG. 78 is a view illustrating a configuration of the connector **2073e** disposed in the common electric substrate. In the same figure, the connector **2073e** includes a plurality of terminal plates **2073e1** disposed in a connector body **2073e0**. As the terminal plate **2073e1**, used is a bent flexible metallic plate having excellent conductivity.

In the connector **2073e**, disposed is a configuration for performing positioning at the time of contact with the ID chip **2035**, which will be described below.

In FIG. 78, on a surrounding area of the terminal plate **2073e1** disposed in the connector **2073e** or part thereof, the wall section **2073g** illustrated in FIG. 77 is disposed, and on part of the wall section **2073g**, formed are positioning pins **2073e3** that are fittable into positioning holes **2035b** and **2035c** (see FIGS. 76A and 76B) disposed at the ID chip **2035** side.

The positioning holes **2035b** and **2035c** are for contact-positioning with the terminal plates **2073e1** at the connector **2073e** on the terminal **2035a** at the ID chip **2035** side. In order to make it easier to fit the positioning pin **2073e3** into, one is formed of a round hole, and the other is formed of an elongate hole. In FIG. 79 which will be described later, the flange pin **2034f** is fixed to a concave section **2035a** and protrudes from a holding member **2034k**. A reference numeral **34q** represents a pedestal of the holding member **2034k**.

The connector **2073e** at the common electric substrate side is connected with the ID chip **2035** in a state illustrated in FIG. 79. In FIG. 79, when the cap section **2034Y** of the developer storage container is inserted into a cap receiving section **2073**, the positioning pins **73a** and **73b** at the cap receiving section **2073** side are inserted into the positioning holes **34a** and **34b** at the cap section **2034Y**, and so the cap section **2034Y** is positioned in the cap receiving section **2073**.

When the cap section **2034Y** is further inserted, the positioning pins **2073e3** of the connector **2073e** moves inside the positioning holes **2035b** and **2035c** at the ID chip **2035** side, and so the position of terminals **2035a1** at the ID chip **2035** side matches with the position of the terminal plates **2073e1** at the connector **2073e**, thereby preventing the contact failure caused by position mismatching.

As the cap section **2034Y** is inserted, the wall section **2073g** positioned around the connector **2073e** covers not only a surrounding area of the connector but also a surrounding area of the ID chip **2035** as indicated by an alternate long and two short dashes line in FIG. 79. Further, the ID chip **2035** is installed at the upper position away from the toner discharge opening W. Thus, it is possible to prevent the toner scattered from the toner discharge opening W from sticking to the contact position between the terminals.

Seventeenth Embodiment

Next, a toner container in which both techniques of the shutter configuration mentioned in the toner container

according to the fifteenth embodiment and the ID chip **535** according to the fifth embodiment are mounted will be described as a seventeenth embodiment.

A target configuration is a configuration related to the body side shutter closing mechanism **73d** that has been described in FIGS. 73 to 75D.

FIG. 80 is a perspective view viewed from the front right side in an insertion direction of a cap **2134Y** in the state in which the shutter **34d** is closed, and FIG. 81 is a perspective view viewed from the front left side in the insertion direction of the cap **2134Y** in the state in which the shutter **34d** is opened. These figures are different from the previous drawings in the following points.

In FIG. 80, unlike the configuration illustrated in FIG. 67, a front cover **2134P** for preventing falling of the ID chip **535** loaded into the concave section **34i** is disposed on the front surface of the cap **2134Y**.

A configuration for mounting the front cover **2134P** includes a heat calking pin **2134P10** disposed, on the front surface of the cap **2134Y**, below the front surface center and a pair of main and sub reference pins **734S3** that are disposed at the positions different from the heat calking pin **2134P10** while sandwiching the concave section **34i** as illustrated in FIG. 82. After the front surface cover **2134P** is fixed, the heat calking pin **2134P10** becomes a state in which the leading end is crushed by a jig while being heated, but a non-crushed state is illustrated in FIGS. 82 to 86.

In the front surface cover **2134P**, holes into which the pins **2134P10** and **734S3** are inserted and an opening that exposes part of the ID chip **535** to the outside are formed, respectively.

By fitting the main reference pin **734S3** and the sub reference pin **734S3** into and inserting the heat calking pin **2134P10** into, the front surface cover **2134P** is positioned in the state in which the ID chip **535** is exposed to the outside. The heat calking pin **2134P10** is heated and compressed, so that the front surface cover **2134P** is fixed to the front surface of the cap **2134Y**.

In the holes, at the front surface cover **2134P** side, into which the pins are fitted, one of the reference pins is a round hole, and the other is an elongate hole, a longitudinal direction of which is horizontal. Further, the insertion hole of the heat calking pin **2134P10** has the diameter slightly larger than the heat calking pin **2134P10**.

By the fixing state, even if the toner container **2132Y** is inserted into or separated from the toner container storage unit, the ID chip **535** does not fall off, and communication or electrical connection of the ID chip exposed to the outside through the opening can be performed.

Meanwhile, a structure related to the body side shutter closing mechanism **73d** includes a guide rail **2134YG** disposed at the side surface of the narrow width box **34Y1** of the cap **2134Y**.

The guide rail **2134YG** has a configuration different from the guide rail **34YG** illustrated, for example, in FIG. 68. As illustrated in FIGS. 80 and 81, the guide rail **2134YG** includes a protruding section **2134YG3** that is configured to protrude to the front side further than the leading end surface of the narrow width box section **34Y1** and have a protruding portion rounding toward the central side. The protruding sections **2134YG3** are symmetrically disposed on both sides of the narrow width box **34Y1**.

Further, as a configuration different from the configurations of the above embodiments, as illustrated in FIG. 83, at the position (the position indicated by a reference numeral **2134P3**) facing the engaged protruding section **34d12b** of the shutter **34d** in the circumferential surface of the medium

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diameter cylindrical section 34Y2, formed is a concave section that has the outer diameter smaller than the outer diameter of the medium diameter cylindrical section 34Y2. The circumferential surface 2134P3 that forms the concave section is configured not to interfere with turning of a sandwiching member 73d2 disposed in the body side shutter closing mechanism 73d illustrated in FIG. 72.

In this configuration, when the cap 2134Y is loaded on the cap receiving section 73 of the apparatus body in the same procedure as illustrated in FIGS. 72 to 74, the cap 2134Y is sandwiched by the body side shutter closing mechanism 73d. FIGS. 84 to 86 are views corresponding to FIGS. 72 to 74 illustrating the loading state of the cap section 34Y used in the above configuration.

At the time of the opening operation of the shutter 34d, first, as illustrated in FIG. 84, in accompany with the mounting operation of the developer storage container 32Y in the white arrow direction, the first sandwiching members 73d1 come in contact with the protruding sections 2134YG3.

Thereafter, as illustrated in FIG. 85, when the mounting operation of the developer storage container 32Y proceeds in the white arrow direction, the body side shutter closing mechanism 73d (the shutter sandwiching mechanism) is pushed by the protruding sections 2134YG3 and so rotates on the support shaft section 73d3.

When the body side shutter closing mechanism 73d rotates, the first sandwiching members 73d1 sandwich the vertical surfaces of the guide rails 2134YG continuing from the protruding section 2134YG3, and the second sandwiching members 73d2 sandwich the side wall surfaces of the main shutter section 34d1 while being engaged with the protrusions 34d1b1 positioned in the engaged protruding section 34d1b of the shutter 34d.

Thereafter, the shutter 34d comes in contact with the wall section (not shown) formed around the toner feeding opening at the cap receiving section 73 side and so stops movement in the mounting direction. At this time, the vertical surfaces of the guide rails 2134YG are sandwiched by the first sandwiching sections 73d1.

In the state in which movement of the shutter 34d is stopped, when the toner container 2132Y moves in the mounting direction, the shutter 34d whose movement in the mounting direction is stopped relatively moves when viewed from the cap section 2134Y, and the narrow width box section 34Y12 of the cap section 2134Y moves to the front side in the mounting direction further than the shutter member 34d. By the relative movement, as illustrated in FIG. 86, the toner discharge opening W is opened.

At this time, as illustrated in FIG. 74, the vertical surfaces of the cap section 2134Y are sandwiched by the first sandwiching members 73d1, and the protrusions 34d1b1 positioned in the engaged protruding section 34d1b of the shutter 34d are engaged by the second sandwiching members 73d2. Since the opening operation of the shutter 34d is performed in the state in which the shutter 34d is sandwiched, the postures of the shutter 34d and the cap section 2134Y in the cap receiving section 73 are decided, and thus the opening/closing operation of the shutter 34d can be smoothly performed.

Meanwhile, when extracting (separating) the developer storage container 2132Y from the developer storage container storage unit 70 (the cap receiving section 73), the operation is performed in a procedure reverse to the mounting procedure. That is, the operation of the body side shutter closing mechanism 73d (the shutter sandwiching mechanism)

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accompanying with the closing operation of the shutter 34d is performed in order of FIGS. 86, 85, and 84.

In the configuration illustrated in FIG. 81, since the protruding section 2134YG3 that is present at the front end of the guide rail 2134YG protrudes to the front side further the front surface of the narrow width box section 34Y12, turning start timing of the body side shutter closing mechanism 73d is delayed. That is, since the protruding section 2134YG3 protrudes from the front surface of the narrow width box section 34Y12 to the outside, when the cap section 2134Y is extracted, a time period when turning of the first sandwiching section 73d1 is stopped by the protruding section 2134YG3 is lengthier, and the shutter 34d remains sandwiched for a longer time compared to when the protruding section 2134YG3 is not disposed.

When the cap section 2134Y moves in the extracting direction, since the first sandwiching section 73d1 faces the engaged protruding section 34d1 of the shutter 34d, the non-turnable state is maintained. For this reason, a protrusion amount of the protruding section 2134YG3 is set so that the body side shutter closing mechanism 73d can be maintained in the non-turnable state until the shutter 34d is completely closed, and sandwiching of the guide rail 2134YG by the first sandwiching section 73d1 can be released when the shutter 34d completely closes the toner discharge opening W.

Since the engaged protruding sections 34d1b at the shutter 34d side are sandwiched by the second sandwiching sections 73d2 until the toner discharge opening W is completely closed by the shutter 34d, when the cap 34Y moves in the extracting direction, the shutter 34d traverses the toner discharge opening W in the sandwiched state and so closes the toner discharge opening W.

Next, a description will be made in connection with features of the toner used in the developer feeding device as follows.

As the toner contained in the toner containers 32Y, 32M, 32C, and 32K, toner formed so that the following relations hold true:

$$3 \square D_v \square 8 \quad (1)$$

$$1.00 \square D_v / D_n \square 1.40 \quad (2),$$

where D_v (μm) represents a volume-average particle diameter, and D_n (μm) represents a number-average particle diameter. A toner particle is selected according to an image pattern in the developing process and excellent image quality is maintained, and satisfactory developing capability is maintained even if the toner is agitated for a long time in the developing device. Moreover, the toner can be efficiently and reliably conveyed without blocking the toner supply path.

The volume average particle diameter and the number average particle diameter of toner can be measured by using a typical device such as a Coulter Counter type particle diameter distribution measuring device: Coulter Counter-TA-II (manufactured by Coulter Electronics Limited); or Coulter Multisizer II (manufactured by Coulter Electronics Limited).

Furthermore, in the present embodiment, as toner contained in the developer storage containers 32Y, 32M, 32C, and 32K, used is substantially spherical toner that is formed so that a shape factor SF-1 is in a range of 100 to 180 and a shape factor SF-2 is in a range of 100 to 180. As a result, high transfer efficiency is maintained, and reduction in cleaning performance is suppressed. Moreover, the toner can

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be efficiently and reliably conveyed without blocking the toner supply path such as the tube **71**.

Here, the shape factor SF-1 represents the sphericity of the toner particle and obtained by the following equation.

$$SF-1=(M^2/S)\times(100/4)$$

In the above equation, M is the maximum particle diameter (the largest particle diameter in uneven particle diameters) in a project plane of the toner particle, and S is a project area of the toner particle. Therefore, the toner particle whose shape factor SF-1 is 100 is perfectly spherical, and the degree of sphericity lowers as it becomes greater than 100.

The shape factor SF-2 represents irregularity of the toner particle and obtained by the following equation.

$$SF-2=(N^2/S)\times(100/4)$$

In the above equation, N is the circumferential length in the project plane of the toner particle, and S is the project area of the toner particle. Therefore, the toner particle whose shape factor SF-2 is 100 has no irregularity and the irregularity becomes larger as it becomes greater than 100.

The shape factor SF-1 and the shape factor SF-2 are obtained by photographing the toner particle by a scanning electron microscope "S-800" (manufactured by Hitachi, Ltd.) and analyzing the photograph of the toner particle by an image analyzer "LUSEX3" (manufactured by Nireco Corp.).

In the first to eight embodiments and the eleventh to seventeenth embodiments, only toner (one component developer) is contained in the toner container (designated as **32Y**, **32M**, **32C**, and **32K**) as the developer container. However, as for an image forming apparatus that appropriately supplies the developing device with a two component developer composed of toner and a carrier, the two component developer can be contained in the toner container (the developer container). Even in these cases, the same effects as in the above embodiments can be obtained.

In the first to eight embodiments and the eleventh to seventeenth embodiments, some or all of image forming units **6Y**, **6M**, **6C**, and **6K** can be replaced with process cartridges. Even in this case, the same effects as in the above embodiments can be obtained.

Further, in the first to sixth embodiments and the eleventh to seventeenth embodiments, by rotatably configuring the container body **33Y**, a configuration has been made to convey the toner contained in the container body **33Y** toward the opening A. On the other hand, as in the seventh embodiment described with reference to FIGS. **48** and **49**, the toner contained in the container body **1033Y** may be conveyed toward the opening A such that the container body **1033Y** is configured to be non-rotatably held on the toner container storage unit **70** together with the cap section **1034Y**, and a conveying member (for example, a conveying member that includes a plurality of conveying blade members installed on a shape section and rotates in a predetermined direction) that conveys the toner toward the opening A inside the container body **1033Y** is installed. Even in this case, the same effects as in the above embodiments can be obtained.

Further, in the above embodiments, in the substrate (designated as **35b** or **535b**) of the ID chip (designated as **35** or **535**), a plurality of metallic pads **35a** have been arranged in line in the vertical direction so that the position in the longitudinal direction is not misaligned. On the other hand, in the substrate of the ID chip, a plurality of metallic pads **35a** may be arranged in the vertical direction so that the position in the longitudinal direction is alternately mis-

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aligned in a zigzag form. In this case, in order to conform to the metallic pads **35a** arranged in the zigzag form, a plurality of body side terminals (designated as **73e2** or **573e2**) in the connector (designated as **73e** or **573e**) are also arranged in the zigzag form. Even in this case, the same effects as in the above embodiments can be obtained.

Furthermore, in the above embodiments, the present invention has been applied to the ID chip (the information storage device) disposed in the toner container **32Y** (the developer container) or the like as the removable device removably installed on the image forming apparatus body **100** or the like. However, the application of the present invention is not limited thereto, and the present invention can be applied even to any other removable device removably installed on the image forming apparatus body **100** or the like as long as the information storage device is installed in the removable device similarly to the above embodiments. For example, in the image forming apparatus **100** illustrated in FIG. **1**, even when the information storage device is installed in the process cartridges **6Y**, **6M**, **6C**, and **6K** as the removable device, the fixing device **20** (the fixing unit) as the removable device, the intermediate transfer unit **15** as the removable device, or the like, the present invention can be applied to each of them similarly to the above embodiments. Even in these cases, the same effects as in the above embodiments can be obtained.

The present invention is not limited to the above embodiments, and it is obvious that the above embodiment can be appropriately changed in addition to what have been suggested in the above embodiments. Further, the number, the position, the shape, and the like of component members are not limited to the above embodiments and may be changed to the number, the position, the shape, and the like suitable for implementing the present invention.

In the present disclosure, there is another invention corresponding to the fifteenth to seventeenth embodiments for solving the third problem. This can be summarized in the form of claims as follows.

1. A developer storage container removably installed in an image forming apparatus body, comprising:
 - a cap section provided with a toner discharge opening configured to discharge toner in a vertical direction, outside the developer storage container; and
 - a shutter that is held on the cap section and is configured to move along an outer surface of the cap section to open and close the toner discharge opening
 wherein, a seal material that is deformable and made of a flexible material is disposed on a surface of the shutter, the surface facing the toner discharge opening, and
 wherein when the shutter moves in a direction in which the shutter moves to close the toner discharge opening, a leading end of the seal material in the direction is rolled up toward the toner discharge opening so that the leading end comes into close contact with a circumferential edge of the toner discharge opening.

What is claimed is:

1. A toner container, comprising:

- a shutter to open and close a toner discharge opening of the toner container; and
- a slide portion to hold the shutter slidable relative to the toner discharge opening and parallel to an attaching direction in which the toner container attaches to an image forming apparatus, wherein

the shutter includes:

- a slider which slides in the slide portion, and
- an elastic seal sandwiched between the toner discharge opening and a main portion of the shutter, and

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the elastic seal is compressed when the shutter closes the toner discharge opening.

2. The toner container according to claim 1, wherein the shutter further includes a stopper to regulate a slide movement of the shutter when the toner container is outside of the image forming apparatus. 5

3. The toner container according to claim 2, wherein the shutter further includes an inclined surface, the inclined surface is inclined relative to the attaching direction, and 10

the inclined surface contacts to the image forming apparatus, during installation of the toner container to the image forming apparatus, to deform the stopper to release a regulation of the slide movement.

4. The toner container according to claim 2, wherein the stopper includes a contact surface, which contacts a portion of the toner container excluding the shutter, to regulate the slide movement of the shutter, and the contact surface is disposed outside of the toner discharge opening in a direction orthogonal to the attaching direction. 15 20

5. The toner container according to claim 1, wherein the shutter further includes the main portion and a stopper, the elastic seal is attached on the main portion, the elastic seal faces and seals the toner discharge opening, and 25

the elastic seal is not attached on the stopper.

6. The toner container according to claim 5, wherein the stopper is integrally connected to an upstream side, in the attaching direction, of the main portion of the shutter. 30

7. The toner container according to claim 6, wherein the elastic seal is attached to a downstream side, in the attaching direction, of the main portion of the shutter.

8. An image forming apparatus, comprising: the toner container according to claim 5; and a toner container receiving connector to receive the toner container. 35

9. An image forming apparatus, comprising: the toner container according to claim 1; and a toner container receiving connector to receive the toner container. 40

10. A toner container, comprising: a shutter to open and close a toner discharge opening of the toner container; and a slide portion to hold the shutter slidable relative to the toner discharge opening and parallel to an attaching direction in which the toner container attaches to an image forming apparatus, wherein the shutter includes: 45

a main portion, an inclined surface inclined relative to the attaching direction, a slider which slides in the slide portion, 50

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an elastic seal sandwiched between the toner discharge opening and the main portion, and

a stopper to regulate a slide movement of the shutter when the toner container is outside of the image forming apparatus,

the elastic seal is compressed when the shutter closes the toner discharge opening,

the inclined surface contacts to the image forming apparatus, during installation of the toner container to the image forming apparatus, to deform the stopper to release a regulation of the slide movement,

the elastic seal is attached on the main portion,

the elastic seal faces and seals the toner discharge opening, and

the elastic seal is not attached on the stopper.

11. An image forming apparatus, comprising: the toner container according to claim 10; and a toner container receiving connector to receive the toner container.

12. A toner container, comprising:

a shutter mechanism to open and close a toner discharge opening of the toner container; and

a slidably holding member portion to hold the shutter mechanism such that the shutter mechanism is slidable relative to the toner discharge opening and parallel to an attaching direction in which the toner container attaches to an image forming apparatus, wherein

the shutter mechanism includes:

a touching and sliding member for touching and sliding in the slidably holding member,

a sealing member for sealing between the toner discharge opening and the shutter mechanism, the sealing member being compressed when the shutter mechanism closes the toner discharge opening, and

a regulating mechanism to regulate a slide movement of the shutter mechanism when the toner container is outside of the image forming apparatus.

13. The toner container according to claim 12, wherein the regulating mechanism includes a contact surface which contacts a portion of the toner container excluding the shutter mechanism to regulate the slide movement of the shutter mechanism.

14. The toner container according to claim 13, wherein the contact surface is disposed outside of the toner discharge opening in a direction orthogonal to the attaching direction.

15. An image forming apparatus, comprising:

the toner container according to claim 12; and

a toner container receiving connector to receive the toner container.

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